

# RAMANUJAN–SATO IDENTITIES: PERIOD LINES AND ARITHMETIC CM CRITERIA

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**ABSTRACT.** A first-order identity for an elliptic period square determines a line in de Rham cohomology. Chudnovsky’s theorem makes this line unique; complex multiplication supplies it as the eigenline complementary to the Hodge filtration. We use this interpretation to classify the CM identities in the four standard Ramanujan–Sato families: there are 36 primitive standard classes and 35 linear Euler companions. Exhaustion outside the CM locus follows from an explicitly stated quadratic-period independence hypothesis. We express that hypothesis in modular jets and distinguish it from the established theorems on complete comparison tensors.

Two unconditional arithmetic criteria complement this complex theory. Weighted congruences at lower prime density greater than  $3/4$  characterize the standard classes, while exact preservation of the Hodge line by crystalline Frobenius at one ordinary place characterizes CM. Variation of the periods also gives divergent denominator gaps for bounded multiplier degree and subpower counts in bounded height windows. The proofs incorporate the precise specialization, algebraization, and counting theorems on which these conclusions depend. Auxiliary constructions, coordinate certificates, further families, and counterexamples are developed in the appendices. The implication from a single complex identity to CM remains unproved.

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## 1. THE CLASSIFICATION PROBLEM

For  $s \in \{1/6, 1/4, 1/3, 1/2\}$  let

$$F_s(z) = {}_3F_2\left(\begin{matrix} 1/2, s, 1-s \\ 1, 1 \end{matrix}; z\right) = \sum_{n \geq 0} c_s(n) z^n, \quad c_s(n) = \frac{(1/2)_n (s)_n (1-s)_n}{(n!)^3}, \quad \vartheta = z \frac{d}{dz}.$$

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The pairs  $(s_\ell, R_\ell)$  attached to levels  $\ell = 1, 2, 3, 4$  are

$$(1/6, 1728), \quad (1/4, 256), \quad (1/3, 108), \quad (1/2, 64).$$

We use the power-series branch at zero and define

$$\mathfrak{S}_\ell(A, B, C) = \sum_{n \geq 0} R_\ell^n c_{s_\ell}(n) (A + Bn) C^{-n} = (A + B\vartheta) F_{s_\ell}(R_\ell/C), \quad (1.1)$$

where the last expression always means the ordinarily convergent sum.

**Definition 1.1.** A primitive standard identity is an ordinarily convergent equality

$$\mathfrak{S}_\ell(A, B, C) = \frac{\alpha}{\pi}, \quad A, B, C \in \mathbb{Z}, \quad C \neq 0, \quad \gcd(A, B) = 1, \quad \alpha \in \overline{\mathbb{Q}}.$$

Classes identify  $(A, B, \alpha)$  with  $(-A, -B, -\alpha)$ , at the same level and denominator. The multiplier is initially allowed to be zero.

The primitive condition concerns the coefficient pair. Requiring only  $\gcd(A, B, C) = 1$  would permit infinitely many multiples of the identity  $(A, B, C) = (1, 6, 256)$  at level four. Analytic continuation is likewise excluded from the definition: at a convergence boundary, the value must be the sum of the original series.

**Hypothesis 1.2** (Quadratic-period independence). *Let  $z = R_\ell/C \in [-1, 1) \setminus \{0\}$ , with  $C \in \mathbb{Z}$ , and put  $x = (1 - \sqrt{1 - z})/2$ . For the algebraic elliptic curve  $E_{s_\ell, x}$  of Section 2.4, use the analytic cycle normalized by*

$$\omega = \frac{2\pi}{\sqrt{6}} {}_2F_1(s_\ell, 1 - s_\ell; 1; x), \quad h = \int_\gamma X \frac{dX}{y}.$$

*If this curve is non-CM, then  $\pi, \omega^2, \omega h$  are linearly independent over  $\overline{\mathbb{Q}}$ .*

**Theorem 1.3** (Conditional classification). *Assume Hypothesis 1.2. Every primitive standard identity has a CM elliptic fiber. There are exactly 36 classes, comprising 11, 12, 9, 4 classes at levels 1, 2, 3, 4, respectively, as listed in Tables 1–4. For each fixed level and denominator, the primitive coefficient pair is unique up to simultaneous sign, and its multiplier is nonzero.*

*For the linear Euler companion  $F_s(z)/(1 - z)$ , the complete list has 35 classes. It is obtained by the coefficient transport and ordinary convergence criterion of Theorem 4.7.*

*Proof.* The elliptic realization and its first derivative identify the coefficient space with de Rham cohomology. In the normalized basis,

$$\pi \mathfrak{S}_\ell(A, B, C) = \frac{3}{2\pi} ((A + Bu_s)\omega^2 + Bv_s\omega h), \quad u_s, v_s \in \overline{\mathbb{Q}}, \quad v_s \neq 0. \quad (1.2)$$

Thus a nonzero coefficient pair gives a nontrivial quadratic-period relation. Hypothesis 1.2 forces CM; the positive endpoint is excluded unconditionally in Section 2.6. CM reciprocity and the finite modular correspondence then bound the class number and give the tabulated parameters. The complementary CM eigenline supplies the identity, and its equivariance supplies a rational coefficient slope. These statements, with the exact finite enumeration, are proved in Section 4. Chudnovsky's theorem gives uniqueness and nonvanishing in Section 2.5. Finally, the invertible Euler transport gives the companion classification.  $\square$

**Theorem 1.4** (Unconditional uniqueness). *At every nonzero algebraic real  $z \in [-1, 1)$ , there is at most one  $[A : B] \in \mathbf{P}^1(\overline{\mathbb{Q}})$  such that  $(A + B\vartheta)F_s(z) \in \overline{\mathbb{Q}}/\pi$ . A nonzero coefficient pair cannot give zero. At  $z = 1$ , no nonzero algebraic pair gives a convergent identity in  $\overline{\mathbb{Q}}/\pi$ .*

The first two assertions are consequences of the identity-line theorem; the endpoint uses CM period independence. In particular, the uniqueness assertion in the leading theorem needs no conjecture.

The main proofs are organized by the information carried by an identity. Section 2 constructs its coefficient line intrinsically. Section 3 describes the precise extra comparison information that would force CM and proves the implications from named period conjectures. Section 4 completes the arithmetic classification within the CM locus. The arithmetic criteria and scarcity theorems then give unconditional conclusions with different hypotheses: they neither assume nor prove the missing single-value implication. The appendices contain coordinate certificates, refinements, further families, and obstructions to strengthening those hypotheses.

## 2. ELLIPTIC PERIOD SQUARES

The distinction between existence and uniqueness becomes transparent in de Rham cohomology. A first-order identity selects a line in a two-dimensional algebraic space. Complex multiplication supplies that line; Chudnovsky's theorem makes it unique. The unproved converse asks whether the complex period pairing can select it in any other way.

**2.1. The determinant and the first jet.** Put  $K = \overline{\mathbb{Q}}$  and  $\Pi = 2\pi i$ . Let  $E \rightarrow X$  be a smooth elliptic family over a smooth algebraic curve over  $K$ , and write

$$\mathcal{H} = H_{\text{dR}}^1(E/X), \quad \mathcal{L} = f_* \Omega_{E/X}^1 \subset \mathcal{H}.$$

Here  $f : E \rightarrow X$  is the structure morphism. Choose a local algebraic frame  $(\lambda, \mu)$  of  $\mathcal{H}$ , with  $\lambda$  spanning  $\mathcal{L}$  and normalized cup product  $\langle \lambda, \mu \rangle = 1$ . At an algebraic fiber  $P \in X(K)$ , viewed over  $\mathbb{C}$ , the comparison isomorphism respects the determinant:

$$\begin{array}{ccc} \bigwedge^2 H_{\text{dR}}^1(E_P) \otimes_K \mathbb{C} & \xrightarrow{\bigwedge^2 \text{comp}} & \bigwedge^2 H_B^1(E_P, \mathbb{Q}) \otimes_{\mathbb{Q}} \mathbb{C} \\ \langle \cdot, \cdot \rangle \downarrow & & \downarrow \text{ev}_{\gamma_1 \wedge \gamma_2} \\ \mathbb{C} & \xrightarrow{\times \Pi} & \mathbb{C}. \end{array}$$

The oriented integral basis  $(\gamma_1, \gamma_2)$  is chosen compatibly with this normalization. Thus the Legendre relation is

$$\omega_1 h_2 - \omega_2 h_1 = \Pi, \quad \omega_j = \int_{\gamma_j} \lambda, \quad h_j = \int_{\gamma_j} \mu. \quad (2.1)$$

The determinant is a Tate period; an individual product  $\omega h$  need not be one.

Let  $D$  be an algebraic derivation,  $\gamma$  a nonzero locally flat integral cycle, and  $k$  a nonzero algebraic function. Suppose the analytic function under consideration has the realization

$$H = k \frac{\omega^2}{\Pi^2}, \quad \omega = \int_{\gamma} \lambda, \quad h = \int_{\gamma} \mu. \quad (2.2)$$

Write

$$\nabla_D \lambda = p\lambda + q\mu, \quad u = D \log k + 2p. \quad (2.3)$$

A specialization  $P \in X(K)$  is *admissible* if all these data are regular there and  $k(P)q(P) \neq 0$ . The condition  $q(P) \neq 0$  says that the Kodaira–Spencer map evaluated on  $D$  is nonzero. At such a point the coefficient map

$$\begin{aligned} T_P : K^2 &\xrightarrow{\sim} \mathcal{H}_P, & (A, B) &\mapsto \nu_{A,B}, \\ \nu_{A,B} &= (A + BD \log k)\lambda + 2B \nabla_D \lambda. \end{aligned} \quad (2.4)$$

is an isomorphism. Differentiating the period pairing gives

$$(A + BD)H(P) = \frac{k\omega}{\Pi^2} \int_{\gamma} \nu_{A,B} = \frac{k}{\Pi^2} ((A + Bu)\omega^2 + 2Bq\omega h). \quad (2.5)$$

All quantities in a fiber formula are henceforth evaluated at  $P$ . This construction is intrinsic: replacing  $\lambda$  by  $a\lambda$  and  $k$  by  $k/a^2$  replaces  $\nu_{A,B}$  by  $a\nu_{A,B}$ , leaving the period pairing and the selected line unchanged.

## 2.2. The transcendence input.

**Theorem 2.1** (Chudnovsky). *Let  $E/K$  be an elliptic curve,  $(\lambda, \mu)$  an algebraic de Rham basis with  $\lambda$  holomorphic, and  $\gamma$  a nonzero integral cycle. Then*

$$R = \frac{h}{\omega}, \quad S = \frac{\pi}{\omega} \quad \text{are algebraically independent over } K. \quad (2.6)$$

We specify the source and normalization, because the power of  $\omega$  in this assertion is essential. For a Weierstrass equation  $y^2 = 4X^3 - g_2X - g_3$  over  $K$ , take  $\lambda = dX/y$  and  $\mu = X dX/y$ . Uniformization gives  $h_j = -\eta_j$ , where  $\eta_j$  is the zeta quasi-period of the full lattice period  $\omega_j$ . Consequently

$$\omega_2 \eta_1 - \omega_1 \eta_2 = 2\pi i. \quad (2.7)$$

Corollary 4 of [1], equivalently Corollary 31 of [2, p. 166], gives precisely (2.6). To see its origin in the elliptic-logarithm form of Chudnovsky's theorem, complete a primitive  $\omega_1$  to a period basis and take the algebraic two-torsion point with logarithm  $v = \omega_2/2$ . Then  $\wp(v) \in K$ ,  $v$  and  $\omega_1$  are  $\mathbb{Q}$ -linearly independent, and

$$\zeta(v) - \frac{\eta_1}{\omega_1} v = \frac{1}{2} \left( \eta_2 - \frac{\eta_1 \omega_2}{\omega_1} \right) = -\frac{\pi i}{\omega_1}.$$

This is the second independent quantity in the cited theorem. Replacing the Weierstrass basis by any algebraic basis as above changes  $R$  by an invertible algebraic affine transformation and  $S$  by a nonzero algebraic factor. Multiplying the cycle by a nonzero integer only rescales  $S$ . These operations preserve algebraic independence. In particular,

$$h/\omega \notin K. \quad (2.8)$$

### 2.3. The identity line.

**Theorem 2.2.** *At an admissible specialization let*

$$\mathcal{R}_P = \{(A, B) \in K^2 : (A + BD)H(P) \in K/\Pi\}.$$

*Then  $\dim_K \mathcal{R}_P \leq 1$ , and a nonzero coefficient pair never has value zero. If  $E_P$  has CM,  $\mathcal{R}_P$  is exactly the inverse image under  $T_P$  of the CM eigenline complementary to  $\mathcal{L}_P$ . More explicitly, if that eigenline is  $K(\mu - r\lambda)$ , then*

$$\mathcal{R}_P = K(-(u + 2qr), 1). \quad (2.9)$$

*Thus a rational coefficient pair exists exactly when  $u + 2qr \in \mathbb{Q}$ , and its primitive integer representative is unique up to sign.*

*If  $\Pi, \omega^2, \omega h$  are  $K$ -linearly independent at every admissible non-CM fiber in question, then  $\mathcal{R}_P \neq 0$  is equivalent to CM.*

*Proof.* The holomorphic period  $\omega$  is nonzero: integration identifies integral homology with a lattice in  $\mathbb{C}$ . By (2.8), evaluation on  $\gamma$  is injective on the two-dimensional  $K$ -space  $\mathcal{H}_P$ . Equation (2.5) and the isomorphism  $T_P$  therefore make  $(A, B) \mapsto (A + BD)H(P)$  injective. Its inverse image of the one-dimensional space  $K/\Pi$  has dimension at most one.

Suppose  $E_P$  has CM. A nonreal endomorphism  $\varphi$  has distinct algebraic eigenvalues  $a, b$  on  $\mathcal{H}_P$ , with  $a$  on the holomorphic line. Its other eigenline has the form  $K\nu$ , where  $\nu = \mu - r\lambda$  and  $r \in K$ . The cycles  $\gamma, \varphi_*\gamma$  are independent over  $\mathbb{Q}$ , since a rational dependence would make the nonreal eigenvalue  $a$  rational. If their determinant in an oriented integral basis is  $m \in \mathbb{Z} \setminus \{0\}$ , compatibility of the endomorphism with the comparison isomorphism gives

$$(b - a)\omega \int_{\gamma} \nu = m\Pi, \quad c_P := \frac{\omega(h - r\omega)}{\Pi} = \frac{m}{b - a} \in K^\times. \quad (2.10)$$

Hence the pair in (2.9) has value  $2kqc_P/\Pi \neq 0$ . Uniqueness identifies its line with all of  $\mathcal{R}_P$ . The rational and primitive assertions follow from its representative with second coordinate one.

Finally a scalar identity is equivalent to

$$k(A + Bu)\omega^2 + 2kBq\omega h = \alpha\Pi. \quad (2.11)$$

The two coefficients on the left cannot both vanish for a nonzero pair. The asserted independence therefore excludes a non-CM fiber.  $\square$

This last independence hypothesis is equivalent, at the specified fiber, to exclusion of *all algebraic* coefficient identities: the triangular coefficient transformation in (2.11) is invertible. For the original problem only rational coefficient directions need be excluded. The distinction will be retained in the converse criteria below.

**Corollary 2.3** (CM period independence). *If  $E/K$  has CM, every nonzero holomorphic period  $\omega$  and  $\pi$  are algebraically independent. For the corresponding second-kind period there are  $c, d \in K$ ,  $d \neq 0$ , such that*

$$h = c\omega + d\frac{\pi}{\omega}. \quad (2.12)$$

*Proof.* Equation (2.10) gives this formula with  $c = r$  and  $d = 2ic_P$ . Consequently

$$K(\omega, \pi) = K(R, S), \quad \omega = \frac{dS}{R - c}, \quad \pi = \frac{dS^2}{R - c}.$$

Chudnovsky's theorem makes this field have transcendence degree two. This proves the exact CM consequence also stated in [1, Corollary 6] and [2, Corollary 32]; the algebraic discriminant factor in the former normalization does not affect the assertion.  $\square$

**2.4. The four standard realizations.** The general construction becomes uniform after the quadratic change

$$x = \frac{1 - \sqrt{1 - z}}{2}, \quad z = 4x(1 - x), \quad F_s(z) = f_s(x)^2, \quad f_s(x) = {}_2F_1(s, 1 - s; 1; x). \quad (2.13)$$

The square root is nonnegative on  $[-1, 1]$ . The equality follows from Clausen's identity and the Gauss quadratic transformation, first as analytic germs and then along this real interval. The models  $E_{s,x}$  and their de Rham calculation are given in Appendix A. With  $\lambda = dX/y$  and  $\mu = X dX/y$  they give the following common normalization.

**Lemma 2.4.** *The cycle analytic at the nodal fiber  $x = 0$  has an orientation for which*

$$\omega_s(x) = \frac{2\pi}{\sqrt{6}} f_s(x), \quad F_s(z) = \frac{3\omega_s(x)^2}{2\pi^2}. \quad (2.14)$$

For  $z \in [-1, 1] \setminus \{0\}$  the fiber is smooth and

$$\begin{aligned} \frac{\vartheta F_s}{F_s} &= u_s + v_s \frac{h_s}{\omega_s}, \\ u_s &= -\frac{x(1-x)}{6(1-2x)} \frac{\Delta'_s}{\Delta_s}, \quad v_s = -\frac{\varepsilon_s}{3(1-2x)} \neq 0, \end{aligned} \quad (2.15)$$

where  $\varepsilon_s = 1$  for  $s = 1/6, 1/4, 1/3$  and  $\varepsilon_{1/2} = 2$ .

*Proof.* The Gauss–Manin connection in Appendix A shows that  $\omega_s$  satisfies the Gauss equation

$$x(1-x)\omega_s'' + (1-2x)\omega_s' - s(1-s)\omega_s = 0.$$

All four cubics specialize to  $(X-1)(2X+1)^2$  at zero. The residue at the node on the normalization is  $\pm 1/\sqrt{-6}$ ; the small cycle therefore has limiting period  $\pm 2\pi/\sqrt{6}$ . The Gauss equation has a unique analytic solution with this chosen positive value. This proves the first formula, and (2.13) proves the second. The same connection and  $\vartheta = x(1-x)(1-2x)^{-1}d/dx$  give (2.15). Euler's integral for  $f_s$  is positive for real  $x < 1$ , so the period does not vanish on the interval in use.  $\square$

Thus  $k = -6$ ,  $D = \vartheta$ , and  $2q = v_s$  in the general theory. In particular its nondegeneracy hypothesis holds at every standard parameter below the positive endpoint.

## 2.5. Value independence and uniqueness.

**Theorem 2.5.** *For algebraic  $z \in [-1, 1] \setminus \{0\}$  the two values  $F_s(z), \vartheta F_s(z)$  are algebraically independent over  $K$ . If  $q_0 = e^{2\pi i \tau}$  is the nome obtained by completing the selected primitive cycle to an oriented period basis, then  $q_0, F_s(z), \vartheta F_s(z)$  are algebraically independent.*

*Proof.* Put  $F = F_s(z)$  and  $G = \vartheta F_s(z)$ . Equations (2.14) and (2.15) give

$$F = \frac{3}{2S^2}, \quad G = F(u_s + v_s R), \quad K(F, G) = K(R, S^2).$$

The last field has transcendence degree two by Chudnovsky's theorem.

For the nome statement use the full conclusion of Nesterenko's theorem, in [3, Theorem 1.1]:

$$\text{trdeg}_K K(q_0, E_2(\tau), E_4(\tau), E_6(\tau)) \geq 3.$$

The period formulas in [3, Proposition 4.7], in our second-kind convention, are

$$E_2 = -2FR, \quad E_4 = \frac{g_2}{3}F^2, \quad E_6 = g_3F^3.$$

They place this field inside  $K(q_0, F, G)$ , a field with three generators. Its transcendence degree is therefore three. If the chosen cycle is an integer multiple of a primitive cycle, passing to that primitive cycle changes these formulas only by nonzero algebraic scaling factors and gives the same conclusion.  $\square$

**Corollary 2.6.** *At every such parameter the space*

$$\{(A, B, \alpha) \in K^3 : AF_s(z) + B\vartheta F_s(z) = \alpha/\pi\}$$

*has dimension at most one, and its nonzero elements have  $\alpha \neq 0$ . In particular there is at most one primitive integer pair up to sign, and*

$$\vartheta F_s(z)/F_s(z) \notin K. \quad (2.16)$$

*Proof.* Theorem 2.2 gives the assertion; the multiplier is uniquely determined by the coefficient pair. Equivalently, linear independence of  $F_s$  and  $\vartheta F_s$  makes projection onto the  $\alpha$  coordinate injective on the displayed relation space.  $\square$

**2.6. Ordinary convergence and the endpoint.** Stirling's formula gives

$$c_s(n) \sim \frac{n^{-3/2}}{\Gamma(1/2)\Gamma(s)\Gamma(1-s)}. \quad (2.17)$$

For a nonzero pair  $(A, B)$ , the series is absolutely convergent for  $|z| < 1$  and divergent for  $|z| > 1$ . At  $z = 1$  it converges precisely when  $B = 0$ . At  $z = -1$  the  $A$  part converges absolutely, while the  $B$  part, if present, converges conditionally: the positive sequence  $nc_s(n)$  tends to zero and its successive ratio is  $1 - 1/(2n) + O(n^{-2})$ . Abel's theorem identifies this sum with the radial value used above.

**Proposition 2.7.** *No nonzero algebraic coefficient pair gives an ordinarily convergent value in  $K/\pi$  at  $z = 1$ .*

*Proof.* Only  $B = 0$  is possible. The four smooth endpoint fibers have  $j$ -invariants 1728, 8000, 54000, 1728, respectively, and all have CM. One may verify this directly:  $j = 1728$  has an order-four automorphism;  $y^2 = x^3 + 4x^2 + 2x$ , of invariant 8000, is 2-isogenous to an algebraically isomorphic curve; and the invariant 54000 occurs on a curve 2-isogenous to  $y^2 = x^3 + 1$ . The isogenies are recorded in Appendix A. Corollary 2.3 makes  $\omega_s(1/2)$  and  $\pi$  algebraically independent. But a nonzero identity at this endpoint would make  $\pi F_s(1) = 3\omega_s(1/2)^2/(2\pi)$  algebraic, a contradiction.  $\square$

The endpoint proposition and Corollary 2.6 prove projective uniqueness throughout the ordinary convergence range. They do not turn the independence of  $h/\omega$  and  $\pi/\omega$  into independence of  $\pi, \omega^2, \omega h$ . The latter is precisely the additional comparison required for CM forcing.

### 3. COMPARISON TENSORS AND THE CM IMPLICATION

Put  $k = \overline{\mathbb{Q}}$ . On a smooth algebraic elliptic curve choose a holomorphic de Rham class  $\Omega$  and a complementary class  $\Xi$ , normalized by the polarization. Integration over an oriented integral basis gives

$$M = \begin{pmatrix} \omega & h \\ \omega_2 & h_2 \end{pmatrix}, \quad \det M = 2\pi i, \quad \tau = \omega_2/\omega \in \mathfrak{H}.$$

For the standard families these are the classes and cycles of Section 2.4. Their period normalization turns an identity into

$$a\omega^2 + b\omega h = \alpha\pi, \quad a = \frac{3}{2}(A + Bu_s), \quad b = \frac{3}{2}Bv_s, \quad (a, b) \neq (0, 0). \quad (3.1)$$

The nonvanishing theorem already proved gives  $\alpha \neq 0$ . This section identifies the precise comparison assertion needed to deduce CM, and explains two equivalent ways to complete the information in (3.1).

**3.1. Quadratic genericity.** Let  $\mathcal{P}_E$  be the torsor of tensor isomorphisms between the de Rham and Betti fiber functors on the de Rham–Betti tensor category generated by  $H^1(E)$  and the Tate object, with algebraic coefficients. At a non-CM elliptic curve its group is  $\mathrm{GL}_2$ ; after choosing the preceding bases its coordinate ring is  $k[w, h, v, k_2, (wk_2 - vh)^{-1}]$ . The determinant is the comparison coordinate of the Tate object. Let  $Z_E$  be the Zariski closure over  $k$  of the comparison point  $M$ , and set

$$U_\gamma = \langle w^2, wh, wk_2 - vh \rangle_k \subset \mathcal{O}(\mathcal{P}_E).$$

The three polynomials defining this space are linearly independent. Evaluation factors through the actual closure:

$$\begin{array}{ccccc} U_\gamma & \hookrightarrow & \mathcal{O}(\mathcal{P}_E) & \xrightarrow{\mathrm{ev}_M} & \mathbb{C} \\ & \searrow \text{restriction} & \downarrow & \nearrow & \\ & & \mathcal{O}(Z_E) & & \end{array}$$

The lower injection is induced by evaluation at  $M$ . Thus Hypothesis 1.2 is exactly injectivity of evaluation on  $U_\gamma$  for the specified non-CM fibers. Full genericity of the comparison point would assert injectivity on the entire coordinate ring.

**Proposition 3.1.** *Hypothesis 1.2 forces every standard identity to have a CM fiber.*

*Proof.* The positive endpoint has already been excluded. If a remaining fiber were non-CM, (3.1) would make the nonzero element

$$aw^2 + b\omega h - \frac{\alpha}{2i}(wk_2 - vh) \in U_\gamma$$

vanish at  $M$ , contrary to injectivity.  $\square$

All results below concern either explicit numerical conjectures or complete comparison tensors. None identifies a functional transcendence theorem with its numerical specialization, nor infers the remaining coordinates of a tensor from one algebraic coordinate. The equivalent restricted rational-direction formulation and the obstructions to such inferences are collected in Appendix D.

**3.2. The modular two-jet.** For this subsection take the Weierstrass model  $y^2 = 4X^3 - g_2X - g_3$ , with  $\Omega = dX/y$  and  $\Xi = X dX/y$ . Write

$$j_0 = j(\tau), \quad P = \frac{\omega^2}{\pi}, \quad Q = \frac{\omega h}{\pi}, \quad R(T) = \frac{2}{3T} + \frac{1}{2(T-1728)}.$$

Primes on  $j$  denote ordinary  $\tau$ -derivatives.

**Proposition 3.2** (Period-jet comparison). *If  $j_0 \notin \{0, 1728\}$ , then  $j'(\tau) \neq 0$  and*

$$j' = -9ij_0 \frac{g_3}{g_2} P, \tag{3.2}$$

$$\frac{j''}{j'} = R(j_0)j' - iQ. \tag{3.3}$$

Consequently quadratic-period independence is equivalent to either of the following linear independence assertions over  $k$ :

$$\{1, j', j''/j'\}, \quad \{j', j'^2, j''\}.$$

Moreover,

$$k(P, Q) = k(j', j''). \tag{3.4}$$

*Proof.* With  $D = (2\pi i)^{-1}d/d\tau$ , the lattice formulas in [3, Proposition 4.7], in the convention  $h = -\eta$ , are

$$E_2 = -\frac{3Q}{\pi}, \quad E_4 = \frac{3g_2P^2}{4\pi^2}, \quad E_6 = \frac{27g_3P^3}{8\pi^3}. \tag{3.5}$$

Ramanujan's differential equations

$$DE_2 = (E_2^2 - E_4)/12, \quad DE_4 = (E_2E_4 - E_6)/3, \quad DE_6 = (E_2E_6 - E_4^2)/2$$

applied to  $j = 1728E_4^3/(E_4^3 - E_6^2)$  give

$$Dj = -jE_6/E_4, \quad \frac{j''}{2\pi ij'} = \frac{E_2}{6} - \frac{2E_6}{3E_4} - \frac{E_4^2}{2E_6}.$$

Substitute (3.5) and  $g_2^3/g_3^2 = 27j_0/(j_0 - 1728)$ . This proves (3.2) and (3.3); every denominator is nonzero. The change from  $(P, Q)$  to  $(j', j''/j')$  is invertible and  $k$ -linear. Multiplication by  $j'$  proves the second independence equivalence, and the same formulas and their inverses prove the field equality.  $\square$

**Corollary 3.3.** *The relation (3.1) is equivalent to*

$$bj'' + \left( \frac{ag_2}{9j_0g_3} - bR(j_0) \right) j'^2 + i\alpha j' = 0. \tag{3.6}$$

Unconditionally,  $j''/j'^2$  is transcendental, and the relation space of  $(j', j'^2, j'')$  over  $k$  has dimension at most one.

*Proof.* The inverse dictionary is  $P = ig_2j'/(9j_0g_3)$  and  $Q = i(j''/j' - R(j_0)j')$ . Substitution proves the first assertion. It also gives

$$\frac{j''}{j'^2} = R(j_0) + \frac{g_2}{9j_0g_3} \frac{h}{\omega}.$$

Chudnovsky's theorem, in the form used in the preceding section, makes  $h/\omega$  transcendental. Thus  $j'^2$  and  $j''$  are linearly independent; adjoining one further number creates at most one relation.  $\square$

For  $b = 0$ , the missing implication already contains the restriction of Lang's derivative conjecture to these points: algebraic  $j(\tau)$  and  $j'(\tau) \neq 0$  should imply transcendental  $j'(\tau)$  [2, p. 167, Conjecture 34]. For  $b \neq 0$  it is the affine quadratic obstruction (3.6).

### 3.3. Sufficient numerical conjectures. Let

$$K_E = k(\omega, \omega_2, h, h_2), \quad q = e^{2\pi i \tau}.$$

At a non-CM algebraic elliptic curve, the following numerical forms of the cited conjectures apply:

- (1) The elliptic period conjecture asserts  $\text{trdeg}_k K_E = 4$  [3, Conjecture 4.5].
- (2) Fonseca's modular-value conjecture asserts

$$\text{trdeg}_k k(2\pi i, \tau, q, E_2, E_4, E_6) \geq 5$$

at nonquadratic  $\tau$  [3, Conjecture 4.8].

- (3) The one-point modular Schanuel conjecture with derivatives asserts

$$\text{trdeg}_{\mathbb{Q}} \mathbb{Q}(\tau, j(\tau), j'(\tau), j''(\tau)) \geq 3$$

at nonquadratic  $\tau$  [10, Conjecture 2.3]. This is the one-point case of the lower bound three times the number of distinct  $\text{GL}_2^+(\mathbb{Q})$ -orbits outside the imaginary quadratic locus.

Only their restrictions to the algebraic  $j$ -values in question are needed. Denote the second numerical bound by  $\text{FV}_{\tau}$  and the third by  $\text{MSCD}_{\tau}$ .

**Proposition 3.4** (Hierarchy of sufficient inputs). *At each such non-CM fiber,*

$$\text{FV}_{\tau} \implies \text{trdeg}_k K_E = 4$$

$$\Downarrow$$

$$\text{MSCD}_{\tau} \implies \text{trdeg}_k k(\tau, j', j'') = 3 \implies QPI.$$

*Each of these conjectural inputs therefore suffices for the CM implication and the conditional classification.*

*Proof.* Legendre's identity and (3.4) give

$$h_2 = \tau h + \frac{2\pi i}{\omega}, \quad K_E = k(\omega, h, \tau, \pi), \quad [K_E : k(\pi, \tau, j', j'')] \leq 2. \quad (3.7)$$

Indeed, adjoining  $\omega$  with  $\omega^2 = \pi P$  suffices. Furthermore (3.5) and its inverses

$$P = \frac{2g_2\pi}{9g_3} \frac{E_6}{E_4}, \quad Q = -\frac{\pi E_2}{3}$$

give the exact equality

$$k(2\pi i, \tau, q, E_2, E_4, E_6) = k(\pi, \tau, q, P, Q). \quad (3.8)$$

Thus the modular-value bound implies  $\text{trdeg}_k K_E(q) \geq 5$ . Adjoining  $q$  raises transcendence degree by at most one, whereas  $K_E$  has four generators; hence  $\text{trdeg}_k K_E = 4$ .

The finite extension in (3.7) then makes  $\pi, \tau, j', j''$  algebraically independent. Deleting  $\pi$  gives the middle assertion; MSCD gives it directly since  $j(\tau) \in k$ . Deleting  $\tau$  makes  $j', j''$  algebraically independent, so the nonzero polynomial (3.6) cannot vanish. Equivalently, (3.4) makes  $1, P, Q$  linearly independent.  $\square$

These implications do not assert equivalence of the named conjectures. In fact the modular-value bound here is equivalent to the elliptic period conjecture together with transcendence of  $q$  over  $K_E$ . The exact equalities above prove this by counting transcendence degree. MSCD requires neither this nome extension nor the additional independence of  $\pi$ .

**3.4. Completing the comparison tensor.** Write  $H = H^1(E)$  and use ordinary symmetric powers, with the usual sign modification of cohomological symmetry. Polarization identifies  $\det H$  with the Tate object of period  $2\pi i$ . There is a canonical isomorphism

$$\Phi : \text{Sym}^2 H \otimes (\det H)^{-1} \xrightarrow{\sim} \text{End}_0(H).$$

For a two-dimensional realization  $V$ , it is defined by

$$\Phi(vw \otimes \delta)(x) = \frac{1}{2} \{ \delta(w \wedge x)v + \delta(v \wedge x)w \}, \quad \delta \in (\det V)^{\vee}.$$

This definition is invariant under change of basis. It proves, in particular, the commutativity of the comparison square

$$\begin{array}{ccc} (\text{Sym}^2 H_{\text{dR}} \otimes (\det H_{\text{dR}})^{-1})_{\mathbb{C}} & \xrightarrow{\Phi_{\text{dR}}} & \text{End}_0(H_{\text{dR}, \mathbb{C}}) \\ \text{comp} \downarrow & & \downarrow T \mapsto MTM^{-1} \\ (\text{Sym}^2 H_{\text{B}} \otimes (\det H_{\text{B}})^{-1})_{\mathbb{C}} & \xrightarrow{\Phi_{\text{B}}} & \text{End}_0(H_{\text{B}, \mathbb{C}}). \end{array}$$

Here a subscript  $\mathbb{C}$  denotes extension of scalars from  $k$  or  $\mathbb{Q}$ , respectively. A *complete algebraic de Rham–Betti tensor* is an algebraic de Rham tensor whose entire comparison image has algebraic Betti coordinates.

The exact external input is as follows.

**Theorem 3.5** (Kreutz–Shen–Vial, specialized). *For one elliptic curve over  $k$  and a coefficient field  $L \subseteq k$ , comparison-compatible endomorphisms with algebraic de Rham matrix and Betti matrix over  $L$  form  $\text{End}(E) \otimes_{\mathbb{Z}} L$ . With algebraic coefficients, the de Rham–Betti group generated by  $H$  is  $\text{GL}_2$  in the non-CM case and the torus of the CM field in the CM case. Every complete algebraic de Rham–Betti tensor on a power of  $E$ , with a Tate twist, is an algebraic linear combination of algebraic cycle classes.*

The endomorphism assertion is [4, Theorem 5.10 and Corollary 5.11]; the tensor and group assertions are [4, Theorem 5.13]. The latter theorem allows algebraic coefficient fields containing at most one of the distinct CM fields among the factors. A power of one elliptic curve satisfies this condition, including for  $L = k$ .

**Proposition 3.6** (Exact completion). *Assume (3.1) and  $b \neq 0$ . Put  $\nu = a\Omega + b\Xi$  and  $c = 2ib/\alpha$ . Let  $\Pi$  be the de Rham projector onto  $k\nu$  along  $k\Omega$ . Then the following are equivalent:*

- (1)  $E$  has CM;
- (2)  $\Pi$  is a complete algebraic de Rham–Betti endomorphism;
- (3) the tensor  $t = \Omega\nu/(\Omega \wedge \nu)$  is complete;
- (4) any one of the three Betti entries of  $\Pi$  other than its upper-right entry is algebraic.

At CM, its Betti entries generate precisely the CM field, and

$$c = \bar{\tau} - \tau, \quad b = -\alpha \text{Im } \tau, \quad (\alpha/b)^2 \in \mathbb{Q}_{>0}. \quad (3.9)$$

*Proof.* Put  $u = a\omega + bh$ . The comparison matrix in the basis  $(\Omega, \nu)$  has columns  $(\omega, \tau\omega)^t$  and  $(u, (\tau + c)u)^t$ : its determinant is  $2\pi ib$ , whereas  $\omega u = \alpha\pi$ . Conjugating  $\text{diag}(0, 1)$  gives

$$\Pi_B = \frac{1}{c} \begin{pmatrix} -\tau & 1 \\ -\tau(\tau + c) & \tau + c \end{pmatrix}. \quad (3.10)$$

Algebraicity of either diagonal entry makes  $\tau$  algebraic; algebraicity of the lower-left entry gives a quadratic equation for  $\tau$ . Conversely, algebraic  $\tau$  makes every entry algebraic. Schneider’s theorem identifies algebraicity of  $\tau$ , since  $j(E) \in k$ , with CM. Alternatively, a complete rank-one projector cannot belong to the non-CM endomorphism algebra  $k$ , by Theorem 3.5.

The defining formula for  $\Phi$  gives  $\Phi(t) = \Pi - \frac{1}{2}I$ , proving the equivalence with tensor completion. At CM the algebra  $\text{End}(E) \otimes k$  splits as  $k \times k$ . Its two rank-one projectors select the CM eigenspaces. Since  $\Pi$  kills the holomorphic eigenspace of ratio  $\tau$ , its image has ratio  $\bar{\tau}$ . Thus  $\tau + c = \bar{\tau}$ , proving the first two identities in (3.9). Its entries generate  $\mathbb{Q}(c, \tau) = \mathbb{Q}(\tau)$ : recover  $c$  from the upper-right entry and  $\tau$  from the upper-left entry. For imaginary quadratic  $\tau$ ,  $(\text{Im } \tau)^2$  is positive rational, proving the last assertion.  $\square$

This argument exhibits the exact missing datum. The scalar identity fixes  $\Pi_{12} = 1/c$ , while completion needs one more algebraic entry. For  $b = 0$  the associated tensor is instead a nonzero nilpotent in  $\text{End}_0(H)$ ; it can never be complete, since neither  $k$  nor  $k \times k$  has a nonzero nilpotent. This identifies the different algebraic type of the constant-weight obstruction.

### 3.5. Propagation to a second cycle.

**Proposition 3.7.** *Under (3.1),  $E$  has CM if and only if the same coefficients  $a, b$  give a value in  $k\pi$  on some integral cycle  $m\gamma_1 + n\gamma_2$  with  $n \neq 0$ .*

*Proof.* Set  $L = m + n\tau$ . Legendre’s identity gives  $h_{m,n} = Lh + 2\pi in/\omega$  and  $\omega_{m,n} = L\omega$ . Hence

$$\frac{\omega_{m,n}(a\omega_{m,n} + bh_{m,n})}{\pi} = \alpha L^2 + 2ibnL. \quad (3.11)$$

Algebraicity of this expression, with  $n \neq 0$ , gives a quadratic equation for  $\tau$  whose leading coefficient is  $\alpha n^2 \neq 0$ . Schneider’s theorem proves CM. Conversely, CM makes  $\tau$  algebraic, so the expression belongs to  $k$  on every integral cycle.  $\square$

The second cycle expresses geometrically what the omitted comparison coordinate expresses algebraically. The original identity supplies neither completion nor propagation; each is an exact additional target for an unconditional CM proof.

## 4. CM ARITHMETIC AND THE CLASSIFICATION

The CM classification rests on two structures. A modular correspondence bounds the orbit of the elliptic invariant; the complementary CM eigenline determines the coefficient slope. The first structure makes the parameter set finite, and the second produces an identity at every admissible CM parameter.

**4.1. The modular correspondence.** Write  $j_s(x) = j(E_{s,x})$  for the elliptic families of Section 2.4. Their invariants are

$$\begin{array}{c|c} \ell & j_{s_\ell}(x) \\ \hline 1 & 1728/[4x(1-x)] \\ 2 & 64(1+3x)^3/[x(1-x)^2] \\ 3 & 27(1+8x)^3/[x(1-x)^3] \\ 4 & 256(1-x+x^2)^3/[x^2(1-x)^2]. \end{array} \quad (4.1)$$

The involution  $\iota : x \mapsto 1-x$  fixes  $C = R_\ell/[4x(1-x)]$ . At levels one and four it fixes  $j_s$  as well. At levels two and three it exchanges the invariants of a cyclically isogenous pair. Indeed, with

$$t_n = \left( \frac{\eta(\tau)}{\eta(n\tau)} \right)^{24/(n-1)}, \quad (n, k_n) = (2, 64), (3, 27),$$

the classical  $X_0(n)$  coordinates give

$$x = \frac{k_n}{t_n + k_n}, \quad C = \frac{(t_n + k_n)^2}{t_n}, \quad w_n(t_n) = \frac{k_n^2}{t_n}.$$

Here  $w_n$  is the Fricke involution, and the two maps to the  $j$ -line are

$$\begin{array}{c|cc} n & j(\tau) & j(n\tau) \\ \hline 2 & (t_2 + 256)^3/t_2^2 & (t_2 + 16)^3/t_2 \\ 3 & (t_3 + 27)(t_3 + 243)^3/t_3^3 & (t_3 + 27)(t_3 + 3)^3/t_3. \end{array}$$

Substitution identifies them with  $j_{s_n}(x)$  and  $j_{s_n}(1-x)$ , respectively. Thus the arithmetic information belongs to the correspondence

$$\begin{array}{ccc} X_0(n) & \xrightarrow{j} & X(1) \\ \downarrow / \langle w_n \rangle & & \\ X_0(n)^+ & \xrightarrow{C} & \mathbf{P}_C^1. \end{array}$$

There is no asserted map from the lower row to  $X(1)$ : the missing descent is precisely why a quadratic equation in  $j$  occurs.

The equations of the four correspondences may be written compactly as

$$\begin{aligned} \Phi_1(J, C) &= C - J, \\ \Phi_2(J, C) &= J^2 - (C^2 - 207C + 3456)J + (C + 144)^3, \end{aligned} \quad (4.2)$$

$$\Phi_3(J, C) = J^2 - C(C^2 - 126C + 2944)J + C(C + 192)^3, \quad (4.3)$$

$$\Phi_4(J, C) = (C - 16)^3 - CJ.$$

Each is monic as a polynomial in  $C$ . For  $C \neq 0$ , its degree in  $J$  is

$$m_\ell = (1, 2, 2, 1)_\ell.$$

These two elementary features encode, respectively, integrality and the CM degree bound.

**Proposition 4.1** (Arithmetic of CM parameters). *Let  $C \in \overline{\mathbb{Q}}^\times$  have degree at most  $d$ , and suppose that the indicated nonsingular elliptic fiber has CM by the order of discriminant  $D$ . Then  $C$  is an algebraic integer and*

$$h(D) \leq m_\ell d.$$

*In particular, a rational CM parameter is an integer; it has  $h(D) \leq 1$  at levels one and four and  $h(D) \leq 2$  at levels two and three. At level four a rational CM parameter also satisfies  $C \mid 4096$ .*

*Proof.* We use the following precise consequence of the main theorem of complex multiplication, including nonmaximal orders [9, Chapters 7 and 11]. If  $\text{End}(E) = \mathcal{O}_D$ , then  $j(E)$  is integral and

$$H_D(J) = \prod_{[\mathfrak{a}] \in \text{Pic}(\mathcal{O}_D)} (J - j(\mathbf{C}/\mathfrak{a})) \in \mathbb{Z}[J] \quad \text{is irreducible,} \quad [\mathbb{Q}(j(E)) : \mathbb{Q}] = h(D). \quad (4.4)$$

Artin reciprocity acts simply transitively on these  $h(D)$  invariants over the quadratic field; complex conjugation permutes the same orbit. The degree of a single invariant over  $\mathbb{Q}$  is therefore  $h(D)$ , whereas the ring class field is obtained after adjoining the quadratic field.

Now  $\Phi_\ell(j, C) = 0$  gives

$$h(D) = [\mathbb{Q}(j) : \mathbb{Q}] \leq [\mathbb{Q}(C, j) : \mathbb{Q}] \leq m_\ell [\mathbb{Q}(C) : \mathbb{Q}].$$

Viewed in the other variable,  $\Phi_\ell(j, C)$  is monic over the ring of algebraic integers. Transitivity of integrality makes  $C$  integral, and a rational algebraic integer belongs to  $\mathbb{Z}$ . For level four the equation also gives

$$4096 = C(C^2 - 48C + 768 - j),$$

whose second factor is an integer when  $C$  is rational.  $\square$

**4.2. The canonical CM coefficient line.** The existence of the coefficient slope is most naturally expressed in de Rham cohomology. At a smooth CM fiber over a number field  $K$ , put

$$H = H_{\text{dR}}^1(E/K), \quad L = F^1 H.$$

The CM algebra acts on  $H \otimes_K \overline{\mathbb{Q}}$  with two eigenlines. One is  $L \otimes_K \overline{\mathbb{Q}}$ . Galois preserves their unordered pair and fixes  $L$ , so its companion is also defined over  $K$ :

$$H = L \oplus L^{\text{CM}}. \quad (4.5)$$

This splitting is intrinsic and isogeny invariant. It does not require all geometric endomorphisms to be defined over  $K$ .

In the standard models write  $\Omega = dX/y$  and  $\Xi = X dX/y$ , and put  $H_x = H_{\text{dR}}^1(E_{s,x}/K)$ . For  $x \neq 0, 1, 1/2$ , define

$$Q(x) = \frac{2x(1-x)}{1-2x}, \quad T_x : K^2 \longrightarrow H_x, \quad (A, B) \longmapsto A\Omega_x + BQ(x)\nabla_x \Omega_x.$$

The Gauss–Manin formula (2.15) shows that  $T_x$  is an isomorphism: in the basis  $(\Omega, \Xi)$  its determinant is  $v_s(x) \neq 0$ . The CM coefficient line is consequently

$$\mathcal{R}_x = T_x^{-1}(L_x^{\text{CM}}). \quad (4.6)$$

It is a  $K$ -line, and it is different from  $B = 0$ .

**Proposition 4.2** (The endomorphism determinant). *Let  $z \in [-1, 1)$  be nonzero and algebraic, and suppose that its indicated elliptic fiber has CM. A nonzero coefficient vector  $(A, B)$  on  $\mathcal{R}_x$  satisfies*

$$(A + B\vartheta)F_s(z) = \frac{\alpha}{\pi}, \quad \alpha \in \overline{\mathbb{Q}}^\times.$$

*More explicitly, choose a nonrational endomorphism  $\varphi$  and write*

$$\varphi^* \Omega = \lambda \Omega, \quad \varphi^* \Xi = r \Omega + \bar{\lambda} \Xi, \quad \kappa = \frac{r}{\bar{\lambda} - \lambda}.$$

*If  $d$  is the signed index of  $(\gamma, \varphi_* \gamma)$  in an oriented integral symplectic homology basis, then*

$$\frac{A}{B} = v\kappa - u, \quad (4.7)$$

$$\alpha = \frac{3iBvd}{\bar{\lambda} - \lambda}. \quad (4.8)$$

*Here  $u, v$  are those of (2.15), and  $d \neq 0$ .*

*Proof.* The vector  $\nu = \kappa \Omega + \Xi$  spans the complementary eigenline. Write  $\omega = \int_\gamma \Omega$  and  $\rho = \int_\gamma \nu$ . The intrinsic determinant identity (2.10) gives  $d \neq 0$  and

$$(\bar{\lambda} - \lambda)\omega\rho = 2\pi id.$$

The condition  $T_x(A, B) \in \overline{\mathbb{Q}}\nu$  says  $A + Bu = Bv\kappa$ . The normalization of the standard period now gives

$$(A + B\vartheta)F_s = \frac{3Bv}{2\pi^2} \omega\rho = \frac{1}{\pi} \frac{3iBvd}{\bar{\lambda} - \lambda}.$$

The multiplier is algebraic and nonzero. The identity-line theorem identifies this line with the unique possible algebraic coefficient slope in the ordinary convergence interval.  $\square$

**Lemma 4.3** (Equivariance of the coefficient line). *The parameter involution  $x \mapsto 1 - x$  identifies projective coefficient vectors under an isogeny of the corresponding smooth elliptic fibers. In particular, it preserves the line  $\mathcal{R}_x$  whenever the fiber has CM.*

*Proof.* The modular correspondence supplies an isogeny  $\psi : E_{s,x} \rightarrow E_{s,1-x}$  of degree 1, 2, 3, 1, respectively, over an algebraic extension of the function field. For levels one and four the equality of the  $j$ -invariants supplies an isomorphism; for the middle levels this is the cyclic isogeny on  $X_0(2)$  or  $X_0(3)$  just described. Write  $\psi^*\Omega_{1-x} = c(x)\Omega_x$ .

The multiplier is constant. Indeed, the period equations for the two chosen differentials are the same Gauss equation

$$y'' + \frac{1-2x}{x(1-x)}y' - \frac{s(1-s)}{x(1-x)}y = 0,$$

which is invariant under  $x \mapsto 1-x$ . On a simply connected open set,  $\psi$  is a constant map of Betti local systems. Multiplication of a solution basis by  $c(x)$  multiplies its Wronskian by  $c(x)^2$ ; both Wronskians are constant multiples of  $[x(1-x)]^{-1}$ . Thus  $c(x) = c \in \overline{\mathbb{Q}}^\times$ .

Functoriality of Gauss–Manin and  $Q(1-x) = -Q(x)$  give  $\psi^*T_{1-x} = cT_x$ . After projectivizing, the square

$$\begin{array}{ccc} \mathbf{P}^1(\overline{\mathbb{Q}}) & \xrightarrow{\mathbf{P}(T_{1-x})} & \mathbf{P}(H_{1-x} \otimes \overline{\mathbb{Q}}) \\ \parallel & & \downarrow \mathbf{P}(\psi^*) \\ \mathbf{P}^1(\overline{\mathbb{Q}}) & \xrightarrow{\mathbf{P}(T_x)} & \mathbf{P}(H_x \otimes \overline{\mathbb{Q}}) \end{array} \quad (4.9)$$

commutes. An isogeny intertwines the CM endomorphism algebras and preserves the Hodge lines, hence also their companions. This proves the assertion. The identity extends to every smooth fiber with  $x \neq 1/2$  by extension of homomorphisms of abelian schemes; the constant nonzero differential multiplier remains nonzero.  $\square$

**Theorem 4.4** (Rational descent of CM identities). *Let  $z \in \mathbb{Q} \setminus \{0\}$  with  $-1 \leq z < 1$ . If its indicated elliptic fiber has CM, there is a primitive integer pair  $(A, B)$ , unique up to sign, for which the ordinarily convergent standard series is a nonzero algebraic multiple of  $1/\pi$ .*

*Proof.* Set  $K = \mathbb{Q}(x)$ , where  $x = (1 - \sqrt{1-z})/2$ . The line  $\mathcal{R}_x$  is defined over  $K$  by (4.5). If  $K = \mathbb{Q}$  there is nothing to descend. Otherwise its nontrivial automorphism  $\sigma$  sends  $x$  to  $1-x$ . Galois functoriality gives  $\mathcal{R}_x^\sigma = \mathcal{R}_{1-x}$ , while the commutative square (4.9) gives  $\mathcal{R}_{1-x} = \mathcal{R}_x$ . Therefore  $\mathcal{R}_x \in \mathbf{P}^1(\mathbb{Q})$ .

Choose its relatively prime integer representative. The endomorphism determinant gives the nonzero algebraic multiplier, and Theorem 1.4 gives uniqueness. The convergence statement follows from the assumed interval, including ordinary alternating convergence at  $z = -1$ .  $\square$

**4.3. Finite exhaustion.** Only orders of class number at most two remain. To make this reduction effective, we use the proved classification of quadratic fields in [8]: a negative fundamental discriminant with class number at most two satisfies  $|d| \leq 427$ . For an arbitrary order of discriminant  $D = f^2d$ , the conductor formula is

$$h(f^2d) = \frac{h(d)f}{[\mathcal{O}_d^\times : \mathcal{O}_{f^2d}^\times]} \prod_{p|f} \left(1 - \frac{(d/p)}{p}\right). \quad (4.10)$$

The order class group surjects onto the maximal-order class group, so  $h(d) \leq h(f^2d)$ . The unit index is at most three and

$$h(f^2d) \geq \frac{h(d)}{3} \varphi(f) \geq \frac{h(d)}{3} \sqrt{f/2}. \quad (4.11)$$

The last inequality follows by multiplying  $p^{a-2}(p-1)^2 \geq 1$  over odd prime powers  $p^a || f$ ; the factor at two is at least  $1/2$ . Thus  $h(D) \leq 2$  implies  $f \leq 72$ . The finite range is a theorem, not a numerical cutoff.

**Lemma 4.5** (Exact arithmetic enumeration). *The rational CM parameters in the closed convergence domain  $|C| \geq R_\ell$  are the  $C$ -columns of Tables 1–4, together with  $C = R_\ell$  at each level.*

*Proof.* Primitive reduced positive definite binary quadratic forms and (4.10), in the proved range  $-427 \leq d \leq -3$ ,  $1 \leq f \leq 72$ , give thirteen orders of class number one and twenty-nine of class number two. For each order compute its integral class polynomial  $H_D$ .

There is a short elimination rule. If  $h(D) = 1$ , substitute its single integral root in  $\Phi_\ell(J, C)$ . If  $h(D) = 2$ , only the middle levels are possible; a shared root forces the monic quadratics  $H_D(J)$  and  $\Phi_\ell(J, C)$  to be equal. Thus, writing them as  $J^2 - T_D J + N_D$  and  $J^2 - T_\ell(C) J + N_\ell(C)$ , the possible  $C$  are exactly the integer roots of

$$\gcd(T_\ell(C) - T_D, N_\ell(C) - N_D).$$

Exact factorization therefore lists every possible parameter without a bound on its magnitude. It gives

| $\ell$            | 1  | 2  | 3  | 4  |
|-------------------|----|----|----|----|
| $ C  \geq R_\ell$ | 12 | 13 | 10 | 5  |
| $C \neq R_\ell$   | 11 | 12 | 9  | 4. |

The complete discriminants, class polynomials, and factorizations are retained in the finite arithmetic certificate described in Appendix B. The class polynomial coefficients were certified by complex interval arithmetic and their integrality; independent PARI calculations gave the same polynomials. Integer roots were recovered both by factorization and, for the middle levels, by elimination resultants.

The correspondence may initially retain either elliptic invariant above  $C$ . They are isogenous, so CM of one is equivalent to CM of the other. Every retained nonsingular fiber is consequently CM on the specified branch. This proves both directions of the enumeration.  $\square$

**Theorem 4.6** (The standard CM classification). *The primitive ordinarily convergent standard identities whose indicated elliptic fibers have CM are exactly the 36 rows of Tables 1–4, up to simultaneous sign. Under Hypothesis 1.2, they exhaust all primitive standard identities. The CM assertion remains unchanged if the nonzero parameter  $C$  is allowed to be rational.*

*Proof.* Integrality reduces rational CM parameters to integer ones. Lemma 4.5 exhausts their possible values in the convergence domain. Theorem 1.4 excludes the four positive endpoints, and rational descent gives an identity at every remaining parameter. It also permits only one primitive pair up to sign. The evaluations below identify those pairs and their multipliers with the tabulated rows. Finally, conditional CM forcing supplies the assertion without the CM premise. The conditional assertion uses the parameter domain of Hypothesis 1.2; enlarging that domain to arbitrary rational  $C$  requires enlarging the hypothesis as well.  $\square$

**4.4. Exact evaluation of the classified lines.** The geometric proof establishes existence and rationality before computing a single singular value. For the displayed coefficients we use the exact modular evaluations in [5, Tables 5–11]. Their role is specific: with the normalizations

$$X_\ell(q) = q + O(q^2), \quad Z_\ell(q) = \sum_{n \geq 0} u_\ell(n) X_\ell(q)^n, \quad G_\ell(X) = 1 - R_\ell X,$$

these tables give the singular value  $X_\ell = 1/C$  and the coefficient slope  $\lambda = A/B$  obtained by differentiating the corresponding modular equation. The modular summation theorem of Chan–Chan–Liu [6, Theorem 2.1], in this normalization [5, Theorem 2.4], applies to the specified weight-two form on the indicated normal extension of  $\Gamma_0(\ell)$ . At the tabulated quadratic nome, inside the open convergence disk, its conclusion is

$$\alpha = \frac{B}{2\sqrt{1 - R_\ell/C}} \begin{cases} \sqrt{\ell/N}, & q = e^{-2\pi\sqrt{N/\ell}}, \\ \sqrt{L/N}, & q = -e^{-2\pi\sqrt{N/L}}, \end{cases} \quad L = \frac{4\ell}{\gcd(4, \ell)}. \quad (4.12)$$

Thus each interior row follows from two exact modular values and this summation theorem; its displayed radical is then fixed by (4.12). These evaluations furnish identities on the CM locus, not a converse for arbitrary algebraic-value identities.

The remaining row is the boundary identity of Bauer,

$$\sum_{n \geq 0} (4n+1) \frac{(1/2)_n^3}{(n!)^3} (-1)^n = \frac{2}{\pi},$$

as recorded with its original source in [5, Section 4.1 and Table 11]. Its ordinary convergence was proved in the analytic part. It is the sole conditionally convergent row; all other rows are absolutely convergent.

**4.5. Linear Euler companions.** Put

$$G_s(z) = {}_2F_1(1 - s/2, (1 + s)/2; 1; z)^2.$$

Euler’s transformation identifies the two branches and their coefficient lines:

$$F_s = (1 - z)G_s, \quad (A + B\vartheta)F_s = (A(1 - z) - Bz + B(1 - z)\vartheta)G_s. \quad (4.13)$$

This is a change of differential basis, whose determinant is  $(1 - z)^2$ . Away from  $z = 1$  it induces an isomorphism of rational projective coefficient lines.

**Theorem 4.7** (Euler transport). *For  $|C| > R = R_\ell$ , define*

$$(a_0, b_0) = ((C - R)A - RB, (C - R)B), \quad d = \gcd(|a_0|, |b_0|).$$

*The primitive Euler companion of a primitive standard identity is*

$$(a, b, \beta) = \varepsilon(a_0/d, b_0/d, C\alpha/d),$$

*where  $\varepsilon = \pm 1$  chooses the representative. This transport is bijective on rational projective slopes. The convergent CM companions are exactly the 35 rows of Tables 5–8; under Hypothesis 1.2, they are all the primitive companions. No nonzero companion pair converges at either endpoint.*

*Proof.* Multiply (4.13) by  $C$ . The coefficient matrix is

$$M_C = \begin{pmatrix} C - R & -R \\ 0 & C - R \end{pmatrix}, \quad \det M_C = (C - R)^2.$$

Its inverse on projective lines is represented by  $(a, b) \mapsto ((C - R)a + Rb, (C - R)b)$ . Both directions preserve algebraicity of the multiplier, and division by the gcd gives exactly the primitive representative.

For convergence, the coefficients of  $G_s = F_s/(1 - z)$  are  $d_s(n) = \sum_{k=0}^n c_s(k)$ . Positivity and convergence of  $F_s(1)$  give  $d_s(n) \rightarrow F_s(1) > 0$ . Hence  $(a + bn)d_s(n)z^n$  fails to tend to zero for every nonzero pair when  $|z| \geq 1$ . In  $|z| < 1$  the series converges absolutely. Thus transport preserves all 35 interior standard rows and loses the single Bauer boundary row. The standard classification and the inverse transport prove exhaustion, with exactly the same conditional qualification.  $\square$

## 5. ARITHMETIC REALIZATIONS OF THE IDENTITY LINE

The coefficient line has a second realization in characteristic  $p$ . The resulting criterion is unconditional, but its hypothesis concerns reductions at primes. We shall keep this hypothesis separate from the complex period relation.

**5.1. From a finite polynomial to a de Rham line.** For  $s \in \{1/6, 1/4, 1/3, 1/2\}$  and  $p > 3$ , put

$$U_p(x) = \sum_{n < p} \frac{(s)_n (1 - s)_n}{(n!)^2} x^n, \quad T_p(z) = \sum_{n < p} c_s(n) z^n, \quad W_p(z) = (A + B\vartheta)T_p(z).$$

Thus  $W_p$  is the canonical truncation of the proposed identity. Let  $z = 4x(1 - x)$ , write  $\Omega = dX/y$  on the standard elliptic model, and put

$$Q(x) = \frac{2x(1 - x)}{1 - 2x}, \quad \nu_{A,B} = A\Omega + BQ(x)\nabla_x \Omega \in H_x, \quad H_x = H_{\text{dR}}^1(E_{s,x}).$$

We exclude finitely many primes at which the parameter, this basis, or a specified nonzero coefficient fails to be integral or a unit. All subsequent density statements are insensitive to these exclusions.

The precise unconditional polynomial input from [16, §7, equation (7.4) and Lemma 7.2] is

$$T_p(4x(1 - x)) \equiv U_p(x)^2 \pmod{p^2 \mathbb{Z}_p[x]}. \quad (5.1)$$

Differentiating this identity gives, after specialization away from  $1 - 2x = 0$ ,

$$W_p(z) \equiv U_p(x)(AU_p(x) + BQ(x)U_p'(x)) \pmod{p^2}. \quad (5.2)$$

In particular, this assertion makes no CM assumption. The source's CM supercongruence is a different result; its hypothesis and first-order term are recorded in Appendix C.6.

**Lemma 5.1** (The Cartier realization). *Let  $k$  be a perfect residue field at a good place, with  $x(1 - x)(1 - 2x) \neq 0$ . The reduction is ordinary if and only if  $U_p(x) \neq 0$ . In that case, for  $B \neq 0$  in  $k$ ,*

$$W_p(z) = 0 \iff k\nu_{A,B} = \ker(\text{Cartier}) = \text{im}(F \bmod p). \quad (5.3)$$

*The displayed line is the unique Frobenius-stable complement to the Hodge line  $F^1 H_x$ . These statements hold over every perfect residue field, including quadratic extensions of  $\mathbf{F}_p$ .*

*Proof.* The Hasse invariant in our differential normalization is  $(-6)^{(p-1)/2} U_p(x)$ . More generally, Cartier evaluation of  $\nu_{A,B}$  is, up to this fixed nonzero factor and the Frobenius twist, the linear functional

$$(A, B) \mapsto AU_p(x) + BQ(x)U_p'(x).$$

The normalization and its compatibility with Gauss–Manin differentiation are proved in Appendix C.2. Conceptually, this calculation identifies the top row of the diagram

$$\begin{array}{ccc} k^2 & \xrightarrow{(A,B) \mapsto \nu_{A,B}} & H_x \\ & \searrow (AU_p + BQU'_p)^{1/p} & \downarrow \text{Cartier} \\ & & H^0(E_{s,x}, \Omega^1)^{(p)}, \end{array}$$

after rescaling its target basis by the fixed nonzero normalization. The diagonal map is semilinear, as is Cartier. The conjugate-filtration exact sequence is

$$0 \longrightarrow H^1(E_{s,x}, \mathcal{O})^{(p)} \xrightarrow{F} H_x \xrightarrow{\text{Cartier}} H^0(E_{s,x}, \Omega^1)^{(p)} \longrightarrow 0. \quad (5.4)$$

Its first image is a line. Since the Gauss–Manin coefficient of  $\Xi$  in  $Q\nabla_x \Omega$  is nonzero,  $\nu_{A,B}$  spans a complement to  $F^1 H_x$  when  $B \neq 0$ . At an ordinary fiber  $U_p(x)$  is a unit, so (5.2) identifies its vanishing with the kernel in (5.4). Finally  $F \bmod p$  kills the Hodge line and has rank one. A stable complement must therefore equal its image; at an ordinary fiber that image is indeed a complement.  $\square$

At a supersingular fiber the first factor in (5.2) vanishes for every pair  $(A, B)$ . Such zeros do not assert stability of a complementary line. This distinction determines the density argument.

**5.2. Algebraization and the density of stable complements.** For a number field  $K$  and a set  $M$  of finite places, define

$$d_K(M; X) = \frac{\sum_{\substack{v \in M \\ p_v \leq X}} [K_v : \mathbb{Q}_{p_v}] \frac{\log p_v}{p_v - 1}}{[K : \mathbb{Q}] \sum_{p \leq X} \frac{\log p}{p - 1}}, \quad \underline{d}_K(M) = \liminf_X d_K(M; X), \quad \bar{d}_K(M) = \limsup_X d_K(M; X). \quad (5.5)$$

Here  $p_v$  is the rational residue characteristic, not the norm of  $v$ . Finite sets have density zero, and

$$\underline{d}_K(M) = 1 - \bar{d}_K(M^c). \quad (5.6)$$

**Theorem 5.2** (Tang’s algebraization inequality). *Let  $X/K$  be a geometrically irreducible quasi-projective variety of dimension  $N$ , let  $P \in X(K)$ , and let  $\mathcal{D} \subset T_X$  be an involutive subbundle of rank  $d$ . Suppose that its smooth formal leaf through  $P$  is Zariski dense. For every embedding  $\sigma : K \hookrightarrow \mathbb{C}$ , assume a holomorphic map  $\gamma_\sigma : \mathbb{C}^d \rightarrow X_\sigma(\mathbb{C})$  takes zero to  $P_\sigma$  and induces that leaf germ biholomorphically. Suppose these maps have finite Nevanlinna orders  $\rho_\sigma$ , and put  $\rho = \max_\sigma \rho_\sigma$ .*

*Spread these data over  $\mathcal{O}_K[1/n]$ . Let  $M_{\text{bad}}$  be the places where the reduced distribution is not closed under restricted  $p_v$ th powers, and put  $\alpha_{\text{bad}} = \bar{d}_K(M_{\text{bad}})$ . Then either  $N = d$ , or*

$$1 \leq \frac{N}{N - d} \rho \alpha_{\text{bad}}. \quad (5.7)$$

This is the quantitative form of [17, §§5.1.1–5.1.4 and Theorem 5.1.5]. The order is computed using an ample Hermitian line bundle on a projective compactification; the needed normalization is given in Appendix C.4. The use of an upper density for the bad places is essential.

**Lemma 5.3** (Naturality of the density). *For a finite extension  $K'/K$ , let  $M'$  consist of all places above  $M$ . Then  $d_{K'}(M'; X) = d_K(M; X)$ . If  $M$  contains every place above a set  $S$  of rational primes of lower natural density  $\delta$ , apart from finitely many exceptions, then  $\underline{d}_K(M) \geq \delta$ .*

*Proof.* The first assertion is the local degree formula

$$\sum_{w|v} [K'_w : \mathbb{Q}_{p_v}] = [K' : K] [K_v : \mathbb{Q}_{p_v}].$$

For the second, sum the local degrees over each rational prime and apply partial summation to the positive decreasing weight  $w(t) = \log t / (t - 1)$ . The bound  $\#\{p \leq t : p \in S\} \geq (\delta - \epsilon)\pi(t)$ , valid for large  $t$ , gives the corresponding weighted bound up to a constant. Division by  $\sum_{p \leq X} w(p) \sim \log X$  and then  $\epsilon \downarrow 0$  proves the assertion.  $\square$

Thus all places over an admitted rational prime must be retained. For instance, an inert prime in a quadratic field contributes its full weight here, although its prime ideal has norm  $p^2$  and the set of such prime ideals has Dirichlet density zero.

**Theorem 5.4** (An algebraic complement cannot be too often stable). *Let  $A/K$  be an abelian variety of positive dimension and let  $W \subset H_{\text{dR}}^1(A/K)$  be a  $K$ -linear complement to  $F^1 H_{\text{dR}}^1(A/K)$ . Let  $M_W$  be the good places where its reduction is stable under crystalline Frobenius modulo the residue characteristic. Then*

$$\underline{d}_K(M_W) \leq \frac{3}{4}. \quad (5.8)$$

*No simplicity or non-CM assumption is needed.*

*Proof.* Let  $g = \dim A$  and let  $G = \mathcal{E}(A^\vee)$  be the universal vector extension. Its defining sequence realizes the Hodge filtration as tangent spaces:

$$\begin{array}{ccccccc} 0 & \longrightarrow & V & \longrightarrow & G & \longrightarrow & A^\vee \longrightarrow 0 \\ & & \text{Lie} & & \text{Lie} & & \text{Lie} \end{array} \quad (5.9)$$

$$0 \rightarrow F^1 H_{\text{dR}}^1(A) \rightarrow H_{\text{dR}}^1(A) \rightarrow H^1(A, \mathcal{O}_A) \rightarrow 0.$$

At good places this identification is integral, and the restricted  $p$ th power on  $\text{Lie } G$  is crystalline Frobenius modulo  $p$ ; these are the comparison facts used in [17, proof of Theorem 2.3.5 and Lemma 5.2.6].

The exponential leaf of  $W$  is Zariski dense in  $G$ . Indeed, let  $H$  be its connected algebraic closure and  $U = H \cap V$ . Surjectivity on tangent spaces shows that  $H \rightarrow A^\vee$  is surjective. A subgroup of a vector group in characteristic zero is a vector group, so  $H/U \simeq A^\vee$  splits the quotient extension  $G/U$ . On extension classes this would make zero the restriction

$$(V/U)^* \hookrightarrow V^* \xrightarrow{\sim} H^1(A^\vee, \mathcal{O}_{A^\vee}).$$

The last map is the universal extension class. Its injectivity forces  $U = V$ , and hence  $H = G$ .

Translation of  $W$  now defines a rank- $g$  involutive distribution on the  $2g$ -dimensional group  $G$ . At every complex embedding its exponential uniformization has order at most two, by the commutative-group estimate in [17, proof of Proposition 2.3.4, footnote 20]. Restricted-power stability of this distribution is precisely the condition defining  $M_W$ . Theorem 5.2 therefore gives

$$1 \leq \frac{2g}{g} \rho \bar{d}_K(M_W^c) \leq 4 \bar{d}_K(M_W^c).$$

Equation (5.6) finishes the proof.  $\square$

**5.3. The arithmetic classification.** We need one Galois-theoretic input that counts rational primes, including those of higher residue degree.

**Theorem 5.5** (Supersingular density for  $\mathbb{Q}$ -curves). *Let  $E$  over a number field be geometrically isogenous to every Galois conjugate. Outside finitely many rational primes, supersingularity is independent of the place above the rational prime. Its rational-prime set has natural density zero if  $E$  is non-CM, and density  $1/2$  if  $E$  is CM.*

*Proof.* For non-CM  $E$ , the one-dimensional rational spaces of isogenies between its conjugates assemble its Tate representation into a projective representation of  $G_{\mathbb{Q}}$ . Serre's open image theorem [18] makes every sufficiently large residual image contain  $\text{PSL}_2(\mathbf{F}_\lambda)$ . At a supersingular rational prime its projective Frobenius has bounded order, with a bound depending only on a fixed Galois field of definition. Elements of bounded order have proportion  $O(1/\lambda)$  in these images. Chebotarev, followed by  $\lambda \rightarrow \infty$ , gives density zero. Appendix C.3 proves both the projective construction and the uniform order bound, so this argument includes primes of higher residue degree.

In the CM case, Deuring's reduction theorem [20] says that good rational primes unramified in the CM field are supersingular exactly when inert in that imaginary quadratic field. Their density is  $1/2$  by Chebotarev. Specialization of the isogenies shows independence of the place in both cases.  $\square$

**Theorem 5.6** (Weighted-density CM criterion). *Let  $z \in \mathbb{Q} \setminus \{0, 1\}$ , let  $x = (1 - \sqrt{1-z})/2$ , and let  $(A, B) \in \mathbb{Q}^2 \setminus \{0\}$ . If*

$$\underline{\delta}\{p : W_p(z) \equiv 0 \pmod{p}\} > \frac{3}{4},$$

*then  $E_{s,x}$  has CM. In particular it suffices that the congruence hold at all but finitely many ordinary rational primes.*

*Proof.* Suppose  $E_{s,x}$  is non-CM. The standard modular correspondences make it a  $\mathbb{Q}$ -curve over  $K = \mathbb{Q}(x)$ , so its supersingular rational primes have density zero by Theorem 5.5. If  $B = 0$ , (5.2) excludes every ordinary zero outside a finite set. If  $B \neq 0$ , the fixed line  $K\nu_{A,B}$  complements the Hodge line. At every admitted ordinary rational prime, Lemma 5.1 makes it Frobenius-stable at *every* place of  $K$  above that prime. Indeed  $W_p(z)$  is rational, and the same polynomial calculation applies at both conjugate parameters

and over their residue fields. Lemma 5.3 consequently gives lower weighted-place density greater than  $3/4$ , contrary to Theorem 5.4. If all but finitely many ordinary primes are admitted, the non-CM supposition gives density one and the same contradiction.  $\square$

**Proposition 5.7** (The CM density dichotomy). *Suppose  $E_{s,x}$  has CM, and denote its complementary CM eigenline by  $L_x^{\text{CM}}$ , as in (4.5). For every fixed nonzero rational pair  $(A, B)$ ,*

$$\delta\{p : W_p(z) = 0 \bmod p\} = \begin{cases} 1, & K\nu_{A,B} = L_x^{\text{CM}}, \\ 1/2, & K\nu_{A,B} \neq L_x^{\text{CM}}. \end{cases}$$

*In the first case the zero set is cofinite; in the second its ordinary part is finite.*

*Proof.* The companion eigenline descends to  $K$  because Galois preserves the pair of CM eigenspaces and fixes the Hodge line. At an ordinary prime  $p$ , the CM field splits. Its two eigenvalues lie in  $\mathbb{Q}_p$  and are fixed by coefficient Frobenius; functoriality therefore makes crystalline Frobenius preserve both eigenlines. Modulo  $p$  it kills the Hodge line and has rank one, whence

$$\text{im}(F \bmod p) = L_x^{\text{CM}} \bmod p.$$

The equality descends from a field defining the CM action. Thus the Cartier criterion identifies ordinary weighted zeros with equality of the reductions of  $K\nu_{A,B}$  and  $L_x^{\text{CM}}$ . Distinct algebraic lines have distinct reductions outside finitely many places: their exterior product is nonzero and has only finitely many prime divisors. At every supersingular prime the Hasse factor  $U_p(x)$  vanishes, so every pair gives a weighted zero. Deuring's density  $1/2$  proves all assertions, including the case  $B = 0$ .  $\square$

**Theorem 5.8** (Arithmetic classification of primitive rows). *Fix  $\ell \in \{1, 2, 3, 4\}$ . Among integer triples with  $\gcd(A, B) = 1$ ,  $|C| \geq R_\ell$  and  $C \neq R_\ell$ , the following conditions are equivalent:*

- (1) *the lower natural density of  $\{p : W_p(R_\ell/C) \equiv 0 \pmod{p}\}$  is greater than  $3/4$ ;*
- (2) *this set of primes is cofinite;*
- (3) *up to simultaneous sign of  $A, B$ , the triple is one of the standard rows at level  $\ell$  in Section 4.4.*

*Proof.* The first condition forces CM. The finite classification in Section 4 leaves the tabulated parameters. At each parameter Proposition 5.7 selects exactly the complementary CM eigenline, which is the complex identity line by Proposition 4.2 and projective uniqueness. Thus the first condition gives the third, and the third gives the second. The remaining implication is immediate.  $\square$

Consequently, on the original convergence domain, CM forcing is equivalent to the assertion that every complex identity has weighted-zero lower density greater than  $3/4$ . The arithmetic criterion describes the missing comparison statement; it does not prove that statement.

**5.4. An exact comparison at one ordinary prime.** There is a second arithmetic endpoint: exact preservation of the characteristic-zero Hodge line by a single ordinary Frobenius already characterizes CM.

**Theorem 5.9** (One-prime CM criterion). *Let  $E/K$  be an elliptic curve over a number field and let  $v \mid p$  be a place of good ordinary reduction. Write  $L = K_v$ , let  $k = \mathbf{F}_{p^f}$  be its residue field, and put*

$$K_0 = \text{Frac } W(k), \quad D = H_{\text{cris}}^1(E_k/W(k))[1/p], \quad \Phi = \varphi^f.$$

*Through crystalline-de Rham comparison, extend the  $K_0$ -linear endomorphism  $\Phi$  to  $H = H_{\text{dR}}^1(E/L)$ . Then*

$$E \text{ has geometric CM} \iff \Phi(F^1 H) \subseteq F^1 H.$$

*This formulation allows  $L/K_0$  to be ramified and selects no Frobenius lift on  $L$ .*

*Proof.* We use the  $p$ -adic Lefschetz theorem in the following precise form [28, Theorem 4.24]. Let  $\mathcal{X}$  be a smooth proper  $p$ -adic formal scheme over the valuation ring of a complete discretely valued characteristic-zero field with perfect residue field. A line bundle  $\mathcal{L}_0$  on its special fiber has some tensor power  $\mathcal{L}_0^{\otimes p^m}$  lifting to  $\mathcal{X}$  if and only if its crystalline Chern class, transported to de Rham cohomology, belongs to  $F^1 H_{\text{dR}}^2(\mathcal{X})$ . Ramification is permitted.

Let  $\mathcal{E}/\mathcal{O}_L$  be the good model and let  $\pi_q : E_k \rightarrow E_k$  be  $q$ -power Frobenius,  $q = p^f$ . Its graph is the divisor  $\Gamma = (\pi_q, -1)^* \{0\}$  on  $E_k \times E_k$ . The polarization identifies the mixed Künneth component of  $c_1(\mathcal{O}(\Gamma))$  with the crystalline action of  $\pi_q$ , namely  $\Phi$ . Under this identification the obstruction to the graph class lying in  $F^1$  is exactly the map

$$F^1 H \xrightarrow{\Phi} H \longrightarrow H/F^1 H. \tag{5.10}$$

Indeed the polarization gives  $F^1 H \simeq (H/F^1 H)^*$  after trivializing the one-dimensional determinant. Projection of the correspondence tensor to  $(H/F^1 H)^{\otimes 2}$  is consequently the same obstruction. The two unmixed Künneth components are divisor classes on the factors and already lie in  $F^1$ .

If (5.10) vanishes, the quoted Lefschetz theorem lifts  $\mathcal{O}(\Gamma)^{\otimes p^m}$  to the formal completion of  $\mathcal{E} \times \mathcal{E}$ . Formal GAGA algebraizes the lift to a line bundle  $\mathcal{L}$  [28, Example 4.5(b)]. For a line bundle on a product of principally polarized elliptic curves, the negative off-diagonal block of its associated homomorphism to the dual is an endomorphism. This construction is additive in the line bundle and commutes with base change:

$$\begin{array}{ccc} \mathrm{Pic}(\mathcal{E} \times \mathcal{E}) & \xrightarrow{\text{correspondence}} & \mathrm{End}(\mathcal{E}) \\ \mathrm{sp} \downarrow & & \downarrow \mathrm{sp} \\ \mathrm{Pic}(E_k \times E_k) & \xrightarrow[\text{correspondence}]{} & \mathrm{End}(E_k). \end{array}$$

For the graph divisor its value is  $\pi_q$ ; hence  $\mathcal{L}$  gives an endomorphism  $u$  specializing to  $[p^m]\pi_q$ . Ordinary Frobenius has eigenvalue valuations 0,  $f$ , so this specialization is nonscalar. A scalar generic endomorphism would specialize to multiplication by an integer, which has equal valuations on both eigenvalues. Thus  $u$  is nonscalar. Geometric homomorphism groups of abelian varieties are unchanged by extension of algebraically closed fields; consequently  $E_{\overline{K}}$  has a nonscalar endomorphism and has CM.

Conversely, suppose  $E$  has CM. After a finite local extension, all its geometric endomorphisms extend to the good model. Specialization embeds its rational geometric endomorphism algebra in that of the ordinary special fiber. Both are imaginary quadratic fields, so the embedding is an isomorphism. Thus  $\pi_q$  is the specialization of a characteristic-zero quasiendomorphism. Its action preserves the Hodge filtration, and crystalline–de Rham comparison is functorial for this action [28, Theorem 4.21]. The vanishing of (5.10) descends from that extension to  $L$ .  $\square$

*Remark 5.10.* The lifting argument applies to any special-fiber endomorphism whose crystalline action preserves the characteristic-zero Hodge filtration. Ordinarity guarantees that Frobenius is nonscalar; at supersingular reduction a power of Frobenius can be scalar even for a non-CM characteristic-zero curve. Furthermore, stability of the complementary line modulo  $p$  in Lemma 5.1 is a different condition from exact stability of the Hodge line here. Appendix C.6 shows that no fixed finite precision substitutes for exactness. No implication from the original complex scalar identity to this exact Frobenius condition is established.

## 6. ARITHMETIC SEPARATION AND COUNTING

The period criteria above concern a single algebraic fiber. We now use the variation of that fiber to obtain unconditional restrictions on the set of possible identities. Two different principles apply. Near the cusp, independent Gauss systems and the Bombieri–André specialization theorem separate identities whose multipliers have bounded degree. On a compact parameter interval, symmetric-square monodromy and Habegger–Pila counting give a subpower bound when all arithmetic heights are bounded. Neither argument assumes the period hypothesis of Theorem 1.3.

For  $d \geq 1$ , let  $\mathcal{E}_d$  be the set of triples  $(\ell, \epsilon, t)$ , with  $1 \leq \ell \leq 4$ ,  $\epsilon \in \{\pm 1\}$ , and  $t \in \mathbb{N}$ , for which there is a primitive standard identity

$$(A + B\vartheta)F_{s_\ell}\left(\frac{\epsilon R_\ell}{t}\right) = \frac{\alpha}{\pi}, \quad [\mathbb{Q}(\alpha^2) : \mathbb{Q}] \leq d. \quad (6.1)$$

Thus  $t$  is the absolute denominator, while  $(\ell, \epsilon)$  retains the level and its sign. By Theorem 1.4, each such triple determines at most one primitive coefficient pair up to sign.

**Theorem 6.1** (Separation in absolute denominator). *Fix  $d \geq 1$ . For every integer  $H \geq 0$ , there exists  $T_H$  such that*

$$\#\{(\ell, \epsilon, t) \in \mathcal{E}_d : M \leq t \leq M + H\} \leq 1 \quad (M \geq T_H).$$

*Equivalently, for fixed levels  $\ell_1, \ell_2$ , signs  $\epsilon_1, \epsilon_2$ , and integer  $D \geq 0$ , there are only finitely many  $t$  for which*

$$(\ell_1, \epsilon_1, t), \quad (\ell_2, \epsilon_2, t + D) \in \mathcal{E}_d,$$

*unless  $D = 0$  and  $(\ell_1, \epsilon_1) = (\ell_2, \epsilon_2)$ . In particular, distinct absolute denominators have gaps tending to infinity, jointly across all four levels and both signs.*

The theorem imposes no height bound on  $A, B$ , or  $\alpha$ , and no fixed field of definition for  $\alpha$ . The square in its degree hypothesis is natural: taking the squared ratio of two identities eliminates their common transcendental factor.

6.1. **The Gauss systems and their products.** Write

$$\varphi_s(z) = {}_2F_1\left(\frac{s}{2}, \frac{1-s}{2}; 1; z\right), \quad F_s = \varphi_s^2, \quad G_s = \vartheta F_s,$$

and let  $\mathcal{V}_s$  denote the rank-two local system of solutions of

$$z(1-z)y'' + \left(1 - \frac{3}{2}z\right)y' - \frac{s(1-s)}{4}y = 0. \quad (6.2)$$

Clausen's identity identifies the third-order system of  $F_s$  with  $\text{Sym}^2 \mathcal{V}_s$ . Its arithmetic and geometric roles will be kept distinct: the Taylor coefficients make  $F_s, G_s$   $G$ -functions, whereas monodromy controls their functional relations.

**Lemma 6.2** (Product monodromy). *The connected algebraic monodromy of  $\mathcal{V}_s$  is  $\text{SL}_2$  for every standard signature  $s$ . Let*

$$r_i = \epsilon_i R_{\ell_i}, \quad s_i = s_{\ell_i}, \quad \rho_1(x) = \frac{r_1}{x}, \quad \rho_2(x) = \frac{r_2}{x+D}, \quad D \in \mathbb{Z}_{\geq 0}.$$

*Unless  $D = 0$  and  $(\ell_1, \epsilon_1) = (\ell_2, \epsilon_2)$ , the connected algebraic monodromy of  $\rho_1^* \mathcal{V}_{s_1} \oplus \rho_2^* \mathcal{V}_{s_2}$  is*

$$\text{SL}_2 \times \text{SL}_2.$$

*Consequently the four functions*

$$F_{s_1}(r_1 u), \quad G_{s_1}(r_1 u), \quad F_{s_2}\left(\frac{r_2 u}{1+Du}\right), \quad G_{s_2}\left(\frac{r_2 u}{1+Du}\right) \quad (6.3)$$

*are algebraically independent over  $\mathbb{C}(u)$ .*

*Proof.* Put  $a = s/2$  and  $b = (1-s)/2$ . Abel's identity gives a Wronskian proportional to  $1/(z\sqrt{1-z})$ , so the monodromy determinant has finite image. At zero the nontrivial unipotent local monodromy fixes the line of  $\varphi_s$ . At one, the connection formula reads

$$\varphi_s(z) = \frac{\Gamma(1/2)}{\Gamma(1-a)\Gamma(1-b)}u(z) + \frac{\Gamma(-1/2)}{\Gamma(a)\Gamma(b)}(1-z)^{1/2}v(z),$$

where  $u(1) = v(1) = 1$ . Both coefficients are nonzero. Hence the reflection at one moves the fixed line of the unipotent at zero. The two conjugate one-parameter unipotent groups generate  $\text{SL}_2$ ; the finite determinant image shows that this is precisely the identity component.

Let  $H$  be the joint algebraic monodromy of the two pullbacks. Its identity component projects onto both copies of  $\text{SL}_2$ . By the Lie-algebra form of Goursat's lemma, a proper connected subgroup with these projections would have Lie algebra

$$\mathfrak{h} = \{(X, \phi X) : X \in \mathfrak{sl}_2\} \quad (6.4)$$

for an isomorphism  $\phi : \mathfrak{sl}_2 \rightarrow \mathfrak{sl}_2$ . Normality of  $H^0$  in  $H$  would then require, for every  $(M, N) \in H$ ,

$$\phi \text{Ad}(M) = \text{Ad}(N)\phi. \quad (6.5)$$

We rule this out by examining the singularities, including the only case in which the two sets of singularities coincide.

The finite singular sets are

$$\{0, r_1\}, \quad \{-D, r_2 - D\}.$$

Every point of either set has noncentral local monodromy. At Gauss argument one this is the reflection with eigenvalues  $1, -1$ . At infinity the exponent difference is  $1/2 - s$ . It is nonintegral unless  $s = 1/2$ . In that remaining case, scalar monodromy at infinity would, by the product relation between the three local monodromies, make the reflection at one a scalar multiple of the unipotent at zero, which is impossible. If the finite singular sets differ, a loop around a point belonging to only one gives  $(M, 1)$  or  $(1, N)$  with a noncentral entry. Equation (6.5) would make that entry central, a contradiction.

If the sets coincide and the systems are not identical, their common configuration is necessarily

$$\ell_1 = \ell_2 = \ell, \quad \epsilon_1 = -1, \quad \epsilon_2 = 1, \quad D = R_\ell.$$

Indeed  $D > 0$  then forces  $r_1 = -D$  and  $r_2 = D$ , and the four radii are distinct. A loop about  $x = 0$  corresponds to Gauss infinity in the first system and Gauss argument one in the second. These local maps

are unramified. For its monodromy pair  $(M, N)$ , the adjoint traces are

$$\begin{aligned} \operatorname{tr} \operatorname{Ad}(N) &= -1, \\ \operatorname{tr} \operatorname{Ad}(M) &= 1 - 2 \cos(2\pi s) = \begin{cases} 0, & s = 1/6, \\ 1, & s = 1/4, \\ 2, & s = 1/3, \\ 3, & s = 1/2. \end{cases} \end{aligned} \quad (6.6)$$

The formula uses only the eigenvalues, so also covers the nondiagonalizable monodromy when  $s = 1/2$ . The unequal traces contradict (6.5). Thus  $H^0$  is the full product in every permitted case.

To pass from local systems to the selected functions, evaluate at an ordinary point. Each group factor moves the nonzero solution column  $(\varphi_s, \varphi'_s)$  through a Zariski-dense subset of  $\mathbb{A}^2$ . The quadratic map

$$(v, w) \mapsto (v^2, 2zvw) \quad (6.7)$$

is dominant for  $z \neq 0$  and sends this column to  $(F_s, G_s)$ . The two group factors act independently. A functional polynomial relation among the four functions would therefore vanish on  $\mathbb{A}^4$  at every ordinary point; all its coefficients would vanish. Substituting  $u = 1/x$  proves the asserted independence.  $\square$

The same argument has a useful one-system consequence.

**Lemma 6.3** (Affine functional independence). *For a standard signature,  $1, F_s, G_s$  are linearly independent over  $\mathbb{C}(z)^{\text{alg}}$ , on every continued branch.*

*Proof.* Make the coefficients of a proposed relation single-valued on a finite algebraic cover. The connected monodromy remains  $\operatorname{SL}_2$ , since passing to a finite-index subgroup does not change its connected Zariski closure. By the dominant map (6.7), continuation of the selected solution makes the proposed affine equation vanish at every point of  $\mathbb{A}^2$ . Its three coefficients are therefore zero. This is the affine form of irreducibility of  $\operatorname{Sym}^2 \mathcal{V}_s$ : its three-dimensional monodromy orbit cannot satisfy an inhomogeneous first-order equation over an algebraic extension of the base function field.  $\square$

**6.2. Specialization and the norm of a period relation.** We use the following form of the Bombieri–André theorem, as stated in [26, Theorem 3.13 and Remark 3.14]. Its coefficient field is fixed; its polynomial coefficients have no height restriction. Both features are essential here.

Recall that a  $G$ -function over  $\mathbb{Q}$  is a power series satisfying a linear differential equation over  $\mathbb{Q}(u)$ , whose coefficient sizes and common denominators up to order  $n$  grow at most exponentially in  $n$ . The standard  $F_s$  have this property: their coefficients become integers after multiplication by  $R_\ell^n$ , and their defining hypergeometric equations are linear. Differentiation and rational substitution vanishing at the origin preserve this class.

**Theorem 6.4** (Bombieri–André, archimedean form). *Let  $g_1, \dots, g_m \in \mathbb{Q}[[u]]$  be  $G$ -functions, and let  $\mathcal{B} \subset \mathbb{P}^1 \times \mathbb{A}^m$  be the Zariski closure over  $\mathbb{Q}$  of their graph. Write*

$$\mu' = \operatorname{trdeg}_{\mathbb{Q}(u)} \mathbb{Q}(u)(g_i^{(j)} : 1 \leq i \leq m, j \geq 0) > 0.$$

*There are constants  $c_1 \geq e$  and  $c_2 > 0$ , depending only on the functions, with the following property. Let  $\delta \geq 1$  and let  $\xi \in \mathbb{Q} \setminus \{0\}$  lie strictly inside all convergence discs. Put  $h = h(\xi)$ , the absolute logarithmic Weil height. If*

$$\begin{aligned} h &\geq c_1 \delta^{\mu'} (1 + \log \delta), \\ |\xi| &\leq \exp \left[ -c_2 \delta^{\mu' / (\mu' + 1)} h^{1 / (\mu' + 1)} \log(h^{1 / (\mu' + 1)}) \right], \end{aligned} \quad (6.8)$$

*there is no polynomial  $P \in \mathbb{Q}[T_1, \dots, T_m]$  of degree at most  $\delta$  such that*

$$P(g_1(\xi), \dots, g_m(\xi)) = 0, \quad \dim(\mathcal{B}_\xi \cap V(P)) < \dim \mathcal{B} - 1.$$

The dimension inequality is the precise meaning of a strongly nontrivial specialized relation. The cited theorem is stated for an arbitrary number field and place; the form above takes  $K = \mathbb{Q}$  and the real place. Its statement for degree  $\delta$  also gives degree at most  $\delta$ , by applying it to each smaller degree. Nonhomogeneous relations are allowed by adjoining the constant 1 to the tuple, as explained in the cited remark.

**Lemma 6.5** (Reciprocal-integer specialization). *Let  $g_1, \dots, g_m$  be  $G$ -functions over  $\mathbb{Q}$ , holomorphic at zero and algebraically independent over  $\mathbb{Q}(u)$ . For each fixed  $\delta \geq 1$ , their values at  $u = 1/t$  satisfy no nonzero polynomial relation over  $\mathbb{Q}$  of total degree at most  $\delta$  for all sufficiently large positive integers  $t$ .*

*Proof.* Functional independence gives  $\mathcal{B} = \mathbb{P}^1 \times \mathbb{A}^m$ , so every nonzero specialized relation is strongly nontrivial. The differential field has finite transcendence degree  $\mu'$ : each  $G$ -function satisfies a linear differential equation over  $\mathbb{Q}(u)$ , so finitely many derivatives generate it. At  $\xi = 1/t$  one has  $h = \log t$  and  $|\xi| = e^{-h}$ . For fixed  $\delta$ , the first inequality in (6.8) eventually holds; the negative logarithm required by the second is  $O(h^{1/(\mu'+1)} \log h) = o(h)$ . The theorem applies, uniformly in the coefficients of the proposed relation.  $\square$

*Proof of Theorem 6.1.* Fix the two signed levels and the gap  $D$ , excluding the identical case. Let  $\mathbf{g} = (g_1, g_2, g_3, g_4)$  be the tuple in (6.3). The rational hypergeometric parameters make  $F_s$  a  $G$ -function. Differentiation and rational substitution vanishing at zero preserve  $G$ -functions, hence every  $g_i$  is a  $G$ -function over  $\mathbb{Q}$ , holomorphic at zero. Lemma 6.2 gives their algebraic independence.

Suppose the two identities occur at absolute denominators  $t$  and  $t + D$ . Their multipliers are nonzero by Theorem 2.5. Set

$$\beta = \frac{\alpha_1^2}{\alpha_2^2}, \quad K = \mathbb{Q}(\beta), \quad e = [K : \mathbb{Q}] \leq d^2,$$

and regard

$$U_1 = A_1 T_1 + B_1 T_2, \quad U_2 = A_2 T_3 + B_2 T_4$$

as linear forms on  $\mathbb{A}^4$ . Cancelling  $\pi$  between the squared identities gives

$$(U_1^2 - \beta U_2^2)(\mathbf{g}(1/t)) = 0.$$

The descent to the fixed coefficient field is the polynomial norm

$$Q = N_{K/\mathbb{Q}}(U_1^2 - \beta U_2^2) = \prod_{\sigma: K \hookrightarrow \mathbb{C}} (U_1^2 - \sigma(\beta) U_2^2) \in \mathbb{Q}[T_1, T_2, T_3, T_4]. \quad (6.9)$$

Each factor is nonzero, since  $U_1, U_2$  are nonzero forms in disjoint pairs of variables. Thus  $Q \neq 0$ , it is homogeneous of degree  $2e \leq 2d^2$ , and  $Q(\mathbf{g}(1/t)) = 0$ . Lemma 6.5 excludes this for all sufficiently large  $t$ .

For fixed  $H$ , there are only finitely many ordered pairs of signed levels and gaps  $0 \leq D \leq H$ . Taking the largest of their thresholds proves the interval assertion. The already established uniqueness of the primitive coefficient direction identifies repeated representations of the same signed-level occurrence.  $\square$

Notice that the norm in (6.9) conjugates only the algebraic coefficient  $\beta$ . It does not assert that a Galois conjugate of a multiplier occurs on the chosen analytic branch. This distinction is what allows an unrestricted coefficient field for  $\alpha$  while applying a specialization theorem over  $\mathbb{Q}$ .

**6.3. Counting in complete height windows.** For a real number  $y$  and  $d \geq 1$ , use the polynomial coefficient height

$$H_d(y) = \min\{\|q\|_\infty : 0 \neq q \in \mathbb{Z}[T] \text{ primitive, } \deg q \leq d, \quad q(y) = 0\},$$

where  $\min \emptyset = \infty$  and  $\|q\|_\infty$  is the largest absolute coefficient. On tuples,  $H_d$  is the maximum coordinate height. This is the  $d$ -height used by Habegger–Pila; in particular  $H_d(y) < \infty$  is equivalent to  $[\mathbb{Q}(y) : \mathbb{Q}] \leq d$ .

**Theorem 6.6** (Habegger–Pila, semirational counting). *Let  $Z \subseteq \mathbb{R}^m \times \mathbb{R}^n$  be definable in a fixed o-minimal structure, with coordinate projections  $\pi_1, \pi_2$ . For every  $d \geq 1$  and  $\varepsilon > 0$ , there is  $c > 0$  such that, whenever  $T \geq 1$  and  $\Sigma \subseteq Z$  satisfy*

$$H_d(\pi_1 P) \leq T \quad (P \in \Sigma), \quad \#\pi_2(\Sigma) > cT^\varepsilon,$$

*there exists a continuous definable path  $\gamma : [0, 1] \rightarrow Z$  such that  $\pi_1 \gamma$  is semialgebraic and real analytic on  $(0, 1)$ , and  $\pi_2 \gamma$  is nonconstant.*

This is [27, Corollary 7.2] with a fixed member of its definable family. Only the first projection is required to be semialgebraic; no degree or height restriction is placed on the second. Our choice of the two projections will use this distinction explicitly.

Fix a compact interval  $I \subset [-1, 1]$  and define the normalized incidence surface

$$X_s = \{(z, a, b) \in (I \setminus \{0\}) \times \mathbb{R}^2 : \pi(aF_s(z) + bG_s(z)) = 1\}. \quad (6.10)$$

The chosen branch is analytic on a neighborhood of  $I$ , so  $X_s$  is definable in  $\mathbb{R}_{\text{an}}$ . Its nonempty fibers over  $z$  are affine lines. These lines fill the surface and therefore prevent a direct application of counting only the usual transcendental part. The relevant assertion concerns paths that move in the parameter.

**Lemma 6.7.** *Every continuous semialgebraic path in  $X_s$  has constant  $z$  coordinate.*

*Proof.* If the parameter varies, semialgebraic cell decomposition and a local inverse give a nonempty interval on which the path is  $(z, a(z), b(z))$ , with  $a, b$  real analytic and algebraic over  $\mathbb{R}(z)$ . Its incidence equation is an affine functional relation between  $1, F_s, G_s$ , contrary to Lemma 6.3. That lemma allows arbitrary complex constants; no algebraicity of the constants defining the semialgebraic path is needed.  $\square$

**Theorem 6.8** (Counting normalized identities). *For a standard signature  $s$ , a compact interval  $I \subset [-1, 1)$ ,  $d \geq 1$ , and every  $\varepsilon > 0$ ,*

$$\#\{(z, a, b) \in X_s : H_d(z, a, b) \leq T\} = O_{s, I, d, \varepsilon}(T^\varepsilon).$$

*Proof.* Duplicate the parameter by setting

$$Z_s = \{((z, a, b), t) : (z, a, b) \in X_s, t = z\}.$$

The two projections fit in the commutative square

$$\begin{array}{ccc} Z_s & \xrightarrow{\pi_1} & X_s \\ \pi_2 \downarrow & & \downarrow p \\ I \setminus \{0\} & \xlongequal{\quad} & I \setminus \{0\}, \end{array} \quad p(z, a, b) = z.$$

If more than  $cT^\varepsilon$  distinct parameters occur, the Habegger–Pila theorem produces a path in  $Z_s$  whose first projection is a semialgebraic path in  $X_s$  and whose parameter is nonconstant. This contradicts Lemma 6.7. Thus the asserted bound holds for the number of parameters.

Above each counted parameter there is at most one algebraic pair  $(a, b)$ . Indeed, two pairs would give a nonzero algebraic linear relation between  $F_s(z)$  and  $G_s(z)$ , excluded by Theorem 2.5. Counting parameters therefore counts the complete normalized triples.  $\square$

**Theorem 6.9** (Counting complete primitive tuples). *Fix a standard pair  $(s, R)$ ,  $d \geq 1$ , and  $\varepsilon > 0$ . The number of ordinarily convergent primitive identities*

$$\pi(A + B\vartheta)F_s(R/C) = \alpha, \quad A, B, C \in \mathbb{Z}, \quad C \neq 0, \quad \gcd(A, B) = 1,$$

*for which*

$$\max\{|A|, |B|, |C|, H_d(\alpha)\} \leq T$$

*is  $O_{s, R, d, \varepsilon}(T^\varepsilon)$ . Both simultaneous signs are included.*

*Proof.* Theorem 1.4 excludes  $z = 1$ . All other ordinarily convergent integer denominators lie in the fixed interval

$$I_R = [-1, R/(R+1)].$$

The multiplier is real and nonzero. Normalize the identity by

$$(z, a, b) = \left(\frac{R}{C}, \frac{A}{\alpha}, \frac{B}{\alpha}\right) \in X_s.$$

If  $q$  is a primitive polynomial of degree at most  $d$  and height at most  $T$  annihilating  $\alpha$ , then, for  $A \neq 0$ ,  $X^d q(A/X)$  is a nonzero integer polynomial annihilating  $A/\alpha$ . Removing its content yields

$$H_d(A/\alpha) \leq T^{d+1}, \quad H_d(B/\alpha) \leq T^{d+1}, \quad H_d(R/C) \leq \max(R, T).$$

The bound for a zero numerator follows from  $H_d(0) = 1$ . Apply Theorem 6.8 with exponent  $\varepsilon/(d+1)$ . It gives  $O(T^\varepsilon)$  normalized triples.

A normalized triple coming from an integer identity determines a rational coefficient direction. Such a direction contains exactly two primitive integer vectors, and fixing either vector determines  $\alpha$ . Thus normalization has fibers of size at most two on the tuples being counted, proving the result.  $\square$

These theorems supply two different forms of arithmetic scarcity. The separation theorem uses only the degree of  $\alpha^2$ ; the counting theorem controls the complete tuple in degree and height. They leave open the possibility of an isolated non-CM identity or an infinite sequence whose denominators and heights grow sufficiently rapidly. In particular, neither conclusion supplies the missing pointwise CM implication.

#### THE REMAINING ASSERTION

The complex identity fixes one value of a quadratic period pairing. At a CM fiber this value, the complementary eigenline, and the arithmetic congruences agree. The converse would require deriving the extra comparison or Frobenius information from that single value. Neither the complete-tensor theorems nor the scarcity theorems provide this step. Accordingly, existence of the tabulated identities and projective uniqueness are unconditional; their unrestricted exhaustion remains conditional on Hypothesis 1.2.

## APPENDIX A. MODELS AND ANCILLARY PERIOD CONSEQUENCES

**A.1. A simultaneous de Rham calculation.** The elliptic models used in the standard realization are  $E_{s,x} : y^2 = 4X^3 - g_2(x)X - g_3(x)$ , with

| $s$ | $g_2$        | $g_3$                | $\Delta = g_2^3 - 27g_3^2$ |
|-----|--------------|----------------------|----------------------------|
| 1/6 | 3            | $1 - 2x$             | $108x(1-x)$                |
| 1/4 | $3(1+3x)$    | $1 - 9x$             | $729x(1-x)^2$              |
| 1/3 | $3(1+8x)$    | $1 - 20x - 8x^2$     | $1728x(1-x)^3$             |
| 1/2 | $3(1-x+x^2)$ | $(1+x)(2-x)(1-2x)/2$ | $729x^2(1-x)^2/4$          |

The discriminants show that the fibers are smooth throughout the real interval used in the paper except at  $x = 0$ .

There is one connection formula for all four rows. For periods of  $\lambda = dX/y$  and  $\mu = X dX/y$ , it is

$$\begin{aligned} \frac{d}{dx} \begin{pmatrix} \omega \\ h \end{pmatrix} &= \begin{pmatrix} p & q \\ -g_2q/12 & -p \end{pmatrix} \begin{pmatrix} \omega \\ h \end{pmatrix}, \\ p &= -\frac{\Delta'}{12\Delta}, \quad q = \frac{3(2g_2g'_3 - 3g_3g'_2)}{2\Delta}. \end{aligned} \tag{A.1}$$

For completeness, this follows by reduction modulo exact differentials, without choosing any analytic periods. Put  $P = 4X^3 - g_2X - g_3$  and  $r = -g_2q/12$ . The elementary identity

$$d\left(\frac{Q}{y}\right) = \left(Q_X P - \frac{1}{2}Q P_X\right) \frac{dX}{y^3}$$

with

$$Q_1 = -2qX^2 + 2pX + qg_2/3, \quad Q_2 = 2pX^2 + 2rX - pg_2/3 - g'_2/12$$

gives

$$\partial_x \lambda = p\lambda + q\mu + d(Q_1/y), \quad \partial_x \mu = r\lambda - p\mu + d(Q_2/y).$$

The displayed  $p, q, r$  are obtained by matching the coefficients of  $X$ ; their common denominator is the discriminant. Integration over a flat cycle proves (A.1).

In the four rows, respectively,

$$q = -\frac{\varepsilon_s}{6x(1-x)}, \quad \varepsilon_s = (1, 1, 1, 2).$$

Eliminating  $h$  in (A.1) gives

$$\omega'' - \frac{q'}{q}\omega' - \left(p' + p^2 - \frac{q'}{q}p - \frac{g_2q^2}{12}\right)\omega = 0.$$

Substitution of the factored discriminants reduces this equation to  $x(1-x)\omega'' + (1-2x)\omega' - s(1-s)\omega = 0$ . This supplies the differential equation used in Lemma 2.4; the residue at the node determines its normalization, so no numerical matching of periods is involved.

At  $x = 1/2$ , the invariants are 1728, 8000, 54000, 1728. The two less immediate CM assertions follow from the standard quotient by the point  $(0, 0)$  of a curve  $y^2 = x^3 + ax^2 + bx$ :

$$(a, b) \mapsto (-2a, a^2 - 4b), \quad j = \frac{256(a^2 - 3b)^3}{b^2(a^2 - 4b)}.$$

For  $(a, b) = (4, 2)$  the quotient has parameters  $(-8, 8)$  and is isomorphic over  $K$  to the original curve by a scaling with square  $-2$ . Composing the 2-isogeny with this isomorphism gives an endomorphism of degree two, which cannot be an integer multiplication. The original curve therefore has CM and invariant 8000. The curve  $y^2 = x^3 + 1$  has CM; translation of its rational two-torsion point gives parameters  $(-3, 3)$ , whose quotient has parameters  $(6, -3)$  and invariant 54000. CM is invariant under isogeny, proving the remaining assertion.

**A.2. What one scalar relation determines.** These consequences describe the value field of a hypothetical identity; none supplies its missing CM converse.

**Proposition A.1.** *Write  $F = F_s(z)$ ,  $G = \vartheta F_s(z)$  at a standard algebraic parameter  $z \in [-1, 1] \setminus \{0\}$ . Suppose  $AF + BG = \alpha/\pi$  with  $A, B, \alpha \in K$  and  $(A, B) \neq (0, 0)$ . For  $L = cF + dG$  with  $Ad - Bc \neq 0$ , the three numbers  $q_0, \pi, L$  are algebraically independent. Moreover*

$$\text{trdeg}_K K(\pi, F, G) = 2, \quad \ker(K[P, X, Y] \rightarrow K[\pi, F, G]) = (P(AX + BY) - \alpha).$$

*Proof.* Corollary 2.6 gives  $\alpha \neq 0$ . The invertible linear change  $(F, G) \mapsto (AF + BG, L)$  and Theorem 2.5 show that  $q_0, AF + BG, L$  are independent. Replacing the nonzero second coordinate by  $\alpha/\pi$  proves the first claim. The relation also gives  $K(\pi, F, G) = K(F, G)$ , of transcendence degree two. Finally

$$K[P, X, Y]/(P(AX + BY) - \alpha) \simeq K[X, Y, (AX + BY)^{-1}],$$

and evaluation on the algebraically independent pair  $(F, G)$  is injective on this localization.  $\square$

**Proposition A.2.** *If  $a\omega^2 + b\omega h = \alpha\pi$ , with  $a, b, \alpha \in K$  and  $\alpha \neq 0$ , at such a standard parameter, then  $q_0, \omega, h$  are algebraically independent.*

*Proof.* Theorem 2.5 makes  $q_0, R, S$  algebraically independent. The scalar relation gives the invertible field change

$$S = \frac{a\omega + bh}{\alpha}, \quad \omega = \frac{\alpha S}{a + bR}, \quad h = R\omega.$$

Its denominators are nonzero because  $\alpha\pi \neq 0$ . Thus  $K(q_0, \omega, h) = K(q_0, R, S)$  has transcendence degree three.  $\square$

**A.3. Functoriality under gauges and pullbacks.** Suppose  $\tilde{H}(t) = g(t)H(\phi(t))$ , with algebraic  $g, \phi$ ,  $\delta = t d/dt$ , and  $D = z d/dz$ . At an algebraic point  $t_0$  where these data are regular, put  $r_\phi = \delta\phi/\phi$  and suppose  $g\phi r_\phi \neq 0$ . The chain rule is the exact coefficient transport

$$(A + B\delta)\tilde{H}(t_0) = g(t_0)(A + B\delta \log g + Br_\phi D)H(\phi(t_0)). \quad (\text{A.2})$$

Its triangular coefficient matrix has determinant  $r_\phi \neq 0$ . It therefore carries algebraic identity lines bijectively, and preserves all rational projective directions when  $\delta \log g, r_\phi \in \mathbb{Q}$ . More invariantly, the condition for preserving  $\mathbf{P}^1(\mathbb{Q})$  is that the induced transformation belong to  $\text{PGL}_2(\mathbb{Q})$ . The algebraic factor  $g(t_0)$  only rescales the multiplier.

These qualifications are essential. At a ramification point with  $r_\phi = 0$ , the nonzero pair  $(-\delta \log g, 1)$  can have value zero; for example  $g = 1$  and  $\phi = z_0 + (t - 1)^2$  give  $\delta\tilde{H}(1) = 0$  even over a non-CM fiber. Also, (A.2) concerns specified analytic branches. Ordinary convergence must be checked for each defining series, and any independence hypothesis must apply at the new image parameter  $\phi(t_0)$ .

#### A.4. A general conditional finiteness principle.

**Theorem A.3.** *Suppose the data of Section 2 descend to a number field  $K_0$ , and let  $\bar{X}/K_0$  be a smooth projective compactification of the base. Let  $t : \bar{X} \rightarrow \mathbf{P}^1$  be a finite  $K_0$ -morphism of degree  $d$ , and assume  $j(E)$  is nonconstant. Fix  $e$ . If the quadratic-period independence in Theorem 2.2 holds for the non-CM fibers at admissible points with  $[\mathbb{Q}(t(P)) : \mathbb{Q}] \leq e$ , only finitely many such points support a nonzero rational coefficient identity. At each one the primitive integer pair is unique up to sign.*

*Proof.* The identity-line theorem forces CM. A fiber of  $t$  has degree  $d$ , so

$$[\mathbb{Q}(j(E_P)) : \mathbb{Q}] \leq [K_0(P) : \mathbb{Q}] \leq de[K_0 : \mathbb{Q}].$$

By the ring class field theorem, a singular modulus for the full endomorphism order of discriminant  $\Delta$  has degree  $h(\Delta)$  over  $\mathbb{Q}$  [9]. The finiteness theorem for imaginary quadratic orders of bounded class number therefore leaves finitely many possible  $j$ -invariants. The nonconstant map  $j : \bar{X} \rightarrow \mathbf{P}^1$  has finite fibers, giving finitely many points. The uniqueness assertion is unconditional by Theorem 2.2.  $\square$

This argument proves finiteness without asserting an effective algorithm at arbitrary degree. Isotrivial CM families and the nonadmissible fibers require separate treatment.

## APPENDIX B. THE FINITE CM CERTIFICATE AND THE IDENTITY TABLES

The arithmetic theorem reduces enumeration to orders of class number at most two. This appendix specifies the resulting finite certificate. For an order of discriminant  $D$ , let  $H_D$  be its monic ring class polynomial. At levels two and three write

$$P_\ell(J, C) = J^2 - T_\ell(C)J + N_\ell(C).$$

If  $h(D) = 2$ , irreducibility gives the particularly short test

$$P_\ell(J, C) = H_D(J) \iff T_\ell(C) = \text{Tr}(j_D), \quad N_\ell(C) = \text{Nm}(j_D).$$

Thus equality of two degree-two characteristic polynomials replaces expanded resultant calculations. For  $h(D) = 1$ , substitution of its integer root gives the corresponding one-variable test. Exact factorization, rather than a bound on the size of  $C$ , exhausts their integer solutions.

The conductor verification and defining correspondences are stated in Section 4. Its finite scan gives exactly

$$\begin{aligned}\mathcal{D}_1 &= \{-3, -4, -7, -8, -11, -12, -16, -19, -27, -28, -43, -67, -163\}, \\ \mathcal{D}_2 &= \{-15, -20, -24, -32, -35, -36, -40, -48, -51, -52, -60, -64, \\ &\quad -72, -75, -88, -91, -99, -100, -112, -115, -123, -147, -148, \\ &\quad -187, -232, -235, -267, -403, -427\}.\end{aligned}$$

Here  $\mathcal{D}_i$  denotes the discriminants of orders with class number  $i$ . Nonmaximal orders are included. The executed certificate uses Arb complex intervals to certify the integer class-polynomial coefficients and independently compares them with PARI. It checks the integer roots both by exact factorization and by resultants. These computations certify the finite algebraic step; the period and modular arguments prove the evaluations.

The finite computation was executed in SageMath 9.3. The complete discriminant list was regenerated using the proved fundamental bound  $|D| \leq 427$  for class number at most two [8]. For each fundamental discriminant  $d$ , the conductor formula

$$h(df^2) = \frac{h(d)f}{[\mathcal{O}_d^\times : \mathcal{O}_{df^2}^\times]} \prod_{p|f} \left(1 - \frac{(d/p)}{p}\right)$$

shows that  $h(df^2) \leq 2$  implies  $\varphi(f) \leq 6/h(d)$ , because the unit index is at most three. The elementary bound  $\varphi(f) \geq \sqrt{f/2}$  gives  $f \leq 72$ . Thus the conductor scan also has a proved stopping bound. Counting primitive reduced forms and comparison with Sage's proved list give the 42 orders above.

All 42 class polynomials were computed with Arb complex interval arithmetic and independently compared with PARI. Integer roots were cross-checked by factorization over  $\mathbb{Q}$ ; the middle-level candidates were also recovered by exact resultants. No cutoff on  $C$  was used. The retained distinct parameter counts are

| $\ell$                      | 1  | 2  | 3  | 4  |
|-----------------------------|----|----|----|----|
| $ C  \geq R_\ell$           | 12 | 13 | 10 | 5  |
| after removing $C = R_\ell$ | 11 | 12 | 9  | 4. |

The remaining sets are the  $C$  columns of Tables 1–4. The negative endpoints  $-1728$ ,  $-256$  and  $-108$  are absent;  $-64$  is present. In level 4 the entire integer root union before the convergence filter is  $\{16, -8, 1, -64, 256, -512, 4096, 64\}$ , which gives a short independent check. The executed source, exact coefficients, roots and transcript are preserved under `problems/ramanujan-sato/verified/`; the transcript ends with `ALL_ASSERTIONS_PASSED` and exit code zero.

**B.1. Standard classes.** The nome and its parameter  $N$  use the convention in (4.12). We choose  $B > 0$ . Every row is ordinarily convergent; the row  $C = -64$  at level four is conditional, and all other rows are absolutely convergent.

TABLE 1. Standard level 1 identities, normalized by  $B > 0$ .

| nome | $N$ | $A$      | $B$       | $C$                 | $\alpha$                  |
|------|-----|----------|-----------|---------------------|---------------------------|
| +    | 2   | 3        | 28        | 8000                | $5\sqrt{5}$               |
| +    | 3   | 1        | 11        | 54000               | $\frac{5\sqrt{15}}{6}$    |
| +    | 4   | 5        | 63        | 287496              | $\frac{11\sqrt{33}}{4}$   |
| +    | 7   | 8        | 133       | 16581375            | $\frac{85\sqrt{255}}{54}$ |
| -    | 7   | 8        | 63        | -3375               | $5\sqrt{15}$              |
| -    | 11  | 15       | 154       | -32768              | $32\sqrt{2}$              |
| -    | 19  | 25       | 342       | -884736             | $32\sqrt{6}$              |
| -    | 27  | 31       | 506       | -12288000           | $\frac{160\sqrt{30}}{9}$  |
| -    | 43  | 263      | 5418      | -884736000          | $\frac{640\sqrt{15}}{3}$  |
| -    | 67  | 10177    | 261702    | -147197952000       | $1760\sqrt{330}$          |
| -    | 163 | 13591409 | 545140134 | -262537412640768000 | $426880\sqrt{10005}$      |

TABLE 2. Standard level 2 identities, normalized by  $B > 0$ .

| nome | $N$ | $A$  | $B$   | $C$         | $\alpha$                 |
|------|-----|------|-------|-------------|--------------------------|
| +    | 2   | 1    | 7     | 648         | $\frac{9}{2}$            |
| +    | 3   | 1    | 8     | 2304        | $2\sqrt{3}$              |
| +    | 5   | 1    | 10    | 20736       | $\frac{9\sqrt{2}}{4}$    |
| +    | 9   | 3    | 40    | 614656      | $\frac{49\sqrt{3}}{9}$   |
| +    | 11  | 19   | 280   | 2509056     | $18\sqrt{11}$            |
| +    | 29  | 1103 | 26390 | 24591257856 | $\frac{9801\sqrt{2}}{4}$ |
| -    | 5   | 3    | 20    | -1024       | 8                        |
| -    | 7   | 8    | 65    | -3969       | $9\sqrt{7}$              |
| -    | 9   | 3    | 28    | -12288      | $\frac{16\sqrt{3}}{3}$   |
| -    | 13  | 23   | 260   | -82944      | 72                       |
| -    | 25  | 41   | 644   | -6635520    | $\frac{288\sqrt{5}}{5}$  |
| -    | 37  | 1123 | 21460 | -199148544  | 3528                     |

TABLE 3. Standard level 3 identities, normalized by  $B > 0$ .

| nome | $N$ | $A$ | $B$   | $C$       | $\alpha$                |
|------|-----|-----|-------|-----------|-------------------------|
| +    | 2   | 1   | 6     | 216       | $3\sqrt{3}$             |
| +    | 4   | 2   | 15    | 1458      | $\frac{27}{4}$          |
| +    | 5   | 4   | 33    | 3375      | $\frac{15\sqrt{3}}{2}$  |
| -    | 9   | 1   | 5     | -192      | $\frac{4\sqrt{3}}{3}$   |
| -    | 17  | 7   | 51    | -1728     | $12\sqrt{3}$            |
| -    | 25  | 1   | 9     | -8640     | $\frac{4\sqrt{15}}{5}$  |
| -    | 41  | 53  | 615   | -110592   | $96\sqrt{3}$            |
| -    | 49  | 13  | 165   | -326592   | $\frac{108\sqrt{7}}{7}$ |
| -    | 89  | 827 | 14151 | -27000000 | $1500\sqrt{3}$          |

TABLE 4. Standard level 4 identities, normalized by  $B > 0$ .

| nome | $N$ | $A$ | $B$ | $C$  | $\alpha$    |
|------|-----|-----|-----|------|-------------|
| +    | 3   | 1   | 6   | 256  | 4           |
| +    | 7   | 5   | 42  | 4096 | 16          |
| -    | 2   | 1   | 4   | -64  | 2           |
| -    | 4   | 1   | 6   | -512 | $2\sqrt{2}$ |

**B.2. Linear Euler companions.** The coefficient transport of Theorem 4.7 gives the following primitive representatives and signed multipliers. Its boundary argument is necessary: it is the reason that 36 standard classes give 35 ordinary Euler classes.

TABLE 5. Euler companions of the standard level 1 identities.

| $a$           | $b$             | $C$                 | $\beta$                                              |
|---------------|-----------------|---------------------|------------------------------------------------------|
| -33           | 196             | 8000                | $\frac{625\sqrt{5}}{14}$                             |
| 7             | 121             | 54000               | $\frac{625\sqrt{15}}{66}$                            |
| 97            | 1323            | 287496              | $\frac{14641\sqrt{33}}{252}$                         |
| 36872         | 614061          | 16581375            | $\frac{52200625\sqrt{255}}{7182}$                    |
| 88            | 189             | -3375               | $\frac{625\sqrt{15}}{63}$                            |
| 159           | 1078            | -32768              | $\frac{16384\sqrt{2}}{77}$                           |
| 77            | 1026            | -884736             | $\frac{16384\sqrt{6}}{171}$                          |
| 7861          | 128018          | -12288000           | $\frac{10240000\sqrt{30}}{2277}$                     |
| 49709         | 1024002         | -884736000          | $\frac{327680000\sqrt{15}}{8127}$                    |
| 6625229       | 170368002       | -147197952000       | $\frac{149923840000\sqrt{330}}{130851}$              |
| 7575892150829 | 303862746112002 | -262537412640768000 | $\frac{64856464530145280000\sqrt{10005}}{272570067}$ |

TABLE 6. Euler companions of the standard level 2 identities.

| $a$     | $b$      | $C$         | $\beta$                               |
|---------|----------|-------------|---------------------------------------|
| -25     | 49       | 648         | $\frac{729}{14}$                      |
| 0       | 1        | 2304        | $\frac{9\sqrt{3}}{32}$                |
| 7       | 80       | 20736       | $\frac{729\sqrt{2}}{40}$              |
| 179     | 2400     | 614656      | $\frac{117649\sqrt{3}}{360}$          |
| 83      | 1225     | 2509056     | $\frac{88209\sqrt{11}}{1120}$         |
| 4014919 | 96059600 | 24591257856 | $\frac{941480149401\sqrt{2}}{105560}$ |
| 7       | 20       | -1024       | $\frac{32}{5}$                        |
| 776     | 4225     | -3969       | $\frac{35721\sqrt{7}}{65}$            |
| 25      | 196      | -12288      | $\frac{256\sqrt{3}}{7}$               |
| 119     | 1300     | -82944      | $\frac{23328}{65}$                    |
| 6605    | 103684   | -6635520    | $\frac{1492992\sqrt{5}}{161}$         |
| 162839  | 3111700  | -199148544  | $\frac{2744515872}{5365}$             |

TABLE 7. Euler companions of the standard level 3 identities.

| $a$   | $b$    | $C$       | $\beta$                          |
|-------|--------|-----------|----------------------------------|
| -5    | 6      | 216       | $6\sqrt{3}$                      |
| 4     | 75     | 1458      | $\frac{729}{20}$                 |
| 32    | 363    | 3375      | $\frac{1875\sqrt{3}}{22}$        |
| 14    | 25     | -192      | $\frac{64\sqrt{3}}{15}$          |
| 10    | 51     | -1728     | $\frac{192\sqrt{3}}{17}$         |
| 10    | 81     | -8640     | $\frac{64\sqrt{15}}{9}$          |
| 268   | 3075   | -110592   | $\frac{98304\sqrt{3}}{205}$      |
| 718   | 9075   | -326592   | $\frac{46656\sqrt{7}}{55}$       |
| 43834 | 750003 | -27000000 | $\frac{375000000\sqrt{3}}{4717}$ |

TABLE 8. Euler companions of the standard level 4 identities.

| $a$ | $b$ | $C$  | $\beta$                |
|-----|-----|------|------------------------|
| -1  | 6   | 256  | $\frac{16}{3}$         |
| 13  | 126 | 4096 | $\frac{1024}{21}$      |
| 5   | 18  | -512 | $\frac{16\sqrt{2}}{3}$ |

## APPENDIX C. INTEGRAL REALIZATIONS AND THE LIMITS OF PRIME TESTS

**C.1. The finite Clausen theorem.** Write  $s = 1/d$ , with  $d \in \{2, 3, 4, 6\}$ , and let  $U_p, T_p$  be the canonical truncations of Section 5. Set

$$k_2 = k_6 = -1, \quad k_3 = -3, \quad k_4 = -2, \quad \chi_{d,p} = \left(\frac{k_d}{p}\right).$$

**Theorem C.1** (Finite Clausen and reflection). *For every  $p > 3$ , the following are polynomial congruences:*

$$T_p(4x(1-x)) \equiv U_p(x)^2 \pmod{p^2\mathbb{Z}_p[x]}, \quad (\text{C.1})$$

$$U_p(x) \equiv \chi_{d,p} U_p(1-x) \pmod{p^2\mathbb{Z}_p[x]}. \quad (\text{C.2})$$

For  $A, B \in \mathbb{Z}_p$ , their differentiated form is

$$(1-2x)W_p(4x(1-x)) \equiv U_p(x)(A(1-2x)U_p(x) + 2Bx(1-x)U_p'(x)) \pmod{p^2}.$$

In particular the weighted congruence requires no inversion until specialization away from  $1-2x=0$ .

*Proof.* The first identity is [16, §7, equation (7.4)], where it is attributed to Chisholm–Deines–Long–Nebe–Swisher, Lemma 18. We explain the reflection step because its polynomial content is needed over residue-field extensions.

For the elliptic families of [16, Proposition 7.1], that proposition gives, over every finite field  $\mathbf{F}_q$  of characteristic greater than three,

$$a_q(\tilde{E}_d(x)) = \chi_q(k_d)a_q(\tilde{E}_d(1-x))$$

at smooth parameters. It does so by a geometric isogeny of degree 1, 3, 2, 1 for  $d = 2, 3, 4, 6$ , respectively, followed by a quadratic twist. The source families have

| $d$ | $j(\tilde{E}_d(t))$           | $t$ corresponding to $E_{1/d,x}$ |
|-----|-------------------------------|----------------------------------|
| 2   | $256(1-t+t^2)^3/[t^2(1-t)^2]$ | $x$                              |
| 3   | $27(9-8t)^3/[t^3(1-t)]$       | $1-x$                            |
| 4   | $64(4-3t)^3/[t^2(1-t)]$       | $1-x$                            |
| 6   | $432/[t(1-t)]$                | $x$ .                            |

Their Hasse polynomials are  $U_p$ , up to nonzero differential normalizations. The trace identity proves the reflection congruence modulo  $p$  at all smooth  $\mathbf{F}_p$ -parameters. The terminating Chu–Vandermonde identity gives  $U_p(0) = 1$  and  $U_p(1) \equiv \chi_{d,p} \pmod{p}$ , including the two omitted parameters. Since the reflected difference has degree less than  $p$ , this proves its vanishing as a polynomial.

Write  $U_p(x) = \chi_{d,p}U_p(1-x) + pN(x)$ . Invariance of the left side of (C.1) under  $x \mapsto 1-x$  gives

$$2\chi_{d,p}N(x)U_p(1-x) = 0 \quad \text{in } \mathbf{F}_p[x].$$

This integral domain has  $U_p \neq 0$ , so  $N = 0 \pmod{p}$ . This is the modulus- $p^2$  argument of [16, Lemma 7.2]. Differentiation preserves the ideal  $p^2\mathbb{Z}_p[x]$  and yields the weighted identity by the chain rule.  $\square$

The reflected product  $\chi_{d,p}U_p(u)U_p(v)$ , with  $u+v=1$  and  $uv=z/4$ , is symmetric in  $u, v$  and therefore belongs to  $\mathbb{Z}_p[z]$ . It equals  $T_p(z)$  modulo  $p^2$ . This descent explains why all these tests remain valid when the two parameters lie in an unramified quadratic extension rather than in  $\mathbb{Q}_p$ .

**C.2. Cartier and Gauss–Manin differentiation.** We give the normalization behind Lemma 5.1. Write

$$P(X) = 4X^3 - g_2(x)X - g_3(x), \quad m = (p-1)/2, \quad H = [X^{p-1}]P^m, \quad J = [X^{p-2}]P^m.$$

For the standard basis  $\Omega = dX/y$ ,  $\Xi = X dX/y$ , write

$$\nabla_x \Omega = a\Omega + b\Xi, \quad \nabla_x \Xi = c\Omega - a\Xi.$$

The exact differential reductions defining this connection imply

$$H' = aH + bJ, \quad J' = cH - aJ. \tag{C.3}$$

Indeed, multiply either reduction identity by  $P^{m-1}$ . Because  $m = -1/2$  in characteristic  $p$ , its exact correction becomes  $\partial_X(RP^m)$ , whose coefficient of  $X^{p-1}$  is zero. Thus the Gauss–Manin system survives coefficient extraction. In particular  $H$  satisfies

$$x(1-x)H'' + (1-2x)H' - s(1-s)H = 0.$$

For  $s = 1/6, 1/4, 1/3$ , the model coefficients give  $\deg_x H < p$ . At  $x = 0$  their common cubic is  $(X-1)(2X+1)^2$ . The coefficient of  $X^{p-1}$  in a polynomial of degree less than  $2p-1$  is translation invariant modulo  $p$ : the possibly interfering coefficients have  $\binom{j}{p-1} = 0 \pmod{p}$  for  $p \leq j < 2p-1$ . Translation by  $-1/2$  therefore gives  $H(0) = (-6)^m$ . The Gauss equation's recurrence

$$(n+1)^2 h_{n+1} = (n+s)(n+1-s)h_n \quad (0 \leq n < p-1)$$

then identifies  $H = (-6)^m U_p$ . For  $s = 1/2$ , the affine change  $X = 3T/2 - (1+x)/2$  turns the cubic into  $(27/2)T(T-1)(T-x)$ . The Legendre Hasse polynomial  $\sum_{n=0}^m \binom{m}{n}^2 x^n$  and the same coefficient invariance give the identical normalization. Hence, in all four families,

$$H = (-6)^m U_p. \tag{C.4}$$

Cartier coefficient extraction now gives

$$\text{Cartier}(\nu_{A,B}) = ((A + BQa)H + BQbJ)^{1/p} \Omega = ((-6)^m (AU_p + BQU_p'))^{1/p} \Omega.$$

There is only one coefficient because  $\deg_X((A + BQa + BQbX)P^m) < 2p-1$ ; the notation incorporates the usual Frobenius twist. Together with the conjugate-filtration sequence, this proves the full Cartier statement over arbitrary perfect residue fields. It also shows why a change of period normalization cannot be ignored in the finite-polynomial test.

**C.3. Projective Frobenius and rational-prime density.** We supply the Galois details used in Theorem 5.5.

**Lemma C.2** (Supersingular projective Frobenius). *For a supersingular elliptic curve over  $\mathbf{F}_{p^f}$ ,  $p \geq 5$ , Frobenius on any auxiliary Tate module has projective order dividing six.*

*Proof.* Write the Frobenius polynomial as  $T^2 - aT + p^f$ . Supersingularity gives slopes  $1/2, 1/2$ , so  $v_p(a) \geq f/2$ ; the Hasse–Weil bound gives  $|a| \leq 2p^{f/2}$ . If  $f$  is odd these inequalities force  $a = 0$ , and the eigenvalue ratio is  $-1$ . If  $f$  is even, then  $a/p^{f/2} \in \{-2, -1, 0, 1, 2\}$ . The middle three values give eigenvalue ratios of order dividing six. At either endpoint Frobenius is scalar: its characteristic equation would otherwise give a nonzero nilpotent in the rational endomorphism division algebra of an elliptic curve. This proves the assertion.  $\square$

For the non-CM  $\mathbb{Q}$ -curve, choose a finite Galois field of definition  $K/\mathbb{Q}$ , write  $d = [K : \mathbb{Q}]$ , and choose isogenies  $\mu_\sigma : {}^\sigma E \rightarrow E$ , with  $\mu_1 = 1$ . They need not be defined over  $K$ . For every sufficiently large auxiliary prime  $\lambda$ , the maps

$$A_g = \mu_{g|_K} \circ g : E[\lambda] \rightarrow E[\lambda]$$

are invertible. The rational Hom spaces between conjugate non-CM curves are one-dimensional, so  $A_g A_h = c(g, h) A_{gh}$  with  $c(g, h) \in \mathbb{Q}^\times$ . Its square is a quotient of the finitely many isogeny degrees; after excluding their prime divisors it is a  $\lambda$ -adic unit. Projectivization therefore gives a continuous representation

$$\bar{\rho}_\lambda : G_{\mathbb{Q}} \rightarrow \mathrm{PGL}_2(\mathbf{F}_\lambda).$$

Continuity follows on adjoining the isogeny coefficients and torsion coordinates to a finite Galois extension. On  $G_K$  this is the usual projective representation. Serre’s open image theorem [18], in its number-field form, says that the residual image on  $G_K$  contains  $\mathrm{SL}_2(\mathbf{F}_\lambda)$  for every sufficiently large  $\lambda$ . Thus the projective image contains  $\mathrm{PSL}_2(\mathbf{F}_\lambda)$ .

Fix such a  $\lambda$  and omit its finite ramification and bad-reduction sets. At a supersingular rational prime, let  $g$  be Frobenius and let  $f \mid d$  be the residue degree in  $K$ . Then  $g^f \in G_K$  acts as Frobenius over  $\mathbf{F}_{p^f}$ . The lemma gives

$$\bar{\rho}_\lambda(g)^{6d} = 1.$$

Put  $M = 6d$  and take  $\lambda > M$ . An element killed by  $M$  is semisimple. The split and nonsplit torus types have orders  $\lambda - 1$  and  $\lambda + 1$ , and each contains at most  $M$  such elements. Hence at most  $2M$  conjugacy classes occur, each of size at most  $\lambda(\lambda + 1)$ . Since  $|\mathrm{PSL}_2(\mathbf{F}_\lambda)| = \lambda(\lambda^2 - 1)/2$ , their proportion in the residual image is at most

$$\frac{4M}{\lambda - 1}.$$

Chebotarev identifies the density of each fixed finite-image condition with its cardinality proportion. The upper natural density of our fixed supersingular prime set is bounded by this quantity for every sufficiently large  $\lambda$ , hence is zero. No uniform Chebotarev estimate is used: the prime-counting limit is taken before  $\lambda$  tends to infinity.

**C.4. The analytic order in Tang’s inequality.** For completeness the order convention in Theorem 5.2 is

$$T_\gamma(r) = \int_0^r \frac{dt}{t} \int_{\|z\| < t} \gamma^* \eta \wedge \omega^{d-1}, \quad \omega = dd^c \log \|z\|^2, \quad \rho_\gamma = \limsup_{r \rightarrow \infty} \frac{\log T_\gamma(r)}{\log r},$$

where  $\eta$  is the curvature of an ample Hermitian line bundle on a projective compactification. The order is independent of these choices in the setting used here.

For an elliptic universal extension the order-two estimate has a direct geometric explanation. Write  $E(\mathbb{C}) = \mathbb{C}/\Lambda$ , and use its Weierstrass functions  $\wp, \zeta, \sigma$  and quasiperiod homomorphism  $\eta_\lambda$ . After rescaling the additive coordinate,

$$G(\mathbb{C}) = \mathbb{C}^2 / \{(\lambda, \eta_\lambda) : \lambda \in \Lambda\}.$$

A complementary line has exponential map  $t \mapsto [(at, bt)]$  with  $a \neq 0$ . On the torsor above  $E \setminus \{0\}$  the rational coordinates  $\wp(u), \wp'(u), v - \zeta(u)$  are invariant under the lattice; the addition theorem for  $\zeta$  gives their rational group law. Since  $\sigma$  is entire of order two, its logarithmic derivatives, and hence these three pulled-back coordinates, have Nevanlinna order at most two. Rational characteristic inequalities give the same bound in any projective embedding.

Conversely the projection  $t \mapsto at \bmod \Lambda$  pulls an ample invariant curvature form on  $E$  back to a positive constant multiple of  $|a|^2 i dt \wedge d\bar{t}$ . Its characteristic is therefore of order exactly two. Comparison of ample bundles on a projective-bundle compactification of  $G$  bounds this characteristic by that of the original map. Thus the order is exactly two. The threshold  $3/4$  cannot be improved solely by sharpening this growth estimate; this is not a claim that the arithmetic threshold itself is optimal.

### C.5. Unweighted zeros and supersingularity.

**Proposition C.3** (An arithmetic classification of parameters). *For  $|C| \geq R_\ell$ ,  $C \neq R_\ell$ , and all sufficiently good primes  $p$ , one has*

$$T_p(R_\ell/C) = 0 \pmod{p} \iff T_p(R_\ell/C) = 0 \pmod{p^2} \iff E_{s,x} \text{ is supersingular.}$$

*Their rational-prime set has natural density  $1/2$  for exactly the 36 level-denominator pairs in the standard tables, and density zero otherwise. At each admitted prime  $W_p = 0 \pmod{p}$  for every coefficient pair. At level four the set is infinite for every parameter in this domain, including the non-CM ones.*

*Proof.* Work in the unramified quadratic algebra containing the roots of  $x^2 - x + (R_\ell/C)/4$ . The Hasse criterion and the finite Clausen square show that vanishing modulo  $p$  is equivalent to supersingularity, and that it implies divisibility by  $p^2$ . Divisibility descends to  $\mathbb{Z}_p$  by unramifiedness; the reverse implication is immediate. Differentiating the square gives  $W_p = 0 \pmod{p}$  when its Hasse factor vanishes. The density assertion now follows from Theorem 5.5 and the finite CM classification.

For infinitude at level four, descend geometrically to the rational curve

$$\mathcal{E}_z : y^2 = 4X^3 - 3(1 - z/4)(1 - z)X - (8 + z)(1 - z)^2/8. \quad (\text{C.5})$$

Its discriminant is  $729z^2(1 - z)^3/64$  and  $j = 64(4 - z)^3/z^2$ , which agrees with that of the Legendre fiber when  $z = 4x(1 - x)$ . Hence geometric supersingularity agrees outside finitely many primes. Elkies' theorem [19] states that every elliptic curve over  $\mathbb{Q}$  has infinitely many supersingular primes, proving the claim.  $\square$

For example,  $C = 72$  at level four has  $j = 21952/9$  and is non-CM, yet has infinitely many such zeros. Thus infinitude alone cannot replace positive density in the unweighted criterion. Nor can finitely many tests distinguish the CM rows: fix finitely many  $p \geq 5$  and exponents  $e_p$ , and put  $M = \prod p^{e_p}$ . Every  $C = 256 + kM > 4096$  has, with  $(A, B) = (1, 6)$ , the same weighted truncations of every length modulo each  $p^{e_p}$  as  $C = 256$ , term by term. Nevertheless

$$j(C) = C^2 - 48C + 768 - 4096/C$$

is nonintegral, so all these parameters are non-CM.

**C.6. Higher precision and the missing comparison.** At an ordinary prime, the weighted factorization defines a canonical local slope, but its lift modulo  $p^2$  differs from the rational CM coefficient slope.

**Proposition C.4** (The local slope). *Let  $p > 3$  and  $z \in \mathbb{Z}_p$ . If  $z, 1 - z$  are units and the fiber is ordinary, then*

$$r_p = -Q(x) \frac{U'_p(x)}{U_p(x)} \pmod{p^2}$$

*is independent of the root  $x$  and lies in  $\mathbb{Z}/p^2\mathbb{Z}$ . For  $A, B \in \mathbb{Z}_p$  with  $B$  a unit,  $W_p(z) = 0 \pmod{p^2}$  if and only if  $A/B = r_p \pmod{p^2}$ . Modulo  $p$  this is the complementary Cartier-kernel line.*

*Proof.* The algebra  $\mathcal{O} = \mathbb{Z}_p[x]/(x^2 - x + z/4)$  is finite etale; its involution has fixed ring  $\mathbb{Z}_p$ . Ordinarity makes  $U_p(x)$  a unit. Reflection and its derivative give  $U'_p(x) \equiv -\chi_{d,p} U'_p(1 - x) \pmod{p^2}$ . Since  $Q(1 - x) = -Q(x)$ , the quotient defining  $r_p$  is invariant, hence rational modulo  $p^2$ ; invariants commute with reduction because two is invertible. The factorization becomes  $W_p \equiv BU_p^2(A/B - r_p) \pmod{p^2}$ , proving the assertions.  $\square$

By contrast, [16, Theorem 2.1], restating the theorem of Chisholm–Deines–Long–Nebe–Swisher, assumes both CM of the source fiber and its identity  $(1 + \beta\vartheta)F_{1/d}(z) = \delta/\pi$  with  $|z| < 1$ . Under its good-reduction, unramifiedness, and  $p$ -adic-unit hypotheses, its conclusion is

$$T_p(z) + \beta z T'_p(z) \equiv \epsilon_p \left( \frac{1 - z}{p} \right) p \pmod{p^2}, \quad \epsilon_p = \begin{cases} 1 & \text{ordinary,} \\ -1 & \text{supersingular.} \end{cases} \quad (\text{C.6})$$

We use this source only in its stated strict convergence range. For a CM identity with  $AB \neq 0$ , putting  $\beta = B/A$  therefore gives, outside finitely many primes,

$$W_p(z) \equiv A\epsilon_p \left( \frac{1 - z}{p} \right) p \pmod{p^2}. \quad (\text{C.7})$$

Thus  $v_p(W_p) = 1$ . At ordinary primes the exact first-order drift of the local slope is

$$r_p \equiv \frac{A}{B} \left( 1 - \frac{\epsilon_p \left( \frac{1 - z}{p} \right) p}{U_p(x)^2} \right) \pmod{p^2}.$$

For the valid row  $(A, B, C) = (1, 6, 256)$ , one obtains  $W_7(1/4) = 42 \bmod 49$ ,  $r_7 = 20 \bmod 49$ , whereas  $1/6 = 41 \bmod 49$ . A transfer demanding  $W_p = 0 \bmod p^2$  would therefore fail even for a tabulated identity. Likewise the universal finite Clausen congruence cannot be strengthened to modulus  $p^3$ : at  $s = 1/2$  and  $p = 5$  its discrepancy is  $75x^6 \bmod 125$ .

**Lemma C.5** (The complex series does not sum directly  $p$ -adically). *At every prime  $p \nmid C$ , a pair-primitive standard series diverges as a  $p$ -adic series.*

*Proof.* Choose  $e \in \{0, 1\}$  with  $A + Be \neq 0 \bmod p$ . For  $n = e + p^r$  and  $p^r > 6$ , the base- $p$  digit blocks of  $kn$  are disjoint for  $1 \leq k \leq 6$ . Legendre's formula for factorial valuations, applied to the balanced factorial ratio  $u_\ell$ , therefore gives

$$v_p(u_\ell(e + p^r)) = v_p(u_\ell(e)) + v_p(u_\ell(1)).$$

The linear weight and  $C^{-n}$  are units. Along this subsequence the summand valuation is constant and finite, so the terms do not tend to zero.  $\square$

Raw finite truncations also need not equal Frobenius unit roots modulo  $p^2$ . At  $s = 1/2$ ,  $x = 1/3$ ,  $z = 8/9$ , the fiber is non-CM and has trace  $-2$  at  $5$ . The unit root of  $T^2 + 2T + 5$  is  $13 \bmod 25$ , while

$$U_5(1/3) = 8, \quad T_5(8/9) = 14 \bmod 25,$$

and  $13^2 = 19 \bmod 25$ . The unit-root statement in [16, Theorem 2.2] retains its CM premise. In that work the CM action first selects a second-kind eigenvector, after which a compatible Frobenius lift yields the supercongruence [16, §2.1, equation (2.3), and Lemma 2.3]. Neither the universal polynomial identity nor this CM construction provides the eigenvector from an isolated complex period product.

### C.7. Exact compatibility, finite precision, and a common tensor.

**Proposition C.6** (Finite precision does not force CM). *There are non-CM elliptic curves over  $\mathbb{Q}$  with a common good ordinary reduction at  $5$  whose Hodge-preservation obstruction in Theorem 5.9 is arbitrarily small  $5$ -adically in fixed analytic local frames.*

*Proof.* Consider  $E_N : y^2 = x^3 - x + 5^N$ ,  $N \geq 2$ . Its discriminant is  $16(4 - 27 \cdot 5^{2N})$ , a  $5$ -adic unit, and its common reduction has eight  $\mathbf{F}_5$ -points, hence ordinary trace  $-2$ . Its invariant  $j(E_N) = 6912/(4 - 27 \cdot 5^{2N})$  is a nonzero rational number of absolute value less than one, hence nonintegral and non-CM. The curves tend  $5$ -adically to the CM curve  $y^2 = x^3 - x$ . On a sufficiently small deformation disk, the family crystalline-de Rham comparison identifies the fixed special-fiber isocrystal with the analytic de Rham bundle and makes the Hodge line analytic [28, §4.6]. The obstruction  $F^1 H \rightarrow H/F^1 H$  therefore tends to zero. It is nonzero at every  $E_N$  by Theorem 5.9.  $\square$

The distinction between local and common invariants is equally sharp on the Galois side.

**Proposition C.7** (A common Galois tensor). *For an elliptic curve  $E/K$  and an auxiliary prime  $\lambda$ ,  $E$  has geometric CM if and only if*

$$\mathrm{Sym}^2 V_\lambda(E) \otimes \det(V_\lambda(E))^{-1}$$

*has a nonzero invariant over some finite extension of  $K$ .*

*Proof.* For every two-dimensional vector space, polarization gives

$$\mathrm{Sym}^2 V \otimes \det(V)^{-1} \simeq \mathrm{End}^0(V),$$

where the superscript denotes trace zero. Faltings' endomorphism theorem [21] identifies the invariant endomorphisms over a finite extension  $L/K$  with  $\mathrm{End}_L(E) \otimes \mathbb{Q}_\lambda$ . Its trace-zero part is zero for a non-CM curve and nonzero once the CM endomorphisms are defined. This proves both directions.  $\square$

For every elliptic curve over  $\mathbb{Q}$ , however, each good Frobenius on the untwisted symmetric square has eigenvalues  $r_p^2, r_p s_p, s_p^2$ , with  $r_p s_p = p$ . The Tate eigenvalue is therefore present at every prime, including for non-CM curves; its eigenline need not be common to the different Frobenii. Factoring off these local eigenvalues cannot by itself prove an analytic pole of an  $L$ -function, since the remaining factor may have a zero. A complex scalar relation, a collection of local eigenvalues, and a common Galois tensor are distinct data.

## APPENDIX D. SCALAR COMPARISON AND ITS REFINEMENTS

Throughout this appendix,  $E/k$  is an algebraic elliptic curve,  $k = \overline{\mathbb{Q}}$ , and the period matrix has determinant  $2\pi i$ . Unless stated otherwise we assume the nonzero scalar relation

$$\omega(a\omega + bh) = \alpha\pi, \quad a, b, \alpha \in k, \quad \alpha \neq 0.$$

The main text identifies its missing comparison coordinate. We record here the exact restricted hypothesis, useful analytic refinements, and the limits of weaker replacement conditions.

**D.1. The restricted directions and known independence.** For the standard real branch put  $t = \sqrt{1-z}$  and  $(e_1, e_2, e_3, e_4) = (0, 1/4, 1/2, 0)$ . The Gauss–Manin coefficients give

$$\left\{ \frac{a}{b} : A/B \in \mathbb{Q} \right\} = t\mathbb{Q} - e_\ell \quad (B \neq 0).$$

Indeed, substituting the explicit logarithmic derivatives of the discriminants into  $a/b = (A/B + u_s)/v_s$  yields a nonzero rational multiple of  $tA/B$ , a rational multiple of  $t$ , and the constant  $-e_\ell$ . Consequently the precise restriction needed to exclude non-CM rational slopes is transcendence of

$$\frac{\omega^2}{\pi}, \quad \frac{\omega h + (tq - e_\ell)\omega^2}{\pi} \quad (q \in \mathbb{Q})$$

at the specified non-CM fibers. Multiplication by the nonzero algebraic coefficient  $b$  proves both directions of this equivalence. This is weaker than QPI, but is an equivalent reformulation of the rational-slope exclusion itself.

Two unconditional independence theorems clarify this restriction. Chudnovsky gives algebraic independence of  $h/\omega$  and  $\pi/\omega$ . Under the scalar relation, write  $r = h/\omega$ ; then

$$\frac{\pi}{\omega} = \frac{\omega(a + br)}{\alpha}, \quad \omega = \frac{\alpha(\pi/\omega)}{a + br}.$$

The denominator is nonzero because  $\alpha \neq 0$ . Hence  $k(r, \pi/\omega) = k(r, \omega) = k(\omega, h)$ , so  $\omega, h$  would be algebraically independent even under a hypothetical non-CM identity. The theorem therefore does not rule out that identity.

Nesterenko's theorem, in [3, Theorem 1.1], states

$$\text{trdeg}_{\mathbb{Q}} \mathbb{Q}(q, E_2(\tau), E_4(\tau), E_6(\tau)) \geq 3 \quad (\tau \in \mathfrak{H}).$$

Put  $\xi = \omega/(2\pi i)$  and  $v = h/(2\pi i)$ . The exact lattice formulas

$$E_2 = 12\xi v, \quad E_4 = 12g_2\xi^4, \quad E_6 = -216g_3\xi^6$$

place the modular quantities in the three-generator field  $k(q, \xi, v)$ . That field therefore has transcendence degree exactly three. The proposed identity becomes

$$4\pi(a\xi^2 + b\xi v) + \alpha = 0,$$

which only puts  $\pi$  in  $k(\xi, v)$ . It does not contradict the independence of  $q, \xi, v$ .

The comparison fields of the main text give a concise account of the conjectural deficit. A relation  $aP + bQ = \alpha$ , where  $P = \omega^2/\pi$  and  $Q = \omega h/\pi$ , implies

$$\begin{aligned} \text{trdeg}_k k(\tau, j', j'') &\leq 2, \\ \text{trdeg}_k K_E &\leq 3, \\ \text{trdeg}_k k(2\pi i, \tau, q, E_2, E_4, E_6) &\leq 4. \end{aligned}$$

Each inequality follows from (3.4)–(3.8), adding at most one transcendence degree for each extra generator. These are precisely the deficits contradicted by the respective numerical conjectures.

**D.2. Functional transcendence and specialization.** The unconditional Ax–Schanuel consequence in [10, Section 4.1] does not provide the numerical MSCD assertion used here. In that reference let  $C_j = \text{jcl}(\emptyset)$ . It is an algebraically closed field containing  $k$ , and Lemma 4.6 gives

$$j(\tau) \in C_j \implies \tau \in C_j \implies j(\tau), j'(\tau), j''(\tau) \in C_j.$$

Its Proposition 4.7 measures transcendence over a  $j$ -closed field. At an algebraic  $j$ -value all the displayed entries lie in  $C_j$ ; both the transcendence degree and the associated dimension in that bound are zero. Thus this established functional theorem gives no positive numerical lower bound at the points required here. A relation holding at one point also supplies no relation among its derivatives on a neighborhood. Similarly, an abstract algebraic conjugation cannot be commuted with an infinite analytic sum without a separate argument.

**D.3. A real refinement of cycle propagation.** Suppose now that the model and the coefficients are real algebraic. Choose a fundamental basis with  $\omega > 0$  and  $\tau = r + iy$ , where  $r \in \{0, 1/2\}$  and  $y > 0$ .

**Proposition D.1** (Mixed-cycle reality). *Under the scalar relation, the following are equivalent:*

- (1)  $E$  has CM;
- (2)  $\omega_{m,n}(a\omega_{m,n} + bh_{m,n})$  is real for some integers  $m, n$  with  $n(m + nr) \neq 0$ .

At CM every such value satisfies

$$\frac{\omega_{m,n}(a\omega_{m,n} + bh_{m,n})}{\pi} = \alpha|m + n\tau|^2 \in k \cap \mathbf{R}.$$

*Proof.* By (3.11), the imaginary part of the value divided by  $\pi$  is

$$2n(m + nr)(\alpha y + b).$$

The reality hypothesis gives  $y = -b/\alpha \in k$ . Thus  $\tau$  is algebraic and Schneider's theorem proves CM.

Conversely, at CM the independence of  $\omega$  and  $\pi$  excludes  $b = 0$ . Proposition 3.6 then gives  $b = -\alpha y$ . For  $L = m + n\tau$  we have  $\bar{L} = L - 2iny$ , so  $\alpha L^2 + 2ibnL = \alpha L\bar{L}$ . This proves the formula and the converse.  $\square$

The condition  $n(m + nr) \neq 0$  is essential. Values on a real or purely imaginary cycle are automatically real in (3.11). In particular, Schwarz reflection, which makes values on conjugate cycles conjugate, supplies no mixed-cycle reality condition.

The same limitation can be seen before forming the quadratic value. Write  $\omega_2 = r\omega + iv$  and  $h_2 = rh + iu$ , with  $u, v$  real. Legendre gives  $\omega u - vh = 2\pi$ , and the scalar relation gives

$$\omega(u - 2a\omega/\alpha) - h(v + 2b\omega/\alpha) = 0.$$

Conjugation repeats this equation. If  $v + 2b\omega/\alpha = 0$ , then  $v/\omega$  is algebraic and Schneider forces CM. Otherwise a purported algebraic linearization must first establish algebraicity of that same period ratio.

For completeness, the modular form of cycle transport is equally exact. If  $\gamma = \begin{pmatrix} u & v \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$  and  $\lambda = c\tau + d$ , then, changing the period basis while keeping the algebraic model fixed,

$$P(\gamma\tau) = \lambda^2 P(\tau), \quad Q(\gamma\tau) = \lambda^2 Q(\tau) + 2ic\lambda.$$

The first equation follows from  $\omega_{d,c} = \lambda\omega$ ; the second follows from Legendre. Hence the transported affine value is  $\alpha\lambda^2 + 2ibc\lambda$ , agreeing with (3.11) and supplying no automatic propagation.

**D.4. The almost-holomorphic coordinate.** In the same real normalization put

$$\kappa = \frac{h}{\omega} + \frac{\pi}{y\omega^2}.$$

Under the scalar relation, algebraicity of  $\kappa$  forces CM. Indeed, Legendre's identity gives the exact linear relation

$$h_2 - \kappa\omega_2 - \left(r + \frac{ib}{\alpha}\right)h + \left(r\kappa - \frac{ia}{\alpha}\right)\omega = 0.$$

If  $\kappa$  were algebraic and  $E$  non-CM, this would contradict the algebraic linear independence of its four period entries [1, Section 5]; the coefficient of  $h_2$  is one. Conversely, CM makes  $\kappa$  algebraic. To see the normalization directly, diagonalize a nonreal CM endomorphism on de Rham cohomology. Its antiholomorphic eigenvector has the form  $\Xi - \kappa_0\Omega$  with  $\kappa_0 \in k$ . Thus  $h_j - \kappa_0\omega_j = K\bar{\omega}_j$ . Taking the determinant with  $(\omega_1, \omega_2)$  yields

$$K = -\frac{\pi}{y\omega^2}, \quad \kappa_0 = \frac{h}{\omega} + \frac{\pi}{y\omega^2} = \kappa.$$

This also proves that  $\kappa$  is real.

For  $g_2g_3 \neq 0$ , the modular interpretation is

$$E_2^* = E_2 - \frac{3}{\pi y}, \quad \frac{E_2^*E_4}{E_6} = -\frac{2g_2}{3g_3}\kappa.$$

This follows by substitution of the lattice formulas. Masser's theorem gives algebraicity of this almost-holomorphic value at CM points [15, Section 5]. Under the scalar relation alone, however, one only has

$$\kappa = -\frac{a}{b} + \left(\frac{\alpha}{b} + \frac{1}{y}\right)\frac{\pi}{\omega^2} \quad (b \neq 0),$$

whose coefficient  $y$  is uncontrolled. The exceptional cases  $g_2g_3 = 0$  already have CM.

**D.5. Coordinates of tensor completion.** Put  $D = 2\pi i$  and  $k_0 = \alpha/(2i)$ . The symmetric bilinear coordinates of  $a\Omega^2 + b\Omega\Xi$ , divided by its determinant period, are

$$s_{ij} = \frac{a\omega_i\omega_j + (b/2)(\omega_i h_j + h_i \omega_j)}{D}.$$

Legendre's identity gives

$$(s_{11}, s_{12}, s_{22}) = (k_0, k_0\tau + b/2, k_0\tau^2 + b\tau), \quad s_{12}^2 - s_{11}s_{22} = b^2/4. \quad (\text{D.1})$$

Thus algebraicity of  $s_{12}$  is equivalent to algebraicity of  $\tau$ , to CM, and to algebraicity of all three coordinates. The determinant relation is automatic and adds no missing coordinate.

When  $b \neq 0$ , the tensor  $t$  of Proposition 3.6 has, in the polynomial Betti basis  $(e_1^2, e_1e_2, e_2^2)/(e_1 \wedge e_2)$ , coordinates

$$t_B = \left( \frac{1}{c}, \frac{2\tau + c}{c}, \frac{\tau(\tau + c)}{c} \right). \quad (\text{D.2})$$

The middle polynomial coefficient is twice the symmetric bilinear coefficient. The canonical adjoint identification sends  $(x, y, z)$  to

$$\begin{pmatrix} -y/2 & x \\ -z & y/2 \end{pmatrix} = \Pi_B - \frac{1}{2}I.$$

This verifies the conventions in the intrinsic comparison square.

At CM the rational endomorphism algebra is the imaginary quadratic field  $K$ , which has no nontrivial idempotent. The complete projector belongs to  $K \otimes k \simeq k \times k$ , and already exists with coefficients in  $K$ ; it is a  $K$ -linear combination of algebraic endomorphisms. Its entry field is exactly  $K$  in a rational Betti basis. There is no assertion that it is a rational endomorphism.

**D.6. The local annihilator condition.** Set  $T = \text{Sym}^2 H \otimes (\det H)^{-1}$ ,  $N = T \oplus \mathbf{1}$ , and  $\xi = (t, c^{-1}) \in N_{\text{dR}}$ . Let  $\beta$  be the algebraic Betti functional which subtracts the  $\mathbf{1}$ -coordinate from the first coordinate of  $T$ . Equation (D.2) gives  $\beta(\text{comp } \xi) = 0$ .

**Proposition D.2** (Local annihilator lift). *Under the scalar identity with  $b \neq 0$ , CM is equivalent to the existence of a de Rham–Betti subobject  $N' \subseteq N$  such that*

$$\xi \in N'_{\text{dR}}, \quad N'_B \subseteq \ker \beta.$$

*Proof.* For non-CM  $E$ , Theorem 3.5 identifies the algebraic-coefficient group with  $\text{GL}_2$ . Its adjoint representation  $T$  is irreducible and nontrivial. Semisimplicity and  $\text{Hom}(T, \mathbf{1}) = \text{Hom}(\mathbf{1}, T) = 0$  show that a subrepresentation of  $T \oplus \mathbf{1}$  containing a vector with nonzero components in both summands is the full object. Both components of  $\xi$  are nonzero. Thus  $N' = N$ , contrary to  $N'_B \subseteq \ker \beta$ . At CM, completion makes all coordinates of  $\text{comp } \xi$  algebraic. The line spanned by  $\xi$  and its comparison is a de Rham–Betti subobject whose Betti line lies in  $\ker \beta$ .  $\square$

Proposition 3.6 of [4] characterizes equality between the numerical Zariski closure of a comparison point and its smallest algebraic torsor by such annihilator lifting for every object in the generated tensor category. Proposition 5.8 proves lifting in the abelian category generated by first cohomology of abelian varieties. The adjoint tensor  $T$  is not a first-cohomology object. Theorem 5.13 identifies the de Rham–Betti torsor with the motivic torsor, without asserting that the numerical Zariski closure equals that torsor. The local lift above is consequently an exact additional premise.

This distinction also persists for the Kummer surface  $S = \text{Km}(E \times E)$ . Its Picard rank is 19 in the non-CM case and 20 in the CM case: the ranks 3 and 4 of  $E \times E$  acquire the sixteen exceptional divisors. Corollary 6.17 of [4] gives, respectively,

$$\Omega_S = \Omega_S^{\text{mot}}, \quad Z_S = \Omega_S^{\text{mot}}.$$

Here  $Z_S$  is the numerical Zariski closure and  $\Omega_S$  is the smallest algebraic subtorsor containing the comparison point. The rank-19 assertion does not identify  $Z_S$  with that torsor.

**D.7. Formal obstructions to automatic completion.** For fixed  $c \in k^\times$ , every trace-one, rank-one projector with upper-right entry  $1/c$  is

$$\begin{pmatrix} x & 1/c \\ cx(1-x) & 1-x \end{pmatrix}, \quad x \in \mathbb{C}.$$

Indeed trace one fixes the second diagonal entry and determinant zero fixes the lower-left entry; the Cayley–Hamilton identity then proves idempotence. Thus these algebraic conditions leave one free coordinate. For example  $c = -2i$ ,  $\tau = iy$ ,  $y > 0$ , gives

$$\begin{pmatrix} y/2 & i/2 \\ i(y^2 - 2y)/2 & 1 - y/2 \end{pmatrix}.$$

Conjugation of its entries followed by conjugation by  $\text{diag}(1, -1)$  fixes the matrix, for every  $y > 0$ . Real structure and positivity therefore leave the same freedom; actual CM completion would require  $y = 1$  by (3.9). These are formal comparison objects, not counterexamples over algebraic elliptic curves.

Connectedness of a torsor is likewise insufficient for genericity: evaluation at 2 on the coordinate ring  $k[t, t^{-1}]$  of the connected torsor  $\mathbf{G}_m$  has kernel  $(t - 2)$ . More directly, set

$$f(M) = aw^2 + bwh - \kappa \det M, \quad \kappa \neq 0, \quad M = \begin{pmatrix} w & h \\ v & k_2 \end{pmatrix}.$$

At a zero of  $f$  in  $\text{GL}_2$ ,  $f(\text{diag}(1, r)M) = \kappa(1 - r) \det M$ . Thus a numerical zero does not make the relation an identity on the torsor.

**D.8. Why logarithm methods need additional data.** The analytic subgroup theorem applies to a logarithm with algebraic exponential lying in an algebraic linear subspace of a commutative algebraic Lie algebra. The universal vector extension of  $E$ , together with  $\mathbf{G}_m$ , provides the period logarithm  $(\omega, h, 2\pi i)$  with exponential the identity. Our relation instead places it on

$$q(w, v, t) = aw^2 + bwh - \frac{\alpha}{2i}t = 0.$$

The tangent hyperplane there is not algebraic. Its normal vector is  $(2a\omega + bh, b\omega, -\alpha/(2i))$ ; since its last entry is nonzero algebraic, an algebraic tangent hyperplane would make the other two entries algebraic. If  $b \neq 0$  this makes  $\omega$  algebraic; if  $b = 0$ , then  $a \neq 0$  and the first entry has the same consequence. Both contradict period transcendence [1, Corollary 3].

A bilinear central extension does not remove the obstruction. Let  $B : V \times V \rightarrow W$  be bilinear in characteristic zero. The associative law

$$(v, t) * (w, s) = (v + w, t + s + B(v, w))$$

is commutative exactly when  $B$  is symmetric, since its commutator is  $(0, B(v, w) - B(w, v))$ . In that case  $(v, t) \mapsto (v, t - B(v, v)/2)$  is an algebraic isomorphism to the additive group. Its differential at zero is the identity; hence the original exponential is  $(v, t) \mapsto (v, t + B(v, v)/2)$ . Algebraicity of its endpoint still requires algebraicity of  $v$ , which fails for the period vector. This calculation is about global bilinear cocycles on vector groups; it neither splits an elliptic universal extension nor classifies general biextensions.

Nor can one repair this construction by reducing a quadratic coordinate modulo a fixed discrete group. Suppose a lattice  $\Lambda$  spans a complex vector space  $L$ , and a homogeneous quadratic form  $q$  obeys

$$q(z + \lambda) - q(z) \in \Gamma \quad (z \in L, \lambda \in \Lambda)$$

for a discrete additive subgroup  $\Gamma \subset \mathbb{C}$ . Writing  $q(z) = B(z, z)$ , the affine function  $2B(z, \lambda) + B(\lambda, \lambda)$  has discrete image, so its linear part is zero. Since  $\Lambda$  spans  $L$ ,  $B = 0$ . The two elliptic period–quasi-period vectors span their complex Lie algebra by Legendre’s determinant, so this obstruction applies to them.

Finally, the tensor itself lies outside first-cohomology logarithm theory. The Hodge types of  $T = \text{Sym}^2 H^1(E)(1)$  are  $(1, -1), (0, 0), (-1, 1)$ . Cohomological realizations of 1-motives have types among  $(0, 0), (1, 0), (0, 1), (1, 1)$ ; their even-weight graded pieces, and those of finite extensions of their Tate twists, are Tate. Strictness of the Hodge and weight filtrations excludes every Tate twist of the full  $T$  as a subquotient of such an object. For non-CM  $E$  there is not even a nonzero Hodge morphism between  $T$  and these objects. Indeed

$$\text{Hom}_{\text{HS}}(\mathbb{Q}(0), T) = \text{End}_{\text{HS}}(H^1(E))_0 = (\text{End}(E) \otimes \mathbb{Q})_0 = 0.$$

The middle identification follows by clearing denominators in a Hodge map and interpreting it as a holomorphic homomorphism of complex tori. Duality excludes Tate quotients as well. Any morphism to or from the stated mixed objects would factor through such a Tate piece by strictness, proving the assertion.

Rigidified  $\mathbf{G}_m$ -biextensions of  $E \times E$  are classified by  $\text{Hom}(E, E^\vee)$ : restrict to  $\{x\} \times E$  to obtain a point of  $E^\vee$ ; the biextension law makes this map a homomorphism, and pullback of the Poincaré bundle is inverse to this construction. Under a principal polarization these are endomorphisms of  $E$ . In the non-CM case their rational first Chern tensors are multiples of the polarization and have zero projection to  $T$ . A nonzero symmetric-square Chern tensor would therefore already force CM. It would be algebraic by the Lefschetz  $(1, 1)$  theorem. Absolute Hodge theory starts with such a complete Hodge class; it does not construct the

class from one scalar period value. Logarithms of algebraic points on a biextension require further data, whose algebraicity must be established before a logarithm theorem applies.

#### APPENDIX E. ANALYTIC REFINEMENTS OF THE SCALAR RELATION

We record three restrictions on a possible infinite set of identities: its coefficients cannot remain close to a prescribed polynomial at the cusp, its complete tuples cannot lie on a fixed algebraic curve, and its fibers cannot remain in a fixed isogeny class. A separate differential statement shows why a scalar identity cannot be differentiated as though it were a functional one. None of these restrictions supplies the missing pointwise CM implication.

**E.1. The cusp and a discrete coefficient lattice.** Fix a standard pair  $(s, R)$ , and put

$$\mathcal{F}(C) = F_s(R/C), \quad \mathcal{L}(C) = \frac{\vartheta F_s(R/C)}{F_s(R/C)}, \quad \kappa = \frac{s(1-s)}{2}.$$

Their expansions at infinity have rational coefficients and begin with

$$\mathcal{F}(C) = 1 + O(C^{-1}), \quad \mathcal{L}(C) = \frac{\kappa R}{C} + O(C^{-2}). \quad (\text{E.1})$$

The rationality of the entire Laurent expansion, rather than its first term alone, yields the following separation theorem.

**Proposition E.1** (Polynomial avoidance). *Let  $P \in \mathbb{Q}[X]$  and let  $\mathcal{A} \subset \overline{\mathbb{Q}} \cap \mathbb{R} \setminus \{0\}$  be finite. There are  $c, C_0 > 0$  such that every integer identity with  $\alpha \in \mathcal{A}$  and  $|C| > C_0$  satisfies*

$$|B - P(C)| \geq c|C|.$$

*Consequently, for fixed  $d, H, K > 0$  and  $0 \leq \nu < 1$ , there are only finitely many ordinarily convergent integer identities satisfying*

$$[\mathbb{Q}(\alpha) : \mathbb{Q}] \leq d, \quad H(\alpha) \leq H, \quad |B - P(C)| \leq K|C|^\nu.$$

*Here  $H$  is the absolute multiplicative Weil height; primitivity is unnecessary.*

*Proof.* Take the polynomial part at infinity:

$$P(C)\mathcal{L}(C) = Q(C) + r(C), \quad Q \in \mathbb{Q}[C], \quad r(C) = O(C^{-1}).$$

Choose  $D \geq 1$  with  $DQ \in \mathbb{Z}[C]$ . Dividing the identity by  $\mathcal{F}(C)$  expresses it as

$$\underbrace{A + Q(C)}_{\in D^{-1}\mathbb{Z}} + (B - P(C))\mathcal{L}(C) = \frac{\alpha}{\pi\mathcal{F}(C)} - r(C). \quad (\text{E.2})$$

Transcendence of  $\pi$  separates the limiting right side from the lattice:

$$\delta = \min_{\alpha \in \mathcal{A}} \text{dist}(\alpha/\pi, D^{-1}\mathbb{Z}) > 0.$$

For sufficiently large  $|C|$ , the right side of (E.2) has distance at least  $\delta/2$  from  $D^{-1}\mathbb{Z}$ , whereas  $|\mathcal{L}(C)| \leq 2\kappa R/|C|$ . Thus  $|B - P(C)| \geq \delta|C|/(4\kappa R)$ .

Northcott's theorem states that algebraic numbers of bounded degree and absolute Weil height form a finite set. It therefore reduces the second assertion to the first, which bounds  $C$  and then  $B$ . The nonvanishing of  $\mathcal{F}(C)$  determines  $A$  for each remaining  $(B, C, \alpha)$ . The zero multiplier and the positive endpoint have already been excluded in Section 2.6.  $\square$

**Corollary E.2.** *For a finite nonzero multiplier set, every identity sufficiently near the cusp satisfies  $|B| \gg |C|$ . In particular, bounded degree and height of  $\alpha$ , together with  $|B| \leq K|C|^\nu$  for some  $\nu < 1$ , imply finiteness.*

*Proof.* Take  $P = 0$  in Proposition E.1.  $\square$

Exactness is essential. At level four, for any integer  $C > 4096$ , take  $A = 1$  and round  $(4/\pi - F_{1/2}(64/C))/\vartheta F_{1/2}(64/C)$  to the nearest integer  $B$ . Since  $c_{1/2}(n) \leq 1/8$  for  $n \geq 1$ ,

$$|\pi\mathfrak{S}_4(1, B, C) - 4| \leq \frac{4\pi}{C(1 - 64/C)^2}.$$

These primitive pairs lie on non-CM fibers: the rational invariant  $j = (C - 16)^3/C$  is not integral, as its nonintegral part is  $-4096/C$ . They provide arbitrarily accurate approximations, not exact identities.

**E.2. Rational multipliers of growing height.** The discrete-lattice argument admits a quantitative version. The irrationality-measure bound of Zeilberger–Zudilin [12, Part II, conclusion] is

$$\mu(\pi) \leq 7.103205334137\dots$$

Thus, for  $\mu_* = 7.104$ , there is  $c_* > 0$  such that every reduced rational number  $u/v$ ,  $v > 0$ , satisfies

$$|\pi - u/v| \geq c_* v^{-\mu_*}. \quad (\text{E.3})$$

The strict excess of  $\mu_*$  over the cited bound gives the inequality for all large denominators; the remaining finitely many denominators are absorbed in  $c_*$ . This is the only quantitative transcendence input below.

**Proposition E.3.** *Fix  $P \in \mathbb{Q}[X]$ ,  $K, L > 0$ , and  $\gamma \geq 0$ ,  $0 \leq \nu < 1$  with  $\mu_*\gamma + \nu < 1$ . Only finitely many ordinarily convergent integer identities with  $\alpha \in \mathbb{Q}^\times$  satisfy*

$$H(\alpha) \leq L|C|^\gamma, \quad |B - P(C)| \leq K|C|^\nu.$$

*Neither primitivity nor a uniform bound on  $|\alpha|$  is needed.*

*Proof.* With the notation of (E.2), write

$$\alpha = \pi \mathcal{F}(C)(m/D + \epsilon_C), \quad m \in \mathbb{Z}, \quad \epsilon_C = O(|C|^{\nu-1}).$$

Let  $\alpha = p/q$  be reduced, with  $q > 0$ . If  $m = 0$ , then

$$L^{-1}|C|^{-\gamma} \leq |\alpha| \ll |C|^{\nu-1},$$

which is impossible for large  $|C|$  because  $\gamma + \nu < 1$ . If  $m \neq 0$ , the rational number  $u/v = Dp/(mq)$  in lowest terms obeys

$$|\pi - u/v| \ll |C|^{\nu-1}, \quad v \leq |m|q \ll |p| + q|\epsilon_C| \ll |C|^\gamma.$$

Here the first estimate uses  $|m/D| \geq 1/D$  and  $\mathcal{F}(C) = 1 + O(C^{-1})$ ; the second follows from the exact identity. Inequality (E.3) now gives  $|C|^{-\mu_*\gamma} \ll |C|^{\nu-1}$ , again impossible for large  $|C|$ . Bounded  $C$  bounds  $B$  and the rational multiplier set, and the identity then determines  $A$ .  $\square$

**E.3. Finiteness on a fixed algebraic curve.** The normalization  $F_s(0) = 1$  has an arithmetic consequence beyond polynomial coefficient bounds. We first isolate its analytic input.

**Lemma E.4.** *For  $0 < s < 1$ , the logarithmic derivative  $F'_s/F_s$  is not algebraic over  $\mathbb{C}(z)$ .*

*Proof.* Clausen's identity gives  $F_s(4x(1-x)) = f_s(x)^2$ , where  $f_s(x) = {}_2F_1(s, 1-s; 1; x)$ . Algebraicity of the first logarithmic derivative would imply that of  $f'_s/f_s$ . The zero-balanced Gauss connection formula and its differentiated convergent expansion give

$$\frac{f'_s(x)}{f_s(x)} \sim \frac{1}{(1-x)(-\log(1-x))} \quad (x \rightarrow 1^-).$$

This cannot be the leading Puiseux term of a nonzero algebraic function.  $\square$

**Theorem E.5** (Algebraic-curve finiteness). *Let  $s \in \overline{\mathbb{Q}} \cap (0, 1)$  and let  $Z \subset \mathbb{A}_{\overline{\mathbb{Q}}}^4$ , with coordinates  $(z, A, B, \alpha)$ , be an irreducible curve. Suppose  $z|_Z$  is nonconstant and  $Z \neq \{A = B = \alpha = 0\}$ . Then only finitely many real points of  $Z$ , with  $z \in [-1, 1]$ , satisfy an ordinarily convergent identity*

$$\pi(AF_s(z) + B\vartheta F_s(z)) = \alpha. \quad (\text{E.4})$$

*The individual points need not be algebraic, and their coefficients need not be bounded.*

*Proof.* Let  $X$  be the smooth projective normalization of the closure of  $Z$ . Its coordinate functions are meromorphic, and the nonconstant map  $z : X \rightarrow \mathbb{P}^1$  is finite. An infinite set of identity points would therefore accumulate at a point over  $[-1, 1]$ .

Below  $z = 1$ , the selected branch  $F_s$  is holomorphic, so the identity theorem, after clearing meromorphic poles on  $X$ , turns this accumulation into a local functional identity. The same holds above  $z = 1$  after a finite local cover: the Gauss function  ${}_2F_1(s/2, (1-s)/2; 1; z)$  has exponents  $0, 1/2$  there, hence is holomorphic in  $\sqrt{1-z}$ , and  $F_s$  is its square. The finite fiber above one cannot alone supply infinitely many points.

Continue the identity inside  $\mathbb{C} \setminus [1, \infty)$  towards zero, avoiding the finitely many poles and branch values of the algebraic coordinates. In a local root  $t = z^{1/e}$ , the resulting branches have convergent Laurent–Puiseux expansions over  $\overline{\mathbb{Q}}$ . Thus

$$\pi(a(t)F_s(t^e) + b(t)\vartheta F_s(t^e)) = c(t), \quad a, b, c \in \overline{\mathbb{Q}}((t)).$$

The Taylor coefficients of  $F_s$  are algebraic. Coefficient comparison and  $\pi \notin \overline{\mathbb{Q}}$  therefore give  $c = 0$  and  $aF_s + b\vartheta F_s = 0$ . If  $b \neq 0$ , this makes the logarithmic derivative algebraic, contrary to Lemma E.4; if  $b = 0$ , then  $a = 0$  because  $F_s(0) = 1$ . All three coordinate functions  $A, B, \alpha$  vanish on the irreducible curve, a contradiction.  $\square$

For integer denominators, an infinite set has  $|C| \rightarrow \infty$  along a subsequence, and the proof reduces directly to coefficient comparison at  $z = R/C = 0$ . The global form also controls accumulation at arbitrary real parameters and at the convergence boundary.

The condition of lying on one algebraic curve is substantial. The set

$$\{(1/n, 2^n, 3^n, 5^n) : n \geq 1\}$$

is Zariski dense in  $\mathbb{A}^4$ , despite reciprocal-integer first coordinates, primitive middle pairs, and rational last coordinates. Indeed, a polynomial relation would yield  $\sum p_{ijk}(n)(2^i 3^j 5^k)^n = 0$  after clearing powers of  $n$ . Distinct triples give distinct positive bases; division by the largest base contradicts the growth of a nonzero polynomial. These points are not claimed to be identities. They show why arithmetic restrictions alone do not provide the curve required by the theorem.

**E.4. The differential of the incidence relation.** Write  $F = F_s$ ,  $G = \vartheta F_s$  and  $K = \overline{\mathbb{Q}}$ , with  $s$  standard. Clausen's identity and the Gauss equation give the rational system

$$F' = G/z, \quad G' = F Q_s(z, G/F), \quad Q_s(z, T) = \frac{T^2}{2z} + \frac{T + s(1-s)}{2(1-z)}. \quad (\text{E.5})$$

For algebraic coefficient germs  $A, B$ , define the polynomial

$$\mathcal{D}_{A,B}(T) = A' + (A/z + B')T + BQ_s(z, T). \quad (\text{E.6})$$

Then  $(AF + BG)' = F\mathcal{D}_{A,B}(G/F)$ . This is the differential of the scalar incidence relation in the projective coordinate  $G/F$ . Its degree makes the contact calculation transparent.

**Proposition E.6.** *Let  $z_0 \in [-1, 1) \setminus \{0\}$  be algebraic and let  $A, B \in K$  be constant and not both zero. Then*

$$(AF + BG)'(z_0) \neq 0.$$

*In particular, every zero at  $z_0$  of  $\pi(AF + BG) - \alpha$ , for constant  $\alpha \in K$ , is simple.*

*Proof.* By Theorem 2.5,  $F(z_0) \neq 0$  and  $r = G(z_0)/F(z_0)$  is transcendental over  $K$ . The polynomial  $\mathcal{D}_{A,B}$  at  $z_0$  has degree two with leading coefficient  $B/(2z_0)$  if  $B \neq 0$ , and equals  $(A/z_0)T$  otherwise. It is nonzero, so it cannot vanish at  $r$ .  $\square$

**Proposition E.7** (Contact with algebraic coefficient germs). *Let  $z_0$  be as above and let  $A, B, \alpha$  be algebraic function germs holomorphic there. Suppose*

$$H = \pi(AF + BG) - \alpha, \quad H(z_0) = 0, \quad \alpha(z_0) \neq 0.$$

*Then  $H'(z_0)$  is algebraic if and only if*

$$B(z_0) = 0, \quad B'(z_0) = -A(z_0)/z_0.$$

*In that case  $H'(z_0) = \alpha(z_0)A'(z_0)/A(z_0) - \alpha'(z_0)$ . Consequently  $H'(z_0) = 0$  holds precisely when*

$$B(z_0) = 0, \quad B'(z_0) = -A(z_0)/z_0, \quad A'(z_0) = \frac{\alpha'(z_0)}{\alpha(z_0)}A(z_0). \quad (\text{E.7})$$

*In all other cases,  $q, \pi, H'(z_0)$  are algebraically independent, where  $q$  is the nome of Theorem 2.5.*

*Proof.* Evaluate all coefficients at  $z_0$  and put  $r = G(z_0)/F(z_0)$ . The identity gives

$$K(q, F(z_0), G(z_0)) = K(q, \pi, r), \quad H'(z_0) = \alpha \frac{\mathcal{D}_{A,B}(r)}{A + Br} - \alpha'.$$

The left field has transcendence degree three by Theorem 2.5. If  $B \neq 0$ , the displayed rational function of  $r$  is nonconstant, since its numerator has degree two and its denominator degree one. If  $B = 0$ , then  $A \neq 0$ , and it is affine in  $r$ , constant exactly when  $B' = -A/z_0$ . In the constant case it has the asserted algebraic value. Otherwise  $r$  is algebraic over  $K(H'(z_0))$ , so  $K(q, \pi, H'(z_0))$  still has transcendence degree three. This proves every assertion.  $\square$

**Corollary E.8.** *At an identity with constant algebraic coefficients,  $q, \pi, (\pi(AF + BG) - \alpha)'(z_0)$  are algebraically independent.*

*Proof.* The exceptional conditions in (E.7) would force  $B = A = 0$  for constant coefficients.  $\square$

At  $z_0 = -1$  these are derivatives of the analytic branch through an ordinary point. Abel's theorem identifies the values with the ordinary sums; no termwise differentiation of the boundary series is used. The variable-coefficient exception is necessary: a hypothetical constant identity with  $B = 0$  would acquire second-order contact on replacing  $B$  by  $-A(z - z_0)/z_0$ . No existence of such an identity is asserted. The fixed-coefficient result applies equally to CM and hypothetical non-CM fibers. Differentiating a scalar equality therefore imposes an additional equation which is, in this setting, false.

### E.5. Finiteness within one isogeny class.

**Theorem E.9.** *Let  $E/K$  be a non-CM elliptic curve over a number field and set*

$$I_E = [\mathrm{GL}_2(\widehat{\mathbb{Z}}) : \rho_E(G_K)] < \infty.$$

*For  $E'$  geometrically isogenous to  $E$ , let  $N(E, E')$  be the least positive isogeny degree from  $E$  to  $E'$ . Then*

$$[K(j(E')) : K] \geq \frac{\psi(N(E, E'))}{I_E}, \quad \psi(N) = N \prod_{p|N} (1 + 1/p).$$

*In particular, the isogeny class contains only finitely many invariants of absolute degree at most  $d$ , all with  $N(E, E') \leq I_E d$ .*

*Proof.* Serre's open image theorem [18] states that the adelic Galois image of a non-CM elliptic curve over a number field has finite index, which is precisely  $I_E < \infty$ . A least-degree isogeny has cyclic kernel: otherwise its kernel contains  $E[m]$  for some  $m > 1$ , and factoring through  $[m]$  lowers the degree.

Let  $\mathrm{Cyc}_N(E)$  be the set of cyclic order- $N$  subgroups of  $E[N]$ . The quotient construction is a  $G_K$ -equivariant injection

$$\mathrm{Cyc}_N(E) \longrightarrow \overline{K}, \quad H \longmapsto j(E/H).$$

Indeed, isomorphic quotients would give two degree- $N$  maps to one target. Since  $\mathrm{Hom}(E, E') \otimes \mathbb{Q}$  is one-dimensional, they differ by a rational scalar of square one, and hence have the same kernel. Equivariance and injectivity identify the two Galois orbits:

$$|G_K \cdot H| = [K(j(E/H)) : K].$$

Thus no descent of an individual isogeny to the  $j$ -field is needed.

The full group  $\mathrm{GL}_2(\mathbb{Z}/N\mathbb{Z})$  acts transitively on  $\mathrm{Cyc}_N(E)$ , which has cardinality  $\psi(N)$ . Its Galois subgroup has index at most  $I_E$ , so every orbit has cardinality at least  $\psi(N)/I_E$ . Apply this to the cyclic kernel of a least-degree isogeny. Finally,

$$N \leq \psi(N) \leq I_E [K(j(E')) : K] \leq I_E [\mathbb{Q}(j(E')) : \mathbb{Q}].$$

Bounded degree therefore bounds  $N$ , and only finitely many kernels remain.  $\square$

**Corollary E.10.** *In any fixed elliptic isogeny class over  $\overline{\mathbb{Q}}$ , each standard family has only finitely many rational nonzero parameters  $C$  with smooth fiber in that class.*

*Proof.* For rational  $z = R_\ell/C$ , the auxiliary parameter  $x = (1 - \sqrt{1 - z})/2$  has degree at most two, and  $j_s(x) \in \mathbb{Q}(x)$ . The theorem treats a non-CM class. In a CM class the quadratic field is fixed, and the order conductor formula gives

$$h(f^2 D_K) \geq h(D_K) \varphi(f)/3 \longrightarrow \infty.$$

Since the degree of a singular modulus equals the order class number, only finitely many  $j$ -invariants of degree at most two occur there as well. The nonconstant rational map  $x \mapsto j_s(x)$  has finite fibers; so does  $C = R_\ell/[4x(1 - x)]$  on the parameters under consideration.  $\square$

## APPENDIX F. SPECIALIZATION BEYOND A FIRST-ORDER WEIGHT

The main separation theorem exploits cancellation of the common Tate period at two parameters. A complementary argument works at one parameter by taking a field norm of the period relation. Its value is the uniformity in arbitrary polynomial weights: their derivatives remain in one symmetric-square differential module.

We use the following fixed-field relation theorem from the preprint [26, Theorem 4.2(3)]. Its uniformity concerns the degree of a relation, not the heights of its coefficients.

**Theorem F.1** (Baldi–Binyamini–Urbanik, Theorem 4.2(3)). *Let  $L$  be a rank- $n$   $G$ -operator over a number field  $K$  on  $\mathbb{P}^1$ , with singular set  $\Sigma$ . Choose any complex fundamental matrix  $\mathcal{P}$  of its solutions and a branch near  $x = \infty$ . Let  $B_{\mathcal{P}}$  be the  $K$ -Zariski closure of its graph. A polynomial relation at  $x \in \mathbb{P}^1(K)$  is nontrivial if it does not vanish identically on any  $K$ -irreducible component of the fiber  $B_{\mathcal{P},x}$  through  $\mathcal{P}(x)$ .*

*For every fixed integer  $\delta \geq 1$  and every integer  $\nu \geq 1$  there exist an integer*

$$1 \leq D \leq n^2(|\Sigma| + 1)^2 + \nu$$

*and a constant  $\kappa$  such that, for every integer  $t > \kappa$ , at least  $\nu$  of the  $n^2 + \nu$  points*

$$t, t + D, \dots, t + (n^2 + \nu - 1)D$$

*satisfy no nontrivial polynomial relation over  $K$  of total degree at most  $\delta$  among the entries of  $\mathcal{P}$ .*

The theorem puts no bound on the heights of the relation coefficients. The fundamental matrix need not have algebraic initial values. The fixed field  $K$  is essential; below a norm polynomial brings every multiplier of bounded degree into the single field  $K = \mathbb{Q}$ .

**Theorem F.2** (Uniformly sparse denominators). *Fix a standard level  $\ell \in \{1, 2, 3, 4\}$  and an integer  $d \geq 1$ . Let  $\mathcal{E}_{\ell,d}$  consist of the nonzero integers  $C$  for which there exist integers  $A, B$ , not both zero, and an algebraic number  $\alpha$  such that the standard series converges ordinarily,*

$$(A + B\vartheta)F_{s_\ell}(R_\ell/C) = \frac{\alpha}{\pi}, \quad [\mathbb{Q}(\alpha^2) : \mathbb{Q}] \leq d.$$

Then  $\mathcal{E}_{\ell,d}$  has upper Banach density zero:

$$\limsup_{N \rightarrow \infty} \sup_{M \in \mathbb{Z}} \frac{\#(\mathcal{E}_{\ell,d} \cap \{M+1, \dots, M+N\})}{N} = 0.$$

In particular,

$$\#\{C \in \mathcal{E}_{\ell,d} : 0 < |C| \leq X\} = o(X).$$

More precisely, for each sign  $\epsilon \in \{1, -1\}$  and each  $\nu \geq 1$  there exist  $D \leq 64 + \nu$  and  $\kappa$  such that every integer  $t > \kappa$  satisfies

$$\#\{0 \leq j \leq \nu + 3 : \epsilon(t + jD) \in \mathcal{E}_{\ell,d}\} \leq 4.$$

Neither  $D$  nor  $\kappa$  depends on  $A, B$ , the field generated by  $\alpha$ , or the height of  $\alpha$ . They may depend on  $\ell, d, \nu, \epsilon$  and the chosen branch. In particular, the density-zero conclusion holds under the more familiar restriction  $[\mathbb{Q}(\alpha) : \mathbb{Q}] \leq d$ .

*Proof.* Put  $s = s_\ell$ ,  $R = R_\ell$ ,  $a = s/2$ , and  $b = (1 - s)/2$ . Thus  $a + b = 1/2$ , and Clausen's formula gives

$$f(z) = {}_2F_1(a, b; 1; z), \quad F_s(z) = f(z)^2.$$

The function  $f$  is annihilated by

$$L_s = z(1 - z)\partial_z^2 + (1 - \frac{3}{2}z)\partial_z - ab. \quad (\text{F.1})$$

It is a rank-two  $G$ -operator: the rational hypergeometric germ is a  $G$ -function, its minimal equation has rank two by Lemma 6.2, and factors of  $G$ -operators remain  $G$ -operators [26, Theorem 3.5].

Choose a fundamental matrix

$$\mathcal{P}(z) = \begin{pmatrix} p_1(z) & p_2(z) \\ p'_1(z) & p'_2(z) \end{pmatrix}, \quad p_1(z) = \pi f(z),$$

and scale the second solution so that

$$\det \mathcal{P}(z) = \frac{\pi}{z\sqrt{1-z}}. \quad (\text{F.2})$$

This is possible because Abel's identity for (F.1) says that every nonzero Wronskian is a constant multiple of  $1/(z\sqrt{1-z})$ . Here  $\sqrt{1-z}$  is the branch equal to one at zero. The same slit punctured neighborhood of zero may contain both real rays; alternatively the two signs may be treated with separate branch choices.

We first prove that the  $\overline{\mathbb{Q}}$ -Zariski closure of the graph of  $\mathcal{P}$  is the whole  $\mathbb{P}^1 \times \text{GL}_2$ . The connected monodromy is  $\text{SL}_2$  by Lemma 6.2.

For completeness, this last assertion about monodromy does not by itself give the claimed arithmetic graph closure: its determinant must also be treated. Suppose a polynomial over  $\overline{\mathbb{Q}}(z)$  vanishes on  $\mathcal{P}(z)$ . Analytic continuation makes it vanish on every  $\text{SL}_2$ -orbit of  $\mathcal{P}(z)$ , hence on the hypersurface with determinant  $\det \mathcal{P}(z)$ . Localize the matrix coordinate ring at  $X_{11}$  and replace  $X_{22}$  with  $(T + X_{12}X_{21})/X_{11}$ , where  $T = \det X$ . Expanding in  $X_{11}^{\pm 1}, X_{12}, X_{21}$  shows that each resulting coefficient polynomial in  $T$  vanishes at  $T = \pi/(z\sqrt{1-z})$ . This value is transcendental over  $\overline{\mathbb{Q}}(z)$ : otherwise the constant  $\pi$  would be algebraic over that field, hence algebraic over  $\overline{\mathbb{Q}}$ , a contradiction. All coefficient polynomials vanish identically. There was therefore no nonzero polynomial relation on the graph. In particular its  $\mathbb{Q}$ -Zariski closure has the same property, and every nonzero polynomial is nontrivial on its smooth fibers.

Let  $z = R/C$  with  $z \neq 0, 1$ . Writing a matrix variable as  $X = (X_{ij})$ , define homogeneous degree-two and degree-four polynomials

$$U_C(X) = AX_{11}^2 + 2BzX_{11}X_{21}, \quad V_C(X) = z^2(1-z)(\det X)^2.$$

The assumed series identity and (F.2) give

$$U_C(\mathcal{P}(z)) = \alpha\pi, \quad V_C(\mathcal{P}(z)) = \pi^2, \quad U_C(\mathcal{P}(z))^2 = \alpha^2 V_C(\mathcal{P}(z)). \quad (\text{F.3})$$

Let  $L = \mathbb{Q}(\alpha^2)$ , with  $e = [L : \mathbb{Q}] \leq d$ . Descent to the fixed coefficient field is the norm

$$Q_C(X) = \text{Nm}_{L/\mathbb{Q}}(U_C(X)^2 - \alpha^2 V_C(X)). \quad (\text{F.4})$$

This polynomial has rational coefficients, total degree  $4e \leq 4d$ , and vanishes at  $\mathcal{P}(z)$ . It is nonzero. Indeed,  $U_C$  is nonzero because  $(A, B) \neq (0, 0)$  and  $z \neq 0$ . It cannot be proportional to  $\det X$ , whose  $X_{12}X_{21}$  term is nonzero. Consequently  $U_C^2/V_C$  is a nonconstant rational function on  $\mathrm{GL}_2$ , and no nonzero polynomial  $m$  with constant coefficients annihilates it. Thus (F.4) is a nontrivial relation in the exact sense of Theorem F.1.

Use the rational base coordinate  $x = C = R/z$ . The pullback of  $L_s$  remains a  $G$ -operator by [26, Theorem 3.5(4)]; it has rank two and singular set  $\{0, R, \infty\}$ , of cardinality three. One may retain the displayed matrix by choosing the differential-module basis corresponding to  $1, \partial_z$ ; equivalently, changing to  $1, \partial_x$  is an invertible rational row change and does not affect the graph-closure or polynomial-degree argument. Apply Theorem F.1 with  $K = \mathbb{Q}$ ,  $\delta = 4d$ , and  $n = 2$ . This gives the asserted positive-denominator block bound. Replacing the base coordinate by  $x = -C$  gives the negative bound, with singular set  $\{0, -R, \infty\}$ . The excluded points  $C = 0, R$  are finite and have no effect on the conclusion. The conditionally convergent point  $z = -1$  is nonsingular and is covered by analytic continuation and Abel's theorem, or simply omitted as one additional finite exception.

Finally fix  $\nu$  and the resulting  $D$ . In any interval of  $N$  consecutive positive integers, discard its intersection with  $[1, \kappa]$  and partition each residue class modulo  $D$  into blocks of  $\nu + 4$  terms, leaving fewer than  $\nu + 4$  terms per residue class. Every full block contains at most four exceptional terms. Thus the number of exceptional terms is at most

$$\frac{4N}{\nu + 4} + D(\nu + 4) + \max(0, \lceil \kappa \rceil).$$

The error is independent of the position of the interval. Apply the same argument to negative denominators. Splitting any integer interval at zero therefore gives an upper Banach density bound  $4/(\nu + 4)$ . Letting  $\nu$  increase proves the claim.  $\square$

**Proposition F.3** (A degree bound for two nearby identities). *Fix two standard levels, a common sign, and a gap  $D \geq 0$ , excluding the identical-level, zero-gap case. There is a constant  $K > 0$ , depending only on these fixed data, such that any two nonzero identities at denominators  $\epsilon t$  and  $\epsilon(t + D)$ , with  $t$  sufficiently large and respective multiplier-square degrees*

$$d_i = [\mathbb{Q}(\alpha_i^2) : \mathbb{Q}],$$

*must satisfy*

$$\log t < K(d_1 d_2)^4 \log(d_1 d_2 + 1).$$

*In particular, for a suitable  $c > 0$ ,*

$$d_1 d_2 \geq c \left( \frac{\log t}{\log \log t} \right)^{1/4}, \quad \max(d_1, d_2) \geq c \left( \frac{\log t}{\log \log t} \right)^{1/8}.$$

*One may replace  $d_1 d_2$  in the first inequality by  $[\mathbb{Q}(\alpha_1^2/\alpha_2^2) : \mathbb{Q}]$ , adjusting  $K$ .*

*Proof.* For  $Y = F_s(z)$  and  $Z = \vartheta F_s(z)$ , the rank-two Gauss equation gives the rational differential relations

$$\vartheta Y = Z, \quad \vartheta Z = \frac{Z^2}{2Y} + \frac{zZ}{2(1-z)} + \frac{s(1-s)zY}{2(1-z)}.$$

Rational composition preserves rational differential closure. Consequently the differential field generated by the four  $G$ -functions in the cross-level proof has transcendence degree  $\mu' = 4$  over  $\mathbb{Q}(u)$ ; their algebraic independence gives equality rather than just an upper bound.

With this  $\mu'$ , the height condition in [26, Theorem 3.13, (3.5)] is  $h \geq c_1 \delta^4 (\log \delta + 1)$ . Its smallness condition at  $u = 1/t$ , where  $h = \log t$ , is implied by

$$h^{4/5} \geq c'_2 \delta^{4/5} \log h.$$

Both conditions therefore hold whenever  $h \geq K_0 \delta^4 \log(\delta + 1)$ , with a fixed sufficiently large  $K_0$ . For example, after increasing a fixed lower bound on  $h$ , the inequality  $\log h \leq h^{3/10}$  shows that the second condition follows already from  $h \geq K_1 \delta^{8/5}$ ; the stated bound dominates this for every  $\delta \geq 1$ .

The polynomial constructed from the multiplier-square ratio has degree at most  $2d_1 d_2$ . If the displayed sufficient height bound held for  $\delta = 2d_1 d_2$ , no such polynomial could vanish. Its contrapositive, with constants absorbed into  $K$ , proves the first assertion. The two lower bounds follow by elementary inversion; if  $d_1 d_2 > \log t$  they are immediate, while otherwise  $\log(d_1 d_2 + 1) \leq \log(\log t + 1)$ . Using the actual degree of the ratio's minimal polynomial gives the final strengthening.  $\square$

What this does and does not prove. All 36 listed standard identities have  $\alpha^2 \in \mathbb{Q}$ , so the theorem already excludes identities of that multiplier type at a density-one set of denominators, with no bound on  $A$  or  $B$ . The fixed-gap proposition further shows that such identities cannot continue to occur in clusters of bounded diameter at arbitrarily large denominators. These results do not prove that the exceptional denominators are finite, identify a particular denominator as nonexceptional, or force CM at an individual exceptional denominator. The union over all  $d$  cannot be taken inside the density conclusion: a countable union of density-zero sets can have density one. In particular the original unrestricted algebraic multiplier problem remains open in this argument. The cited input is a recent preprint; this deduction is unconditional relative to its stated theorem, without a transcendence conjecture.

The density conclusion also holds for the union over the four standard levels, since this is a finite union. The separate cross-level proof establishes the stronger divergent-gap property for that union. Consequently, if a set of integer denominators of positive upper Banach density supports standard identities with selected algebraic multipliers  $\alpha_C$ , then the degrees  $[\mathbb{Q}(\alpha_C^2) : \mathbb{Q}]$  cannot be uniformly bounded. More precisely, for every fixed  $d$ , the denominators for which some identity has degree at most  $d$  form a set of upper Banach density zero.

**F.1. All polynomial weights and fixed elliptic pullbacks.** The rank-two formulation gives more than a density theorem for linear weights. It controls every rational polynomial weight at once, without bounding its degree or coefficient height. The multiplier must be nonzero; otherwise Picard–Fuchs annihilators give zero identities at every parameter.

**Lemma F.4** (Quadratic-form exclusion). *Let  $L$  be a fixed rank-two  $G$ -operator over  $\mathbb{Q}(t)$ , with fundamental matrix  $X(t)$ , such that its arithmetic graph closure is  $\mathbb{P}^1 \times \mathrm{GL}_2$  and*

$$(\det X(t))^2 = \pi^2 r(t), \quad r \in \mathbb{Q}(t)^\times.$$

*Fix positive integers  $e, d$ . Consider regular reciprocal integer parameters  $t = 1/N$  at which*

$$Q_N(X_{11}(t), X_{21}(t)) = \alpha\pi \tag{F.5}$$

*for a homogeneous quadratic form  $Q_N$  whose coefficients belong to some number field of degree at most  $e$ , and some  $\alpha \in \overline{\mathbb{Q}}^\times$  with  $[\mathbb{Q}(\alpha^2) : \mathbb{Q}] \leq d$ . These positive integers  $N$  have natural density zero. For every  $\nu \geq 1$  there are integers*

$$1 \leq D \leq 4(|\Sigma(L)| + 1)^2 + \nu$$

*and a constant  $\kappa$  such that every block  $N, N + D, \dots, N + (\nu + 3)D$ , with  $N > \kappa$ , contains at least  $\nu$  parameters admitting no identity (F.5). The analogous assertion holds for negative integers, ordered by absolute value.*

*Proof.* Let  $E_N$  contain the coefficients of  $Q_N$  and put  $T_N = E_N \mathbb{Q}(\alpha^2)$ , so  $[T_N : \mathbb{Q}] \leq ed$ . Square (F.5) and take the coefficient-field norm of

$$r(1/N)Q_N(X_{11}, X_{21})^2 - \alpha^2(\det X)^2.$$

This gives a rational polynomial relation of degree at most  $4ed$ . Each conjugate factor is nonzero on  $\mathrm{GL}_2$ : its quadratic-form term depends only on the first column, whereas its nonzero determinant term varies with the second column. The product is therefore nonzero and, by the graph-closure hypothesis, nontrivial in the sense of Theorem F.1. That theorem gives the block assertion after the reciprocal change of coordinate.

Write  $k = \nu + 4$ . Summing the bound of at most four exceptional entries per block over all starting integers up to  $Y$  counts each exceptional integer  $k$  times, apart from  $O(kD)$  boundary terms. The upper density is thus at most  $4/k$ . Letting  $\nu$  increase proves density zero. The same count is uniform on arbitrary intervals, so the upper Banach density is also zero. Replacing  $t$  by  $-t$  proves the negative version.  $\square$

If  $\det X = \pi w(t)$  with  $w \in \mathbb{Q}(t)^\times$ , one may instead norm  $w(1/N)Q_N - \alpha \det X$ . A bound  $[\mathbb{Q}(\alpha) : \mathbb{Q}] \leq d$  then requires relation degree only  $2ed$ .

**Theorem F.5** (Unrestricted polynomial weights). *Under the matrix hypotheses of Lemma F.4, suppose an analytic germ has the representation*

$$H(t) = k(t)X_{11}(t)^2/\pi^2,$$

*where  $k$  lies in a fixed algebraic extension of  $\mathbb{Q}(t)$  of degree at most  $e$ . Assume  $X' = M(t)X$  with  $M \in \mathrm{M}_2(\mathbb{Q}(t))$ . For each fixed  $d$ , the reciprocal integer parameters admitting*

$$P\left(t \frac{d}{dt}\right) H(1/N) = \frac{\alpha}{\pi}, \quad P \in \mathbb{Q}[T], \quad 0 \neq \alpha \in \overline{\mathbb{Q}}, \quad [\mathbb{Q}(\alpha^2) : \mathbb{Q}] \leq d,$$

have density zero and satisfy the block conclusion of Lemma F.4. The polynomial  $P$  may vary without any restriction on its degree or coefficient height. At points of ordinary interior convergence this includes all polynomially weighted series  $\sum_{n \geq 0} h_n P(n) N^{-n}$  for  $H(t) = \sum h_n t^n$ .

*Proof.* Differentiation preserves the three-dimensional space of homogeneous quadratic forms in the first column of  $X$ . Its coefficients remain in the fixed field  $\mathbb{Q}(t, k)$ , since differentiation of an algebraic function stays in the same function field. Induction gives

$$\pi^2 \left( t \frac{d}{dt} \right)^j H = Q_j(t; X_{11}, X_{21}), \quad j \geq 0,$$

with  $Q_j$  homogeneous quadratic. Outside a fixed finite pole and ramification locus, all coefficients specialize into one field of degree at most  $e$ . Their linear combination for any  $P$  therefore satisfies Lemma F.4. The argument does not bound  $j$ , and the excluded locus is independent of  $j$ .  $\square$

For the four standard families this assertion can be seen directly. Put  $f = {}_2F_1(s/2, (1-s)/2; 1; z)$ ,  $v = \vartheta f$ , and

$$c = \frac{s(1-s)z}{4(1-z)}, \quad b_0 = \frac{z}{2(1-z)}.$$

The Gauss equation gives  $\vartheta v = cf + b_0 v$ , hence

$$\vartheta(f^2) = 2fv, \quad \vartheta(fv) = cf^2 + b_0 fv + v^2, \quad \vartheta(v^2) = 2cfv + 2b_0 v^2.$$

Every polynomial weight reduces to a rational quadratic form in  $(f, v)$ . Thus Theorem F.2 holds simultaneously for all rational polynomial weights with a nonzero multiplier of bounded square-degree.

*Remark F.6* (A necessary nonvanishing hypothesis). For any rational  $z_0$  with  $0 < |z_0| < 1$ , the nonzero polynomial

$$P_{z_0}(T) = T^3 - z_0(T + \tfrac{1}{2})(T + s)(T + 1 - s)$$

has an absolutely convergent weighted series of value zero. This is the hypergeometric differential equation

$$[\vartheta^3 - z(\vartheta + \tfrac{1}{2})(\vartheta + s)(\vartheta + 1 - s)]F_s = 0$$

specialized at  $z_0$ . Clearing denominators and dividing their gcd gives a primitive integral polynomial weight. Allowing  $\alpha = 0$  in Theorem F.5 would therefore make every rational parameter exceptional. This does not affect the original first-order conjecture.

**Corollary F.7** (Gauges and rational pullbacks). *Let  $\phi \in \mathbb{Q}(t)$  be nonconstant of degree  $r$  and let  $g$  be a fixed algebraic function of degree at most  $e$  over  $\mathbb{Q}(t)$ . For a specified analytic germ*

$$H(t) = g(t)F_s(\phi(t)), \quad s \in \{\tfrac{1}{6}, \tfrac{1}{4}, \tfrac{1}{3}, \tfrac{1}{2}\},$$

*Theorem F.5 holds, with relation-degree bound  $4ed$  and block step*

$$D \leq 4(3r + 1)^2 + \nu.$$

*One omits the finite set of undefined matrix specializations. In particular rational Euler gauges leave the underlying rank and singularity bound unchanged.*

*Proof.* A rational pullback is a  $G$ -operator by [26, Theorem 3.5(4)]. After deleting branch values it induces a finite-index subgroup on fundamental groups, so the connected algebraic monodromy remains  $\mathrm{SL}_2$ . The determinant in the derivative frame is the old determinant composed with  $\phi$ , multiplied by  $\phi'$ . Its square is still  $\pi^2$  times a nonzero rational function. The monodromy and transcendental-determinant argument in the proof of Theorem F.2 therefore gives the full arithmetic graph closure.

The non-apparent singularities lie above  $\{0, 1, \infty\}$  and number at most  $3r$ . A critical point above an ordinary point has two holomorphic pulled-back solutions and is apparent in the scalar frame, so does not enlarge the set of Definition 3.7 of the cited source. The algebraic gauge changes the quadratic coefficients, not the rank-two differential module. Apply Theorem F.5.  $\square$

The safe bounds for the Euler, quadratic-companion, Domb, and level-nine transformations are respectively  $64 + \nu$ ,  $196 + \nu$ ,  $400 + \nu$ , and  $1444 + \nu$ . The level-nine gauge has degree two, giving relation-degree bound  $8d$ ; its rational parameter map has degree six. These are densities of integer parameters. They make no claim about the prime density of raw weighted Euler truncations at one fixed parameter.

## F.2. A general elliptic-family version.

**Proposition F.8.** *Let  $E \rightarrow U$  be a fixed nonisotrivial elliptic family over  $\mathbb{Q}$ , where  $U$  is a nonempty open subset of  $\mathbb{P}^1$ . Choose an algebraic de Rham frame over  $\mathbb{Q}(t)$  and one locally flat nonzero integral cycle, whose holomorphic period is  $\omega(t)$ . Suppose*

$$H(t) = k(t)\omega(t)^2/\pi^2,$$

*with  $k$  algebraic of degree at most  $e$  over  $\mathbb{Q}(t)$ . For each  $d$ , the integers  $N$  admitting a rational polynomial weight and a nonzero multiplier  $\alpha$  of degree at most  $d$  with  $P(t d/dt)H(1/N) = \alpha/\pi$  have density zero. A degree-2ed relation and a fixed rank-two geometric  $G$ -operator suffice.*

*Proof.* Extend the chosen nonzero integral cycle to a rational Betti basis whose intersection pairing is one. This is possible even for a nonprimitive first cycle, by dividing the second cycle by the first cycle's divisibility. Normalize the cup product of the de Rham basis so that its period matrix has determinant  $2\pi i$ , and multiply its second Betti column by  $1/(2i)$ . The resulting fundamental matrix has determinant  $\pi$ , while its first column is unchanged. It is a matrix for the same rank-two differential module; arbitrary complex solution bases are permitted in the cited theorem. A cyclic scalar equation for this Gauss–Manin module is a  $G$ -operator by [26, Theorem 3.10].

Nonisotriviality gives connected monodromy  $\mathrm{SL}_2$ : after a finite level-structure cover, the nonconstant moduli map is a finite map of curves over suitable open subsets, giving finite-index modular monodromy. The graph-closure proof used above now applies with the constant determinant  $\pi$ . More explicitly, a rational polynomial relation on the graph vanishes on each full  $\mathrm{SL}_2$  orbit. At an algebraic regular base point it therefore vanishes on the determinant- $\pi$  hypersurface; the transcendence of  $\pi$  forces that specialization to be the zero polynomial. Infinitely many such base points force the original polynomial to vanish identically. The graph closure is the full product. Use the degree-two norm variant of Lemma F.4 and the same quadratic-form reduction.  $\square$

This proposition applies to any higher-level elliptic period-square realization on a rational base after its normalization is verified. It supplies no realization for general triangle-group or compact Shimura functions. Nor does density zero imply finiteness, identify the exceptional parameters, or control the union over all multiplier degrees. The individual non-CM exclusion remains the unresolved step.

**Corollary F.9** (Divergent gaps for all polynomial weights). *For  $d \geq 1$ , let  $\mathcal{E}_{\ell,d}^{\mathrm{poly}}$  be the integer denominators  $C$  in the interior convergence domain at which*

$$P(\vartheta)F_{s_\ell}(R_\ell/C) = \frac{\alpha}{\pi}, \quad P \in \mathbb{Q}[T], \quad 0 \neq \alpha \in \overline{\mathbb{Q}}, \quad [\mathbb{Q}(\alpha^2) : \mathbb{Q}] \leq d.$$

*For each fixed positive integer  $D$ , there are only finitely many positive  $t$  for which both  $\epsilon t$  and  $\epsilon(t + D)$  lie in  $\mathcal{E}_{\ell,d}^{\mathrm{poly}}$ , for either fixed sign  $\epsilon$ . The same conclusion holds with two different standard levels, also allowing  $D = 0$ . Consequently the successive gaps in the positive part, and in the absolute values of the negative part, of  $\bigcup_{\ell=1}^4 \mathcal{E}_{\ell,d}^{\mathrm{poly}}$  tend to infinity. Neither polynomial degree nor coefficient height is bounded.*

*Proof.* Fix the levels, sign, and gap, excluding the equal-level zero-gap case. Write  $r_i = \epsilon R_{\ell_i}$  and  $f_i(z) = {}_2F_1(s_{\ell_i}/2, (1 - s_{\ell_i})/2; 1; z)$ . The four genuine  $G$ -functions

$$f_1(r_1 u), \quad (\vartheta f_1)(r_1 u), \quad f_2\left(\frac{r_2 u}{1 + Du}\right), \quad (\vartheta f_2)\left(\frac{r_2 u}{1 + Du}\right)$$

are holomorphic at zero and algebraically independent over  $\mathbb{Q}(u)$ . Indeed the product-monodromy proof of Theorem 6.1 acts independently on their two nonzero rank-two solution columns, and each  $\mathrm{SL}_2$  orbit is dense in its affine two-space. Here  $\vartheta$  differentiates the displayed hypergeometric argument.

At  $u = 1/t$ , any two polynomially weighted identities reduce by the quadratic recursions above to

$$Q_1(G_1, G_2) = \alpha_1/\pi, \quad Q_2(G_3, G_4) = \alpha_2/\pi,$$

with rational homogeneous quadratic forms  $Q_i$ . They are nonzero because  $\alpha_i \neq 0$ . Put  $\beta = \alpha_1^2/\alpha_2^2$ , so  $[\mathbb{Q}(\beta) : \mathbb{Q}] \leq d^2$ . Writing  $m_\beta(T) = \sum_{j=0}^e c_j T^j$  for its minimal polynomial, the polynomial

$$\sum_{j=0}^e c_j Q_1(T_1, T_2)^{2j} Q_2(T_3, T_4)^{2(e-j)}$$

has rational coefficients, total degree at most  $4d^2$ , and vanishes at the four specialized  $G$ -functions. It is nonzero: each factor  $Q_1^2 - \sigma(\beta)Q_2^2$  involves nonzero homogeneous forms in disjoint variable sets and  $\sigma(\beta) \neq 0$ . Lemma 6.5 excludes this relation for all sufficiently large  $t$ , uniformly in its coefficients. Taking the maximum of the finitely many thresholds for ordered level pairs and  $0 \leq D \leq H$  gives the asserted divergent-gap statement.  $\square$

## APPENDIX G. FROBENIUS QUOTIENTS AND CANONICAL NORMALIZATION

A first jet governs transport of a complex identity. Arithmetic transport depends on a different object: the quotient of a normalized series by its Frobenius pullback. The quotient is functorial; its raw Taylor truncation need not be. This distinction both explains the finite transformations and gives a precise rigidity theorem for the standard normalization.

**G.1. Lucas congruences as a polynomial Frobenius structure.** For  $h \in 1 + t\mathbb{F}_p[[t]]$ , write

$$\mathcal{Q}_p(h) = \frac{h(t)}{h(t^p)}, \quad [h]_{<p} = \sum_{n < p} [t^n] h t^n.$$

The coefficient congruences  $h_{pa+b} = h_a h_b$  are equivalent to

$$\mathcal{Q}_p(h) = [h]_{<p}. \quad (\text{G.1})$$

Indeed  $h(t^p) = 1 + O(t^p)$ , so this equality says exactly that  $\mathcal{Q}_p(h)$  is a polynomial of degree less than  $p$ . For normalized  $g$  and  $r \in t\mathbb{F}_p[[t]]$ , Frobenius gives the commutative diagram

$$\begin{array}{ccc} h & \xrightarrow{h \mapsto g(h \circ r)} & g(h \circ r) \\ \mathcal{Q}_p \downarrow & & \downarrow \mathcal{Q}_p \\ \mathcal{Q}_p(h) & \xrightarrow[\mathcal{Q}_p \mapsto g^{1-p}(\mathcal{Q}_p(h) \circ r)]{} & g^{1-p}(\mathcal{Q}_p(h) \circ r). \end{array}$$

Thus the obstruction to transporting Lucas congruences is simply failure of the lower-right expression to remain a polynomial of controlled degree.

**Proposition G.1** (The standard Frobenius quotient). *Put  $u_\ell(n) = R_\ell^n c_{s_\ell}(n)$ , as in (1.1), and  $k_\ell = 6, 4, 3, 2$  for  $\ell = 1, 2, 3, 4$ . For every prime  $p$ ,*

$$u_\ell(pa + b) \equiv u_\ell(a)u_\ell(b) \pmod{p} \quad (a \geq 0, 0 \leq b < p).$$

*For  $p > k_\ell$ , the polynomial  $T_{\ell,p}(t) = \sum_{n < p} u_\ell(n)t^n$  has degree  $d_\ell(p) = \lfloor (p-1)/k_\ell \rfloor$ . Consequently*

$$G_\ell(t) = T_{\ell,p}(t)G_\ell(t^p), \quad G_\ell(t) = \sum_{n \geq 0} u_\ell(n)t^n, \quad (\text{G.2})$$

*in  $\mathbb{F}_p[[t]]$ . The same assertions hold for  $F_{s_\ell}(z) = G_\ell(z/R_\ell)$  when  $p \nmid R_\ell$ .*

*Proof.* The factorial ratios have the integral presentations

$$\binom{6n}{3n} \binom{3n}{n, n, n}, \quad \binom{4n}{n, n, n, n}, \quad \binom{3n}{n, n, n} \binom{2n}{n}, \quad \binom{2n}{n}^3.$$

If  $b < p/k_\ell$ , their lowest base- $p$  digits have no carries, so Lucas's multinomial theorem separates the digit  $b$  from the higher digits. If  $b \geq p/k_\ell$ , Legendre's valuation formula makes both sides zero: its periodic summands on  $[0, 1)$  are, respectively,

$$\lfloor 6x \rfloor - \lfloor 3x \rfloor, \quad \lfloor 4x \rfloor, \quad \lfloor 3x \rfloor + \lfloor 2x \rfloor, \quad 3\lfloor 2x \rfloor,$$

all nonnegative and positive on  $[1/k_\ell, 1)$ . This proves Lucas. For  $0 \leq n < p/k_\ell$  all factorial arguments are below  $p$ , so the coefficient is a unit; the remaining coefficients below  $p$  vanish. This proves the degree formula. Digit decomposition proves (G.2), and Fermat's theorem permits the rescaling by  $R_\ell$ .  $\square$

**Proposition G.2** (Rational gauge rigidity). *Suppose  $f \in 1 + t\mathbb{F}_p[[t]]$  has polynomial Frobenius quotient  $T$  with  $1 \leq \deg T < p-1$ . If  $g \in \mathbb{F}_p(t)$  is regular at zero,  $g(0) = 1$ , and  $gf$  is Lucas, then  $g = 1$ . Hence a nonconstant rational gauge over  $\mathbb{Q}$ , normalized at zero, destroys the Lucas property of each standard series at every sufficiently large prime.*

*Proof.* Write  $g = P/Q$  in lowest terms. Polynomiality of

$$\mathcal{Q}_p(gf) = Tg^{1-p} = \frac{TQ^{p-1}}{P^{p-1}} \quad (\text{G.3})$$

forces  $P^{p-1} \mid T$ , hence  $P$  is constant. The degree bound then forces  $\deg Q = 0$ , and normalization gives  $g = 1$ . A fixed nonconstant rational function remains nonconstant and in lowest terms outside finitely many primes, which proves the last assertion.  $\square$

The covariance identity also distinguishes the quotient from its truncation. If  $g(t_0) = 1$  and  $g'(t_0) = 0$  at a regular  $t_0 \in \mathbb{F}_p$ , then the first jets of  $\mathcal{Q}_p(gf)$  and  $T$  agree there. Their raw truncations can disagree: truncation discards the rational Frobenius tail. This is an identity in characteristic  $p$ , not a comparison with a complex period value.

**G.2. Finite transport.** Put  $\delta = t\partial_t$ , and retain the branch and nondegeneracy conditions of (A.2).

**Lemma G.3** (Polynomial transport criterion). *Let  $p > 3$ ,  $g \in 1 + t\mathbb{Z}_p[[t]]$ ,  $r \in t\mathbb{Z}_p[[t]]$ , and  $H = gF_s(r)$ . If the reduction of  $\Lambda_p = g^{1-p}T_p(r)$  is a polynomial of degree less than  $p$ , then*

$$[H]_{<p} \equiv \Lambda_p \pmod{p}. \quad (\text{G.4})$$

Writing  $u = \delta \log g$  and  $v = \delta \log r$ , one also has

$$(A + B\delta)[H]_{<p} \equiv g^{1-p}((A + Bu)T_p(r) + Bv r T_p'(r)) \pmod{p}. \quad (\text{G.5})$$

The weighted equality is evaluated only where its expressions are regular.

*Proof.* The standard Lucas identity and the diagram identify  $\Lambda_p$  with  $\mathcal{Q}_p(H)$ . The degree condition and (G.1) prove the first equality. The second is its derivative, since  $\delta \log(g^{1-p}) = u$  in characteristic  $p$ .  $\square$

**Proposition G.4** (Three finite transformations). *The quadratic, Domb, and level-nine transformations in Appendix H have the following exact reductions for every  $p > 3$ :*

$$[\mathcal{G}]_{<p}(t) \equiv (1 + Kt)^{p-1} T_p\left(\frac{Rt}{(1 + Kt)^2}\right), \quad R = 4K, \quad (\text{G.6})$$

$$[H_{\text{Domb}}]_{<p}(t) \equiv (1 - 4t)^{p-1} T_{p,1/3}\left(\frac{108t^2}{(1 - 4t)^3}\right), \quad (\text{G.7})$$

$$[H_9]_{<p}(t) \equiv D(t)^{(p-1)/2} T_{p,1/6}\left(\frac{1728t^3}{D(t)^3}\right), \quad D(t) = 1 - 6t - 27t^2. \quad (\text{G.8})$$

Here  $H_{\text{Domb}}$  and  $H_9$  denote the respective generating functions in that appendix, and  $T_{p,s} = [F_s]_{<p}$ . All right sides are polynomials of degree less than  $p$ , and their weighted forms use the same logarithmic derivatives as the analytic transport.

*Proof.* The degree of  $T_{p,s}$  is  $\lfloor p/k_\ell \rfloor$ . Thus the  $n$ th terms on the right have the forms

$$t^n(1 + Kt)^{p-1-2n}, \quad t^{2n}(1 - 4t)^{p-1-3n}, \quad t^{3n}D(t)^{(p-1)/2-3n},$$

respectively. Every exponent is nonnegative, and the corresponding degrees are at most  $p - 1$ . The lemma applies to the three defining germ identities.  $\square$

**Theorem G.5** (Arithmetic classification after finite transport). *For the ordinarily convergent quadratic-companion, Domb, and level-nine series, fix an integer denominator  $C \neq 0$  and primitive integer weights  $(A, B)$ . Omitting finitely many bad primes, put*

$$Z = \{p : (A + B\delta)[H]_{<p}(1/C) = 0 \pmod{p}\}.$$

The following conditions are equivalent:

- (1)  $Z$  has lower natural prime density greater than  $3/4$ ;
- (2)  $Z$  contains every sufficiently large prime;
- (3) the data, up to simultaneous sign, are one of the five quadratic-companion, two Domb, or two level-nine CM rows.

*Proof.* At a regular convergent parameter the coefficient map is

$$(A, B) \mapsto (A + Bu, Bv),$$

with rational entries and nonzero determinant. The proposition identifies its standard weighted truncation with the original one, up to a nonzero factor. Their zero sets therefore differ by a finite set. Theorem 5.6 forces CM, the exact CM classification determines the parameter, and Proposition 5.7 selects the complementary CM eigenline. The period characterization of that line and invertible analytic transport identify precisely the listed weights. Those weights give cofinite zeros, proving both directions.

The only convergent parameters omitted by this regular argument are  $C = K$  for quadratic companions and  $C = 16$  for Domb. Convergence forces  $B = 0$ ; finite transport then gives the zero condition  $T_p(1) = U_p(1/2)^2 = 0$ . The elliptic fiber at  $x = 1/2$  is smooth and CM, so Deuring's theorem gives density  $1/2$ . These endpoints contribute no further row.  $\square$

This is an unconditional arithmetic theorem. It does not infer its prime-density hypothesis from a complex identity at an unknown non-CM fiber.

**G.3. Algebraic gauges and the half-density threshold.** Rational rigidity admits one algebraic exception: an inverse square root of a linear polynomial. The degree of the function-field extension explains why a majority of primes already detects it.

**Theorem G.6** (Classification of algebraic Lucas gauges). *Let  $s \in \{1/6, 1/4, 1/3, 1/2\}$  and let  $g \in \mathbb{Q}[[z]]$  be algebraic over  $\mathbb{Q}(z)$  with  $g(0) = 1$ . Let  $\mathcal{P}_g$  be the primes at which  $gF_s$  is integral and Lucas. The following are equivalent:*

- (1)  $\mathcal{P}_g$  has upper natural prime density greater than  $1/2$ ;
- (2)  $\mathcal{P}_g$  contains every sufficiently large prime;
- (3)  $g = (1 - az)^{-1/2}$  for some  $a \in \mathbb{Q}$ , on the branch at zero.

*Under these conditions, either  $g'(0) = 0$ , or  $g(z_0) = 1$  at a nonzero regular point of an analytic continuation, forces  $g = 1$ .*

*Proof.* Let  $d = [\mathbb{Q}(z, g) : \mathbb{Q}(z)]$ . The inclusion  $\mathbb{Q}(z, g) \subset \mathbb{Q}((z))$  makes this function field regular over  $\mathbb{Q}$ : a Laurent series algebraic over  $\mathbb{Q}$  has derivative zero and is a rational constant. A primitive polynomial for  $g$  is therefore absolutely irreducible. Outside finitely many primes its reduction remains absolutely irreducible with the same degrees. For completeness, reducible forms of fixed degree form a closed set: they are the finite union of the proper images of multiplication maps between projective spaces of forms. A polynomial avoiding that set in characteristic zero meets it at only finitely many primes. Eisenstein's algebraic-power-series theorem also makes  $g$  integral outside finitely many primes. Hence

$$[\mathbb{F}_p(z, g_p) : \mathbb{F}_p(z)] = d \quad (\text{G.9})$$

for every sufficiently large  $p$ .

At a Lucas prime, write  $T_p = \mathcal{Q}_p(F_s)$  and  $U_p = \mathcal{Q}_p(gF_s)$ . Covariance gives

$$g_p^{p-1} = T_p/U_p. \quad (\text{G.10})$$

Since  $\mu_{p-1} \subset \mathbb{F}_p$ , the extension generated by  $g_p$  is cyclic of degree dividing  $p-1$ : its automorphisms act by multiplying  $g_p$  by  $(p-1)$ st roots of unity. Thus  $\mathcal{P}_g$  is, up to finitely many exceptions, contained in  $p \equiv 1 \pmod{d}$ . The prime number theorem in arithmetic progressions gives this progression density  $1/\varphi(d)$ . The strict half-density hypothesis forces  $d \leq 2$ .

If  $d = 1$ , rational gauge rigidity applies. If  $d = 2$ , the nontrivial automorphism sends  $g_p$  to  $-g_p$  at every sufficiently large Lucas prime. The rational trace of  $g$  therefore vanishes modulo infinitely many primes and is zero. Consequently  $g^2 = R \in \mathbb{Q}(z)$ . It remains to bound the divisor of  $R$ .

Every root of  $T_p$  has multiplicity at most two. Indeed its degree  $d_p = \lfloor (p-1)/k_\ell \rfloor$  is at most  $(p-1)/2$ , and

$$L_s T_p = 0, \quad L_s = \vartheta^3 - z(\vartheta + 1/2)(\vartheta + s)(\vartheta + 1 - s),$$

because derivatives kill the Frobenius factor of  $F_s$ . At an ordinary point, a root of multiplicity  $r \geq 3$  contradicts the nonvanishing indicial coefficient  $r(r-1)(r-2)$ , since  $r < p$ . There is no root at zero. At  $z = 1$  the indicial expression is  $r(r-1)(r-1/2)$ ; its additional positive residue  $(p+1)/2$  exceeds  $d_p$ , so the multiplicity there is at most one.

Write  $R = P/Q$  in lowest terms, with  $P(0) = Q(0) \neq 0$ . The identity

$$U_p = T_p(Q/P)^{(p-1)/2}$$

forces  $P$  to be constant: otherwise, for large  $p$ , a zero of  $P$  would require a zero of  $T_p$  of multiplicity at least  $(p-1)/2$ . Normalize  $R = 1/Q_0$ ,  $Q_0(0) = 1$ . The remaining divisor is bounded at infinity:

$$\deg U_p = d_p + \frac{p-1}{2} \deg Q_0 \leq p-1.$$

Since  $d_p > 0$ , this forces  $\deg Q_0 \leq 1$ , proving (3).

Conversely,  $g = (1 - az)^{-1/2}$  gives

$$\mathcal{Q}_p(gF_s) = T_p(z)(1 - az)^{(p-1)/2},$$

a polynomial of degree at most  $p-1$  at every sufficiently large prime. This proves (2), hence (1). Finally  $g'(0) = a/2$ , while  $g(z_0) = 1$  implies  $az_0 = 0$ .  $\square$

*Remark G.7* (Sharpness). For  $a \neq 0$ , the cubic gauge  $(1 - az)^{-1/3}$  has Lucas-prime density exactly  $1/2$ . Its degree forces  $p \equiv 1 \pmod{3}$ ; at every sufficiently large prime in that progression its quotient is  $T_p(z)(1 - az)^{(p-1)/3}$ , of degree at most  $5(p-1)/6$ . The theorem therefore has a sharp density threshold. Its extra normalization selects the standard section among algebraic scalar gauges; it supplies no complex-to-Frobenius comparison.

**G.4. Geometric hypotheses in the transport theorem.** At a regular algebraic point, the first-jet transformation is

$$\begin{pmatrix} H \\ \delta H \end{pmatrix} = g \begin{pmatrix} 1 & 0 \\ \delta \log g & \delta \log \phi \end{pmatrix} \begin{pmatrix} F \circ \phi \\ (\vartheta F) \circ \phi \end{pmatrix}, \quad H = gF(\phi). \quad (\text{G.11})$$

Its determinant is  $g^2 t \phi' / \phi$ . The following examples isolate what fails at a zero of this determinant, at a singular fiber, or under a change of coefficient field. They concern transformed families, not the original standard branches.

**Proposition G.8** (A critical non-CM pullback). *For an integer  $m > 3$  with  $m \nmid 64$ , set*

$$H_m(t) = F_{1/2}((2t - t^2)/m) = \sum_{k \geq 0} h_k t^k.$$

*The series and its derivative converge absolutely at  $t = 1$ . The fiber there is smooth and non-CM, yet*

$$\delta H_m(1) = 0, \quad \sum_{k < p} k h_k \equiv 0 \pmod{p^3} \quad (p > 2, p \nmid m).$$

*Proof.* Choose  $r > 1$  with  $2r + r^2 < m$ . The pullback lies in the disk of convergence on  $|t| \leq r$ , and its derivative at 1 is zero. The invariant  $64(4m - 1)^3/m$  is rational and nonintegral; a CM invariant is an algebraic integer, so the fiber is non-CM. For the congruence truncate first in the source coordinate:

$$V_p(t) = \sum_{n < p} \frac{\binom{2n}{n}^3}{64^n} ((2t - t^2)/m)^n.$$

For  $n \leq (p-1)/2$  each term has degree below  $p$ ; for larger  $n < p$  its coefficient is divisible by  $p^3$ . Therefore  $V_p - [H_m]_{<p} \in p^3 \mathbb{Z}_{(p)}[t]$ . As  $V_p'(1) = 0$  exactly, differentiation proves the assertion.  $\square$

The primitive pair is  $(0, 1)$  and the multiplier is zero. Here the proposed de Rham direction itself vanishes; it is not a complement to the Hodge line. Multiplying by  $1 + t$  also gives the complex zero identity with primitive weights  $(-1, 2)$ .

A vanishing gauge destroys uniqueness more completely:  $(1 - t)^2 F_{1/2}(t/5)$  and its derivative both vanish at 1, although its smooth fiber has non-CM invariant  $438976/5$ . Every primitive pair then gives zero. A singular fiber creates a different failure. For  $m > 2$ , the noncritical pullback  $K_m(t) = F_{1/2}(t(1 - t)/m)$  has

$$K_m(1) = 1, \quad \delta K_m(1) = -1/(8m), \quad (1 + 8m\delta)K_m(1) = 0;$$

its image is the nodal parameter zero.

**Proposition G.9** (Rational coefficients need not preserve rational slopes). *The normalized rational-coefficient germ*

$$L(t) = \frac{1 + \sqrt{1 + t/3}}{2} F_{1/2}(t/4)$$

*has an absolutely convergent first jet at  $t = 1$  and a CM fiber there, but no nonzero rational weight pair gives an inverse- $\pi$  identity.*

*Proof.* Both factors are analytic on a disk of radius greater than one. At  $z = 1/4$ , the standard identity is  $(1 + 6\vartheta)F_{1/2} = 4/\pi$  and  $j = 54000$ . Projective uniqueness makes  $1/6$  the only algebraic slope. Since  $\delta \log g(1) = (2 - \sqrt{3})/4$ , jet transport gives the unique slope

$$A/B = -1/3 + \sqrt{3}/4 \notin \mathbb{Q}.$$

For  $B = 1$  the multiplier is  $2g(1)/3 \neq 0$ , so this is an actual algebraic identity line with no rational point.  $\square$

At a singular source value even the chain rule must be interpreted through analytic germs. Clausen gives

$$M(t) = F_{1/2}(t(2 - t)) = {}_2F_1(1/2, 1/2; 1; t/2)^2,$$

whose first jet converges at 1. The Legendre relation at parameter  $1/2$  gives  $\delta M(1) = 2/\pi$ ; the fiber has  $j = 1728$ . Thus the vanishing pullback derivative does not force a zero value when the source derivative is singular. This also shows how a pullback can improve ordinary convergence.

**G.5. Boundary terms and finite-jet obstructions.** Raw truncation does not commute with the Euler gauge. Direct coefficient summation yields

$$\left[ \frac{F_s}{1-t} \right]_{<p} = \frac{T_p(t) - t^p T_p(1)}{1-t} \pmod{p}.$$

The boundary term explains the following failure of a raw arithmetic criterion for a genuine complex identity.

**Proposition G.10** (A CM Euler identity with zero density  $1/2$ ). *For  $H = F_{1/2}/(1-t)$ , the identity  $(-1+6\delta)H(1/4) = 16/(3\pi)$  has raw weighted-truncation zeros of natural prime density exactly  $1/2$ .*

*Proof.* Differentiating the boundary formula in characteristic  $p$  gives

$$(-1+6\delta)[H]_{<p}(1/4) \equiv \frac{4}{3}(1+6\vartheta)T_p(1/4) - \frac{1}{3}T_p(1).$$

The first term vanishes outside finitely many primes by the standard CM criterion. Finite Clausen identifies the second with  $U_p(1/2)^2 = P_{(p-1)/2}(0)^2$ . For a Legendre polynomial,  $P_m(0) = 0$  when  $m$  is odd and  $P_m(0) = (-1)^{m/2}2^{-m}\binom{m}{m/2} \not\equiv 0 \pmod{p}$  when  $m$  is even. Thus the zero primes are, up to finitely many, exactly  $p \equiv 3 \pmod{4}$ .  $\square$

**Proposition G.11** (Finite jets do not determine prime-zero density). *Let  $M \geq 2$ ,  $N \geq 1$ ,  $F = F_{1/2}$ , and*

$$g_{M,N}(z) = 1 + \frac{z^N(4z-1)^M}{3-4z}, \quad H = g_{M,N}F.$$

*Then  $H(192t) \in \mathbb{Z}[[t]]$ , and the minimal differential module of  $H$  is isomorphic over  $\mathbb{Q}(z)$  to the irreducible rank-three module of  $F$ . The difference  $H - F$  vanishes to order  $N$  at zero and order  $M$  at  $1/4$ . Hence the ordinarily absolutely convergent identity  $(1+6\vartheta)H(1/4) = 4/\pi$  holds. Nevertheless the primes at which  $(1+6\vartheta)[H]_{<p}(1/4) = 0$  have natural density zero.*

*Proof.* Put  $P = 3-4z+z^N(4z-1)^M$ . For  $p \geq 2(M+N)+1$ ,  $\deg(PT_p) < p$ , so finite geometric summation gives

$$[H]_{<p}(z) = \frac{P(z)T_p(z) - (4z/3)^p P(3/4)T_p(3/4)}{3-4z} \quad \text{in } \mathbb{F}_p[z]. \quad (\text{G.12})$$

The quotient is polynomial because its numerator vanishes at  $3/4$ ; its germ agrees with  $H$  through degree  $p-1$ . At  $1/4$  the first jets of  $H$  and  $F$  agree. Frobenius kills the derivative of  $(4z/3)^p$ , while  $(1+6\vartheta)(3-4z)^{-1}|_{z=1/4} = 2$ . Thus

$$(1+6\vartheta)[H]_{<p}(1/4) = (1+6\vartheta)T_p(1/4) - 2^{M+1-2N}3^{N-1}T_p(3/4). \quad (\text{G.13})$$

For  $N=1$  the coefficient is  $2^{M-1}$ . The first term is cofinally zero by CM. Finite Clausen identifies  $T_p(3/4)$  with the square of the Hasse polynomial of the Legendre curve at parameter  $1/4$ . Its invariant is  $35152/9$ , so this curve is non-CM. Serre's open-image theorem and Chebotarev, in the form of Theorem 5.5, give supersingular-prime density zero. This proves the density assertion.

The remaining properties express elementary functoriality. Rational multiplication by  $g_{M,N} \neq 0$  is an invertible change of differential module; the numerator gives the two vanishing orders; and all factors are analytic on  $|z| < 3/4$ . Finally  $F(192t)$  has integral coefficients, and the denominator of the added gauge term becomes  $3(1-256t)$ , whose factor 3 divides  $192^N$ . Hence the stated integral rescaling holds.  $\square$

The construction preserves any prescribed finite jets at both the cusp and the identity point. It changes the generating function, while retaining its CM fiber; it is not a counterexample to standard CM forcing.

**Proposition G.12** (Finite jets with Lucas congruences on a progression). *For  $M \geq 2$ ,  $N \geq 1$ , put  $D = M+N$  and choose an integer  $m \geq 2D$ . The branch*

$$g(z) = (1+2^{-M-1}z^N(4z-1)^M)^{-1/m}, \quad g(0) = 1,$$

*is nonrational. For  $H = gF_{1/2}$ , the difference  $H - F_{1/2}$  vanishes to orders at least  $N$  at zero and  $M$  at  $1/4$ ;  $(1+6\vartheta)H(1/4) = 4/\pi$  converges absolutely. Moreover  $H$  is Lucas at every sufficiently large prime  $p \equiv 1 \pmod{m}$ .*

*Proof.* Writing the polynomial inside parentheses as  $P$ , one has  $|P-1| \leq 2^{-1}4^{-N} < 1$  on  $|z| \leq 1/4$ . The normalized root therefore extends beyond that disk and takes value one at  $1/4$ . Its vanishing orders prove the complex assertions. It cannot be rational, since its order at infinity would imply  $m \mid D$ , whereas  $0 < D < m$ . At a good prime  $p \equiv 1 \pmod{m}$ ,

$$\mathcal{Q}_p(H) = T_p P^{(p-1)/m}, \quad \deg \mathcal{Q}_p(H) = \frac{p-1}{2} + D \frac{p-1}{m} \leq p-1.$$

The polynomial quotient criterion proves Lucas.  $\square$

The progression restriction is essential by the half-density theorem. Together these results identify arithmetic normalization as additional data that finite complex jets cannot recover. They leave open the comparison of the standard complex identity with exact Frobenius preservation.

#### APPENDIX H. FUNCTORIALITY AND FURTHER ELLIPTIC BRANCHES

The classification is stable under an algebraic change of variable only after its branch, derivative, and convergence domain have been specified. The following elementary observation keeps these three issues separate. For a germ  $F$  and a pullback  $G(t) = r(t)F(\phi(t))$ , put

$$\delta = t \frac{d}{dt}, \quad u = \delta \log r, \quad v = \delta \log \phi.$$

At a regular point with  $rv \neq 0$ , the first jets satisfy

$$\begin{pmatrix} G \\ \delta G \end{pmatrix} = r \begin{pmatrix} 1 & 0 \\ u & v \end{pmatrix} \begin{pmatrix} F \circ \phi \\ (\vartheta F) \circ \phi \end{pmatrix}. \quad (\text{H.1})$$

Thus pullback induces an isomorphism of coefficient lines,

$$[A : B] \mapsto [A + Bu : Bv].$$

Algebraicity, the period reduction, and projective uniqueness transport through this map. An identity of germs transports analytic values; ordinary convergence of the resulting series is a separate assertion.

**H.1. Quadratic companions.** Put  $K = R_\ell/4$ ,  $\mathcal{F}(X) = F_{s_\ell}(R_\ell X) = \sum u_\ell(n)X^n$ , and define

$$U(n) = \sum_{k=0}^n u_\ell(k) \binom{n+k}{n-k} (-K)^{n-k}, \quad \mathcal{G}(Y) = \sum_{n \geq 0} U(n)Y^n.$$

The binomial generating function gives

$$\mathcal{G}(Y) = \frac{1}{1 + KY} \mathcal{F}\left(\frac{Y}{(1 + KY)^2}\right). \quad (\text{H.2})$$

For  $c \neq 0, -K$  set  $J = (c + K)^2/c$ . Formula (H.1) becomes

$$(A + B\delta)\mathcal{G}(1/c) = \frac{1}{J}((c + K)A - KB + (c - K)B\Theta)\mathcal{F}^{\text{an}}(1/J), \quad \Theta = X\partial_X. \quad (\text{H.3})$$

The continuation is along the real interval from zero. For  $|c| > K$  the coefficient map is invertible. Its image has  $J > 4K$  if  $c > K$  and  $J < 0$  if  $c < -K$ ; in the latter case the standard power series need not converge.

**Proposition H.1** (The five quadratic companions). *Among ordinarily convergent quadratic-companion identities with  $c \in \mathbb{Z} \setminus \{0\}$ ,  $\gcd(A, B) = 1$ , and a CM elliptic fiber, the following are precisely the classes up to simultaneous sign:*

| $\ell$ | $A$ | $B$ | $c$   | $\alpha$        |
|--------|-----|-----|-------|-----------------|
| 2      | 2   | 9   | 512   | 4               |
| 2      | 0   | 1   | -256  | $4\sqrt{3}/9$   |
| 2      | 1   | 9   | -4096 | $64\sqrt{7}/49$ |
| 3      | 1   | 8   | -243  | $3\sqrt{3}$     |
| 4      | 0   | 1   | -32   | 2.              |

Here the value is  $\alpha/\pi$ , and every displayed sum is absolutely convergent.

*Proof.* Clausen's and Pfaff's transformations give

$$\mathcal{G}(Y) = (1 + KY)^{2s-1} {}_2F_1(s, s; 1; -KY)^2, \quad s = s_\ell.$$

The only singularity on  $|KY| = 1$  is  $KY = -1$ . The Gauss connection formula there gives, for  $s < 1/2$ ,

$$\frac{U(n)}{(-K)^n} = \frac{\Gamma(1 - 2s)}{\Gamma(1 - s)^4} n^{-2s} + O(n^{-1-\epsilon_s}), \quad \epsilon_s > 0;$$

for  $s = 1/2$  it gives  $(d \log n + e)/n + O(\log n/n^2)$  with  $d > 0$ . Consequently a nonzero linear weight converges absolutely for  $|c| > K$ , diverges for  $|c| < K$  or  $c = -K$ , and at  $c = K$  converges exactly when  $B = 0$ . Abel's theorem identifies the last value with  $\mathcal{G}(1/K) = F_s(1)/2$ , excluded by the positive-endpoint theorem.

At CM, Proposition 4.1 makes the rational standard denominator  $J$  integral. Since

$$J = c + 2K + \frac{K^2}{c},$$

we have  $c \mid K^2$ . Thus integrality converts the classification into a finite divisor problem. The signed divisors satisfying  $|c| > K$  number 62, 12, 6, 8 at the four levels. Applying the class-polynomial criterion of Section 4 gives exactly

$$(\ell, c, J) = (2, 512, 648), (2, -256, -144), (2, -4096, -3969), (3, -243, -192), (4, -32, -8).$$

Their selected elliptic invariants are 1728, 0, -3375, 0, 1728, respectively. This is the same exact finite test as for the standard tables, applied before a standard convergence filter.

Three evaluations follow from (H.3): the transported weights are respectively 8/9, 64/441, and 9/4 times the standard pairs (1, 7), (8, 65), and (1, 5), at  $J = 648, -3969, -192$ . The two remaining evaluations follow from the endomorphisms of the equianharmonic and square lattices. The period normalization and determinant identity continue along the nonsingular real interval  $z < 1$ , so their use does not require convergence of the transported standard power series. The complete data, in the notation of (4.8), are

|             |             |              |           |          |     |                                      |
|-------------|-------------|--------------|-----------|----------|-----|--------------------------------------|
| $(\ell, c)$ | $(a, b)$    | $(u, v)$     | $\lambda$ | $\kappa$ | $d$ | $\zeta = \frac{-1 + i\sqrt{3}}{2}$ . |
| (2, -256)   | (4/9, 20/9) | (-1/5, -1/5) | $\zeta$   | 0        | 1   |                                      |
| (4, -32)    | (2, 6)      | (-1/3, -2/9) | $i$       | 0        | 1   |                                      |
|             |             |              |           |          |     |                                      |

Indeed the fibers are  $y^2 = 4X^3 - 4$  and  $y^2 = 4X^3 - 9X$ , with endomorphisms  $(X, y) \mapsto (\zeta X, y)$  and  $(X, y) \mapsto (-X, iy)$ . Their standard cycles are the positive primitive real cycles, transported from the primitive vanishing cycle; the oriented lattice bases are  $(\omega, \zeta\omega)$  and  $(\omega, i\omega)$ . Since  $a + bu = 0$ , the determinant formula yields  $4\sqrt{3}/9$  and 2.

Finally, the period-ratio proof of uniqueness extends over the whole real interval  $z < 1$ ,  $z \neq 0$ : here  $x = (1 - \sqrt{1 - z})/2 < 1/2$ , the Euler integral is positive, and the derivative coefficient is nonzero. The invertible transport therefore excludes every additional slope. The continuation used to evaluate two fibers does not alter the ordinary convergence already proved.  $\square$

These five classes are distinct from the 35 linear Euler companions. They agree with [22, Tables 13 and 15]; for the last row the coefficient convention is

$$U_4(n) = (-1)^n \sum_{k=0}^n \binom{2k}{k}^2 \binom{2n-2k}{n-k}^2,$$

which explains the negative denominator. Removing the CM premise requires quadratic-period independence at the additional fibers  $z = R_\ell c / (c + K_\ell)^2$ ,  $c \in \mathbb{Z}$ ,  $|c| > K_\ell$ . The restricted standard hypothesis does not contain that assertion.

**H.2. The level-nine pullback.** Let

$$D(t) = (1 - 9t)(1 + 3t), \quad H(t) = D(t)^{-1/2} F_{1/6} \left( \frac{1728t^3}{D(t)^3} \right) = \sum_{n \geq 0} h_n t^n,$$

where the square root is one at zero. This is [5, Theorem 9.2,  $\ell = 1$ ], with the factor 1728 converting its factorial variable to the present hypergeometric variable.

**Theorem H.2.** *For a nonzero pair  $(a, b)$  and integer  $C \neq 0$ , the weighted series  $\sum h_n(a + bn)C^{-n}$  converges exactly when  $|C| \geq 20$ ; its convergence is then absolute. At each such  $C$  there is at most one algebraic projective slope with value in  $\overline{\mathbb{Q}}/\pi$ , and a nonzero pair cannot have value zero. The complete primitive CM list, with  $b > 0$ , is*

|     |     |     |               |
|-----|-----|-----|---------------|
| $a$ | $b$ | $C$ | $\alpha$      |
| 1   | 4   | 27  | $9\sqrt{3}/2$ |
| 7   | 22  | -27 | $9\sqrt{3}$ . |

*It is exhaustive without a CM premise if quadratic-period independence holds for the selected non-CM fibers at  $z = 1728t^3/D(t)^3$ ,  $t = 1/C$ ,  $|C| \geq 20$ .*

*Proof.* The radius is  $\rho = (9 + 6\sqrt{3})^{-1}$ . In fact  $z(\rho) = 1$ ; the nonnegative coefficient expansions give  $|z(t)| \leq z(|t|) < 1$  for  $|t| < \rho$ , whereas  $H'(t)$  diverges as  $t \uparrow \rho$ . A nonzero polynomial weight preserves the radius, and no integer reciprocal has modulus  $\rho$ . This proves convergence.

The transport (H.1) has

$$r = D^{-1/2}, \quad u = \frac{3t(1 + 9t)}{D}, \quad v = \frac{3(1 + 27t^2)}{D}.$$

For real  $0 < |t| < \rho$  the derivative  $v$  is nonzero and  $z \in (-1, 1) \setminus \{0\}$ . The unconditional period theorem gives uniqueness and nonvanishing, and the stated additional independence hypothesis gives CM.

At CM, integrality of

$$j = \frac{D(t)^3}{t^3} = \left( C - 6 - \frac{27}{C} \right)^3$$

forces its rational cube root to be integral; hence  $C \mid 27$ . Convergence leaves  $C = \pm 27$ , with  $j = 8000, -32768$ . At these two parameters, respectively, the transported coefficient pair and scalar are

$$((a + bu, bv), r) = \left( \frac{3}{5}(3, 28), \frac{3\sqrt{15}}{10} \right), \quad \left( \frac{3}{8}(15, 154), \frac{3\sqrt{6}}{8} \right).$$

The two standard evaluations prove the displayed identities.  $\square$

The extra hypothesis cannot be suppressed: before CM is known,  $j = (C - 6 - 27/C)^3$  is generally rational rather than integral.

**H.3. Domb numbers.** For

$$D_n = \sum_{k=0}^n \binom{n}{k}^2 \binom{2k}{k} \binom{2n-2k}{n-k}, \quad H(t) = \sum_{n \geq 0} D_n t^n,$$

Rogers's transformation [7] is the germ identity

$$H(t) = \frac{1}{1-4t} F_{1/3} \left( \frac{108t^2}{(1-4t)^3} \right). \quad (\text{H.4})$$

Its equivalent coefficient formula is

$$D_n = \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{2k}{k}^2 \binom{3k}{k} \binom{n+k}{3k} 4^{n-2k}.$$

The singular expansion at  $1/16$  gives  $D_n \sim 2 \cdot 16^n / (\pi^{3/2} n^{3/2})$ , with a full expansion in descending powers of  $n$ . Thus a nonzero linear weight has absolute convergence for  $|C| > 16$ , divergence for  $|C| < 16$ , convergence at  $C = 16$  exactly for  $B = 0$ , and conditional convergence at  $C = -16$  when  $B \neq 0$ . The expansion makes  $nD_n/16^n$  eventually decreasing, so the alternating endpoint is an ordinary sum.

Writing  $Y = (C - 4)^3/C$ , transport gives

$$(A + B\delta)H(1/C) = \frac{C}{C-4} \left( A + \frac{4B}{C-4} + \frac{2(C+2)B}{C-4} \vartheta \right) F_{1/3}(108/Y). \quad (\text{H.5})$$

For  $|C| \geq 16$ ,  $C \neq 16$ , this map is invertible and  $0 < 108/Y < 1$ . The period theorem, with Abel's theorem at  $C = -16$ , gives unconditional projective uniqueness. At  $C = 16$ , the only convergent weight is constant and  $H(1/16) = \frac{4}{3} F_{1/3}(1)$ , so the endpoint theorem applies.

**Proposition H.3.** *The primitive Domb identities with integer denominator and a CM fiber are, up to simultaneous sign,*

| $A$ | $B$ | $C$ | $D$ | $\alpha$        |
|-----|-----|-----|-----|-----------------|
| 1   | 3   | -32 | -48 | 2               |
| 1   | 5   | 64  | -60 | $8\sqrt{3}/3$ . |

Here  $D$  is the endomorphism-order discriminant. Both sums converge absolutely.

*Proof.* The two invariants above the level-three parameter  $Y$  are 3-isogenous; hence at CM their trace is an integer. That trace is the monic polynomial  $Y^3 - 126Y^2 + 2944Y$ . Since  $Y \in \mathbb{Q}$ , it follows that  $Y \in \mathbb{Z}$ , and

$$Y = C^2 - 12C + 48 - \frac{64}{C} \implies C \mid 64.$$

The convergence filter leaves  $C \in \{\pm 16, \pm 32, \pm 64\}$ . Their images form the set

$$\{Y\} = \{108, 500, 686, 1458, 3375, 4913\};$$

the exact level-three CM test retains 108, 1458, 3375, of discriminants  $-12, -48, -60$ . The first is the excluded positive endpoint. Transporting the established standard pairs  $(2, 15)$  and  $(4, 33)$  at the other two parameters gives the stated values. Uniqueness excludes any other coefficient pair.  $\square$

The list is unconditional within the CM locus. Exhaustion of all Domb identities would additionally require quadratic-period independence at  $z = 108C/(C - 4)^3$ , for integer  $|C| \geq 16$ ,  $C \neq 16$ . Integrality of the intermediate denominator was a consequence of CM, and cannot be used to remove this extra hypothesis.

## APPENDIX I. MODULAR COORDINATES AND RETURNING CORRESPONDENCES

For  $q = e^{2\pi i\tau}$ , put

$$(k_\ell, d_\ell) = (24, 64), (12, 27), (8, 16), \quad t_\ell = \left( \frac{\eta(\tau)}{\eta(\ell\tau)} \right)^{k_\ell} \quad (\ell = 2, 3, 4).$$

The Hauptmoduls of [5, Table 2] express the denominator as the Fricke norm coordinate

$$C_1 = j, \quad C_\ell = t_\ell + 2d_\ell + d_\ell^2/t_\ell.$$

Indeed Fricke sends  $t_\ell$  to  $d_\ell^2/t_\ell$ . In the normalized McKay–Thompson convention  $T_g = q^{-1} + O(q)$  [14],

$$C_1 = T_{1A} + 744, \quad C_2 = T_{2A} + 104, \quad C_3 = T_{3A} + 42, \quad C_4 = T_{4A} + 24. \quad (\text{I.1})$$

This follows from  $t_\ell = q^{-1} - k_\ell + O(q)$ . At level four the elliptic model has invariant  $j(2\tau) = (C_4 - 16)^3/C_4$ ; the rescaling of  $\tau$  preserves CM.

**Proposition I.1** (Integral coordinates and CM). *Every integer  $C > 64$  with  $C \nmid 4096$  occurs on the positive convergent level-four branch and corresponds to a non-CM point.*

*Proof.* The positive continuous function  $t_4(iy)$ ,  $y \geq 1/2$ , takes the value 16 at  $y = 1/2$  and tends to infinity. It therefore takes  $t = (C - 32 + \sqrt{C(C - 64)})/2 > 16$ , at which  $C_4 = C$ . But

$$j(2\tau) = C^2 - 48C + 768 - 4096/C$$

is rational and nonintegral, whereas a singular modulus is an algebraic integer.  $\square$

For instance,  $C = 65$  gives  $T_{4A} = 41$  and  $j(2\tau) = 117649/65$ . These points satisfy the functional Moonshine identities; no inverse- $\pi$  evaluation at them is asserted.

**Proposition I.2** (A returning correspondence). *Let  $C_\ell$  have invariance group  $\Gamma_\ell$ . If*

$$C_\ell(g\tau) = C_\ell(\tau), \quad g \in \text{GL}_2^+(\mathbb{Q}),$$

*and the projective action of  $g$  does not belong to  $\Gamma_\ell$ , then  $\tau$  is imaginary quadratic.*

*Proof.* A Hauptmodul separates points of its quotient, so  $g\tau = \gamma\tau$  for some  $\gamma \in \Gamma_\ell$ . These groups have rational projective representatives. The nonscalar rational matrix  $\gamma^{-1}g$  fixes a nonreal point; its fixed-point equation is therefore a rational quadratic with negative discriminant.  $\square$

Replication relates Hecke translates; it does not supply the returning equality. Deriving such an equality from the scalar period relation would be a genuine converse theorem.

For precision, the corresponding differential problem has an exact form. Choose the lift  $\tau(z)$  continued from the cusp, with  $z = R_\ell/C_\ell(\tau) \sim R_\ell q$ . Set  $D = q d/dq$  and  $r = \sqrt{1 - R_\ell/C}$ , continued from  $r = 1$ . The modular parameterization [5, Theorem 2.1] gives

$$F_{s_\ell}(R_\ell/C) = -\frac{DC}{Cr}, \quad \frac{\partial F}{F} = 1 + \frac{R_\ell}{2(C - R_\ell)} - \frac{CD^2C}{(DC)^2}.$$

Consequently, at an ordinary point with  $DC \neq 0$ , the original identity is equivalent to

$$-\frac{C'}{2iCr} \left[ A + B \left( 1 + \frac{R_\ell}{2(C - R_\ell)} - \frac{CC''}{(C')^2} \right) \right] = \alpha, \quad (\text{I.2})$$

where primes mean  $d/d\tau$ . The chosen lift is part of this statement. Differentiated replication does not by itself impose an additional relation at the original point.

## APPENDIX J. ARITHMETIC CORRESPONDENCES AND COMPACT SHIMURA CURVES

**J.1. The finite-map principle.** The arithmetic part of the argument belongs to finite morphisms of curves, rather than to any particular hypergeometric equation.

**Proposition J.1** (Arithmetic correspondence criterion). *Let  $k$  be a number field and let a fixed curve carry an elliptic invariant  $j$  and a parameter  $t$  satisfying  $P(t, j) = 0$ , with  $P \in k[T, J]$  of  $J$ -degree at most  $m$ . Exclude a specified finite exceptional set, require  $P(t, J) \neq 0$  at every allowed parameter, and assume each fixed  $j$  has only finitely many allowed parameters. If  $[k(t) : k] \leq d$  and the fiber has CM by the order of discriminant  $D$ , then*

$$h(D) \leq [k : \mathbb{Q}]md.$$

*There are finitely many such CM parameters. If every admitted identity forces CM and has at most one rational coefficient line at each parameter, there are finitely many primitive integer pairs up to sign, apart from the separately specified exceptional contributions.*

*Proof.* CM reciprocity gives  $h(D) = [\mathbb{Q}(j) : \mathbb{Q}]$ . The tower of residue fields gives

$$[\mathbb{Q}(j) : \mathbb{Q}] \leq [k(t, j) : \mathbb{Q}] \leq [k : \mathbb{Q}] [k(t) : k] [k(t, j) : k(t)] \leq [k : \mathbb{Q}] dm.$$

Siegel's lower bound implies that only finitely many imaginary quadratic fields have bounded class number. For an order of conductor  $f$  in a field of discriminant  $d_0$ , the conductor formula gives

$$h(f^2 d_0) \geq \frac{h(d_0)}{3} \varphi(f) \geq \frac{h(d_0)}{3} \sqrt{f/2}.$$

Thus bounded order class number also bounds  $f$ . There are finitely many singular moduli, hence finitely many parameters by the finite-fiber hypothesis. A rational projective line contains exactly one primitive integer pair up to sign.  $\square$

For a modular curve with an honest  $\mathbb{Q}$ -defined  $j$ -map, a rational point has rational  $j$ , independently of the map's degree. A quotient requires the degree of the forgotten data. The diagram

$$\begin{array}{ccc} X_0(N) & \xrightarrow{j} & \mathbf{P}_j^1 \\ \pi \downarrow & & \\ X_0(N)/W & \xrightarrow[t]{} & \mathbf{P}_t^1 \end{array}$$

encodes a correspondence: the norm  $\text{Nm}_{\mathbb{Q}(X_0(N))/\mathbb{Q}(X_0(N)/W)}(J - j)$  is monic of degree at most  $|W|$  in  $J$ . At rational points of the quotient this gives  $h(D) \leq |W|$ ; if only the value of a degree- $\delta$  coordinate  $t$  is rational, the bound is  $|W|\delta$ . A Fricke quotient therefore gives two, whereas larger Atkin-Lehner quotients need not. The maps must be defined over the asserted number field. Integrality of a particular coordinate additionally requires its monic arithmetic equations and is not preserved by every fractional linear change.

## J.2. A class-number-four identity.

**Proposition J.2.** *Let*

$$u_6(n) = \binom{2n}{n} \sum_{k=0}^n \binom{n}{k}^3, \quad \tau_0 = \frac{-3 + \sqrt{-21}}{6}.$$

*Then*

$$\sum_{n \geq 0} (2 + 9n) \frac{u_6(n)}{(-112)^n} = \frac{2\sqrt{7}}{\pi} \quad (\text{J.1})$$

*is an absolutely convergent primitive identity. The elliptic parameter has endomorphism discriminant  $-84$  and class number four; its rational coordinate lies on  $X_0(6)/\langle w_2, w_3 \rangle$ .*

*Proof.* The exact modular inputs in [5, Tables 2, 4, 15 and 55] are

$$X_6 = \frac{w}{(1-w)^2}, \quad w = \left( \frac{\eta(\tau)\eta(6\tau)}{\eta(2\tau)\eta(3\tau)} \right)^{12}, \quad G_6(X) = (1+4X)(1-32X).$$

The associated weight-two form has coefficients  $u_6(n)$ . At  $q = e^{2\pi i \tau_0} = -e^{-2\pi\sqrt{7/12}}$ , the tabulated exact values are  $X_6 = -1/112$  and  $\lambda_6 = 2/9$ . Thus their Theorem 2.4, equation (2.14), yields

$$\sum_{n \geq 0} (2 + 9n) u_6(n) X_6^n = \frac{9}{2\pi \sqrt{G_6(X_6)}} \sqrt{\frac{12}{7}} = \frac{2\sqrt{7}}{\pi},$$

since  $\sqrt{G_6(-1/112)} = 9\sqrt{3}/14$ . The bound  $u_6(n) \leq 4^n (\sum_k \binom{n}{k})^3 = 32^n$  proves absolute convergence, and  $\gcd(2, 9) = 1$ .

The primitive equation  $6\tau_0^2 + 6\tau_0 + 5 = 0$  gives discriminant  $-84$ . Its reduced forms are

$$[1, 0, 21], \quad [2, 2, 11], \quad [3, 0, 7], \quad [5, 4, 5];$$

the bound  $a \leq \sqrt{84/3} < 6$  proves completeness. Hence its order  $\mathcal{O} = \mathbb{Z}[\sqrt{-21}]$  has four classes.

The quotient explains the rational coordinate intrinsically. Put  $s = \sqrt{-21}$  and let

$$\mathbf{p}_2 = (2, 1 + s), \quad \mathbf{p}_3 = (3, s), \quad \mathbf{n} = \mathbf{p}_2 \mathbf{p}_3 = (6, 3 + s).$$

The two ramified ideal classes have order two and are independent: the norm  $a^2 + 21b^2$  represents none of  $2, 3, 6$ . They therefore generate the class group. On  $\mathbb{C}/\mathfrak{a}$ , the unique endomorphism-stable subgroup of order six is  $\mathbf{n}^{-1}\mathfrak{a}/\mathfrak{a}$ ; quotienting by its ramified  $p$ -part realizes  $w_p$  as multiplication by  $[\mathfrak{p}_p^{-1}]$ . Hence all four CM pairs have the same image on the quotient. Their set is Galois stable, so this image is rational. For

the standard subgroup at  $\tau_0$ , stability follows from  $s = 6\tau_0 + 3$ , which preserves  $\langle 1/6 \rangle$  modulo the period lattice.  $\square$

This example makes the quotient-degree bound necessary; it gives no converse for arbitrary level-six identities.

**J.3. Local monodromy and compact triangle systems.** The compact Shimura curve  $X_6^*$  has triangle signature  $(4, 2, 6)$ . Its Clausen identity is

$${}_2F_1(1/24, 5/24; 3/4; z)^2 = {}_3F_2(1/12, 1/4, 5/12; 1/2, 3/4; z).$$

The rank-two equation has finite local projective monodromy of orders 4, 2, 6, and infinite global hyperbolic triangle monodromy. This distinguishes its realization from the elliptic families above.

**Proposition J.3** (The compact monodromy obstruction). *Let  $L$  be a rank-two complex local system on a smooth connected algebraic curve, with finite projective local monodromy at every point of the boundary in its smooth compactification. After any finite algebraic base change, no positive symmetric power of  $L$  is projectively equivalent, on a nonempty algebraic open set, to the same symmetric power of  $R^1 f_* \mathbb{C}$  for a non-isotrivial elliptic family. Rank-one twists and algebraic meromorphic gauges do not alter this conclusion.*

*Proof.* Pass to a common finite cover and compactify. The nonconstant elliptic invariant  $j$  has a pole. After a finite local base change the elliptic family has multiplicative semistable reduction there, with monodromy  $\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$ ,  $n \neq 0$ . Every positive symmetric power is a nonidentity unipotent matrix, hence has infinite projective order in characteristic zero. The pulled-back  $L$  has finite projective local monodromy at that point: it is a power of an original boundary monodromy, or trivial over an interior point. This is a contradiction. Twists disappear projectively, and gauges conjugate monodromy.  $\square$

An isotrivial elliptic family also cannot realize this compact triangle system: its projective monodromy is finite. For comparison, the standard Gauss equation has a repeated exponent zero at the cusp and Wronskian  $cz^{-1}(1-z)^{-1/2}$ ,  $c \neq 0$ , so it has nontrivial unipotent monodromy there. Every nonconstant finite cover retains such a monodromy above the cusp. Thus the compact and standard systems, or their equal positive symmetric powers, cannot become projectively equivalent on a common finite algebraic cover. The obstruction is global; it neither excludes an isolated special-value identity nor a quaternionic abelian-surface realization.

#### J.4. CM degrees without an elliptic invariant.

**Proposition J.4** (Compact CM degree bound). *Let  $D > 1$  be the reduced discriminant of an indefinite quaternion algebra over  $\mathbb{Q}$ , and let  $N$  be square-free and coprime to  $D$ . On the canonical model  $X = X_0(D, N)$ , let  $W$  be an Atkin–Lehner subgroup,  $Y = X/W$ , and let  $t : Y \rightarrow \mathbb{P}^1$  be a nonconstant  $\mathbb{Q}$ -morphism of degree  $\delta$ . If  $P \in X(\overline{\mathbb{Q}})$  is CM by an optimally embedded order  $R$ , and  $Q = \pi(P)$ , then*

$$h(R) \leq |W| \delta [\mathbb{Q}(t(Q)) : \mathbb{Q}]. \quad (\text{J.2})$$

*Bounded parameter degree therefore gives finitely many CM points and parameters on this fixed curve.*

*Proof.* Put  $K = \text{Frac}(R)$  and let  $H_R$  be its ring class field. For the stated canonical models and optimal embeddings, González–Rotger [11, Theorem 5.8(1)] prove

$$H_R = K\mathbb{Q}(P), \quad [H_R : K] = h(R). \quad (\text{J.3})$$

Their reciprocity identifies this Galois action with the invertible ideal-class action, free on each fixed local-orientation branch by their Proposition 5.6. It is a statement about the field of the individual coarse point, including elliptic stabilizers. The field diagram

$$\begin{array}{ccc} \mathbb{Q}(P) & \longleftrightarrow & H_R = K\mathbb{Q}(P) \\ \uparrow & & \uparrow \\ \mathbb{Q}(Q) & & K \\ \uparrow & \nwarrow & \uparrow \\ \mathbb{Q}(t(Q)) & & \mathbb{Q} \end{array}$$

and the two finite maps give

$$h(R) \leq [\mathbb{Q}(P) : \mathbb{Q}] \leq |W| [\mathbb{Q}(Q) : \mathbb{Q}] \leq |W| \delta [\mathbb{Q}(t(Q)) : \mathbb{Q}].$$

There is no extra factor two from adjoining  $K$ . Bounded class number gives finitely many orders, and each order has finitely many optimal embedding classes by the same Proposition 5.6.  $\square$

For  $X_6^* = X_0(6, 1)/W_6$ , one has  $|W_6| = 4$ . The coordinate taking  $0, 1, \infty$  at its unique CM points of discriminants  $-4, -24, -3$  is defined over  $\mathbb{Q}$ : these singleton CM sets are Galois stable. It has degree one, so  $h(R) \leq 4[\mathbb{Q}(t) : \mathbb{Q}]$ . Equality occurs at a rational parameter. Indeed  $R = \mathbb{Z}[\sqrt{-30}]$  has discriminant  $-120$  and reduced forms

$$[1, 0, 30], \quad [2, 0, 15], \quad [3, 0, 10], \quad [5, 0, 6].$$

The ramified classes above 2 and 3 independently generate its four-element class group, since  $a^2 + 30b^2$  represents none of 2, 3, 6. Their ideal actions are the Atkin–Lehner actions by [11, Lemma 5.9]. Here the local-orientation factors are trivial, so these four points form the entire CM set. Its quotient is therefore a Galois-stable singleton. This is the  $-120$  row of [24, Theorem 1].

The optimal-embedding hypothesis concerns the order itself. For example,  $\mathbb{Q}(\sqrt{-3})$  embeds in the discriminant-six algebra, but its order of conductor two does not optimally embed in a maximal order: at a ramified quaternion prime the intersection with an embedded quadratic field is its maximal local order. No assertion for arbitrary levels, orders, or totally real ground fields is implicit here.

**J.5. The period target belongs to the realization.** Suppose a specified rank-two comparison matrix is

$$P = \begin{pmatrix} \omega & \omega_2 \\ \eta & \eta_2 \end{pmatrix}, \quad \det P = c(2\pi i), \quad c \in \overline{\mathbb{Q}}^\times.$$

For a quaternionic abelian surface this must be a specified multiplicity summand after splitting the quaternion action over  $\overline{\mathbb{Q}}$ , together with its algebraic polarization. A differential equation alone does not provide this realization.

**Proposition J.5** (Conditional rank-two mechanism). *Assume that at every non-CM fiber the numerical comparison point is Zariski generic over  $\overline{\mathbb{Q}}$  in its full  $\mathrm{GL}_2$  comparison torsor, with the determinant giving the Tate comparison coordinate. Then*

$$a\omega^2 + b\omega\eta = \alpha\pi, \quad a, b, \alpha \in \overline{\mathbb{Q}}, \quad (a, b) \neq (0, 0),$$

*forces CM.*

*Proof.* Trivialize the torsor over  $\overline{\mathbb{Q}}$ . The left side is a nonzero quadratic function of the first column; the right side is  $\alpha \det(P)/(2ic)$ . These cannot agree identically on  $\mathrm{GL}_2$ : if  $\alpha \neq 0$ , fix the first column and vary the second; if  $\alpha = 0$ , vary the first column. Genericity excludes their equality at the comparison point.  $\square$

The determinant varies on the torsor; it is not a fixed numerical  $2\pi i$  defining an algebraic variety over  $\overline{\mathbb{Q}}$ . Genericity is the extra period-conjecture input, beyond the computation of the Mumford–Tate group or of its tensor invariants.

Now suppose the chosen branch has a verified normalization

$$\mathcal{F}(t) = \frac{r(t)\omega(t)^2}{\Omega_0^2}, \quad \frac{\vartheta \mathcal{F}}{\mathcal{F}} = u(t) + v(t) \frac{\eta(t)}{\omega(t)}, \quad rv \neq 0, \quad (\text{J.4})$$

with algebraic  $r, u, v$  at the parameter. Transport identifies its coefficient space with  $a\omega^2 + b\omega\eta$ , and the natural target is

$$(A + B\vartheta)\mathcal{F}(t) \in \overline{\mathbb{Q}} \frac{\pi}{\Omega_0^2}. \quad (\text{J.5})$$

Conditional CM forcing and the compact degree bound give finitely many bounded-degree parameters on a fixed compact curve. For finiteness of primitive pairs one must also prove slope uniqueness. It follows if  $\eta/\omega$  is transcendental: two independent pairs eliminate  $\pi$  and give an algebraic linear relation between  $\omega$  and  $\eta$ . Chudnovsky supplies this when the CM realization reduces algebraically to elliptic de Rham cohomology, with  $\omega$  a nonzero holomorphic period on a rational homology cycle and  $\eta$  an independent de Rham class evaluated on that same cycle. An arbitrary algebraic combination of CM cycles may instead be an endomorphism eigencycle; the rational-cycle condition must be retained. Singular parameters and vanishing derivative coefficients require separate treatment.

**Proposition J.6** (Weight obstruction). *Assume  $\omega, \eta, \Omega_0$  are actual degree-one periods of abelian varieties over  $\overline{\mathbb{Q}}$ , and assume genericity of their numerical comparison point in the tensor torsor generated by these varieties and the Tate object. Then no value of (J.4) with a linear weight can equal  $\alpha/\pi$  for  $\alpha \in \overline{\mathbb{Q}}^\times$ .*

*Proof.* Such an equality says  $\pi(a\omega^2 + b\omega\eta) = \alpha\Omega_0^2$ . The central weight homotheties multiply degree-one periods by  $\lambda$  and the Tate period by  $\lambda^2$ . The two sides have weights four and two. Genericity would turn the equality into an identity on the torsor, incompatible with these distinct characters and the nonzero right side.  $\square$

For the standard cusp normalization, the denominator is  $\pi^2$ , of weight four, so this obstruction does not apply. For a compact normalization by an actual degree-one period squared, both  $\mathcal{F}$  and its natural target  $\pi/\Omega_0^2$  have weight zero.

This difference is already visible unconditionally in Yang's example. Let

$$d_n = \frac{(1/12)_n(1/4)_n(5/12)_n}{(1/2)_n(3/4)_nn!}, \quad \Omega_{-4} = \sqrt{\pi} \frac{\Gamma(1/4)}{\Gamma(3/4)}.$$

**Proposition J.7** (A compact CM value outside  $\overline{\mathbb{Q}}/\pi$ ). *The primitive, absolutely convergent identity*

$$\sum_{n \geq 0} (9 + 598n) d_n \left( \frac{8748}{15625} \right)^n = 90\sqrt{5} \frac{\pi}{\Omega_{-4}^2} \quad (\text{J.6})$$

has right side outside  $\overline{\mathbb{Q}}/\pi$ .

*Proof.* The  $D = -52$  row of [24, Theorem 1] gives  $(M, N, R_1, R_2, R_3) = (2^2 3^7, 5^6, 64584, 972, 1)$  and multiplier  $4M^{3/4}N^{1/4}\pi/(12^{1/4}\Omega_{-4}^2)$ . Dividing the coefficients and multiplier by 108 gives (J.6). Its parameter is in  $(0, 1)$ . The beta integral identifies the normalization as an actual CM period:

$$\Omega_{-4} = B(1/4, 1/2) = 2 \int_1^\infty \frac{dX}{\sqrt{X^3 - X}}.$$

Chudnovsky's theorem, in [1, Corollary 6], makes this nonzero period of  $y^2 = X^3 - X$  algebraically independent of  $\pi$ . Membership of the right side in  $\overline{\mathbb{Q}}/\pi$  would contradict that independence.  $\square$

**J.6. The exact role of the construction theorems.** The external constructions enter the theory at distinct stages. Chen–Glebov–Goeanka [22, Theorems 4.1–4.2 and 6.1–6.9] identify the elliptic Picard–Fuchs equations and their hypergeometric periods on specified simply connected domains. Their specialization theorems require a quadratic parameter  $\tau = (-b + \sqrt{-d})/(2a)$ , as stated in their equation (22). For such  $\tau$  with  $E_6(\tau) \neq 0$ , their Theorem 8.1 proves

$$\frac{E_4(\tau)}{E_6(\tau)} \left( E_2(\tau) - \frac{3}{\pi \operatorname{Im} \tau} \right) \in \mathbb{Q}(j(\tau)),$$

and supplies an integer clearing the denominator. At  $j = 1728$  this quotient is undefined; their precursor identity (21) is used instead.

The arithmetic mechanism is a torsion trace. On an algebraic model, for the nonreal endomorphism  $\mu = a\tau$  of its period lattice, their equations (28)–(33) give

$$\mu(\mu - \bar{\mu})e_2^*(\Lambda) = \sum_{0 \neq P \in E[(\mu)]} x(P).$$

The trace is algebraic, the nonzero multiplier permits solving for  $e_2^*$ , and conjugation descends the coefficient to  $\mathbb{Q}(j)$ . On a non-CM curve every multiplier is an integer, so this argument loses precisely the factor  $\mu - \bar{\mu}$ . It computes the CM coefficient; it does not prove the converse.

Yang's compact theorem instead supplies the first-solution constant

$$C_1 = \frac{4}{12^{1/4}} \frac{\pi}{\Omega_{-4}^2},$$

and  $C_1^{-1}$  for the second local solution. Its Hecke and algebraic equations evaluate derivative coefficients at specified CM points. The reciprocal constants make the local branch part of the problem, and (J.6) prevents replacing this target by  $\overline{\mathbb{Q}}/\pi$  automatically.

Takeuchi's classification [23] classifies arithmetic triangle groups, rather than the CM points at rational uniformizer values. Similarly, [25, Theorem 2.2] uses three compatible equations on Frobenius solutions  $w_0 = y_0^2$ ,  $w_1 = y_0 y_1$ ,  $w_2 = y_1^2/2$ :

$$aw_0 + b\vartheta w_0 = 1/\pi, \quad aw_1 + b\vartheta w_1 = ir, \quad aw_2 + b\vartheta w_2 = -\pi(\beta + r^2)/2.$$

Here  $\beta$  denotes their auxiliary parameter, distinct from the multiplier of our original identity. The first equation does not imply the other two. Their  $1/\pi^2$  Calabi–Yau construction uses a different system and explicitly conjectural specialization properties after their equation (41). In each case the valid sequence is realization, a separate CM criterion, reciprocity, and CM evaluation; no construction theorem removes the missing scalar CM implication.

## APPENDIX K. TWO CERTIFICATE MECHANISMS

The two constructions below impose arithmetic restrictions through explicit auxiliary identities. Padé approximation separates two rationally related values; rational telescoping classifies identities within two fixed deformation families. Neither construction is known to be necessary for an arbitrary standard identity.

**K.1. An explicit exclusion for two rationally proportional multipliers.** The preceding specialization arguments exclude nearby pairs asymptotically. A direct Padé construction gives an explicit threshold when the ratio of the two multipliers is rational. We use the polynomial construction of David–Hirata–Kohno–Kawashima [29, Proposition 3.6], and prove the determinant and all growth estimates needed here.

**Theorem K.1** (Explicit two-point independence). *For the four standard levels, define*

$$(\Lambda_1, \Lambda_2, \Lambda_3, \Lambda_4) = \left( \frac{46}{5}, \frac{22}{3}, \frac{13}{2}, 6 \right), \quad U = 6 \log 2 + 3(7 \log 7 - 6 \log 6),$$

and put  $T_\ell = U + 6\Lambda_\ell + 6 \log R_\ell$ . Let  $a_1, a_2$  be distinct nonzero integers and  $M = \max(|a_1|, |a_2|)$ . If the integer  $b$  satisfies

$$b > M^7 \exp(T_\ell), \quad (\text{K.1})$$

then the following seven numbers are linearly independent over  $\mathbb{Q}$ :

$$1, \quad F_s(a_i/b), \quad (\theta F_s)(a_i/b), \quad (\theta^2 F_s)(a_i/b) \quad (i = 1, 2), \quad s = s_\ell.$$

In particular, this concerns ordinary absolutely convergent series.

We first establish the required determinant independently.

**Lemma K.2** (The two-point Padé determinant). *Fix  $s \in \{1/6, 1/4, 1/3, 1/2\}$  and put*

$$d_j = \frac{(1/2)_j (s)_j (1-s)_j}{(j!)^3}, \quad c_k = d_{k+1}.$$

Let  $a_1, a_2$  be distinct nonzero rational numbers,  $n \geq 1$ , and  $0 \leq h \leq 6$ . Write

$$G_h(t) = t^h (t - a_1)^{3n} (t - a_2)^{3n} = \sum_k g_{h,k} t^k, \quad P_h(t) = \sum_k \frac{g_{h,k}}{c_k} \binom{k+n}{n-1} t^k.$$

For  $q = 0, 1, 2$ , define the rational linear functional

$$\psi_{i,q}(t^k) = (k+1)^q c_k a_i^{k+1}, \quad Q_{h,i,q}(z) = \psi_{i,q} \left( \frac{P_h(z) - P_h(t)}{z - t} \right).$$

Then the seven columns  $(P_h(z), Q_{h,i,q}(z))$ ,  $0 \leq h \leq 6$ , form an invertible matrix for every  $z$ . Moreover, if  $\mathcal{F}_0 = F_s - 1$ ,  $\mathcal{F}_1 = \theta F_s$ , and  $\mathcal{F}_2 = \theta^2 F_s$ , then

$$P_h(z) \mathcal{F}_q(a_i/z) - Q_{h,i,q}(z) = \sum_{j \geq 0} \frac{\psi_{i,q}(t^{n+j} P_h(t))}{z^{n+j+1}}. \quad (\text{K.2})$$

The equality is formal at infinity and convergent for  $|z| > \max(|a_1|, |a_2|)$ .

*Proof.* Write  $\theta = t \, d/dt$  and

$$A(X) = (X + 3/2)(X + 1 + s)(X + 2 - s), \quad B(X) = (X + 1)^3.$$

The recurrence  $c_{k+1}/c_k = A(k)/B(k+1)$  gives the construction in [29, Proposition 3.6]. We verify the features needed here. Expanding the divided difference gives the right side of (K.2) initially with  $t^j$  and  $z^{j+1}$ , starting at  $j = 0$ . For  $0 \leq j < n$ , the recurrence and  $t^j H(\theta) = H(\theta - j)t^j$  give the exact identity

$$\psi_{i,q}(t^j P_h) = \frac{a_i}{(n-1)!^3} \text{ev}_{a_i}(\theta + 1)^q \prod_{r=1}^j A(\theta - r) \prod_{r=1}^{n-1-j} B(\theta + r) [t^j G_h(t)].$$

The differential operator has order at most

$$q + 3j + 3(n-1-j) \leq 3n-1$$

on  $t^{h+j}(t - a_1)^{3n}(t - a_2)^{3n}$ . It vanishes, proving the formal equality. Since  $c_k$  and  $c_k^{-1}$  have polynomial growth, the stated convergence follows.

Order the lower rows by  $(i, q) = (1, 2), (1, 1), (1, 0), (2, 2), (2, 1), (2, 0)$  and denote the determinant by  $\Delta(z)$ . Replacing these six rows by their remainders does not change the determinant. Each remainder has

order at least  $n + 1$  at infinity, whereas  $\deg P_h = 6n + h$ . Thus  $\Delta(z)$  is constant. Only the leading term in column  $h = 6$  can contribute, so

$$\Delta = L_n \det(\psi_{i,q}(t^n P_h(t)))_{(i,q), 0 \leq h \leq 5}, \quad L_n = c_{6n+6}^{-1} \binom{7n+6}{n-1}^3. \quad (\text{K.3})$$

Put  $H(X) = \prod_{k=1}^n A(X - k)$  and  $V_h = t^{n+h}(t - a_1)^{3n}(t - a_2)^{3n}$ . The same recurrence, coefficient by coefficient, gives

$$\psi_{i,q}(t^n P_h(t)) = \frac{a_i}{(n-1)!^3} \text{ev}_{a_i} H(\theta)(\theta + 1)^{q-3} V_h.$$

Expand  $H(X)$  in powers of  $X + 1$ . Terms with a nonnegative power of  $\theta + 1$  vanish at  $a_i$ , since their differential order is at most  $3n - 1$ . The remaining three rows are a triangular combination, with diagonal coefficient  $H(-1)$ , of the functionals

$$\varphi_{a,r}(t^k) = \frac{a^k}{(k+1)^{r+1}}, \quad r = 0, 1, 2, \quad H(-1) = \prod_{k=1}^n (1/2 - k)(s - k)(1 - s - k) \neq 0.$$

For integers  $u \geq 0$ , define

$$C_{n,u}(a, b) = \det(\varphi_{a_i,r}(t^{u+h}(t - a)^{3n}(t - b)^{3n}))_{(i,r), 0 \leq h \leq 5}, \quad (a_1, a_2) = (a, b).$$

It follows that the moment determinant in (K.3) equals

$$\frac{(a_1 a_2)^3 H(-1)^6}{(n-1)!^{18}} C_{n,n}(a_1, a_2). \quad (\text{K.4})$$

Here is a direct factorization of  $C_{n,u}$ . In the ordered variables  $(X_0, X_1, X_2, Y_0, Y_1, Y_2)$ , let

$$\mathcal{P} = \prod_Z Z^u (Z - a)^{3n} (Z - b)^{3n} \prod_{i < j} (Z_j - Z_i).$$

Expanding the determinant expresses  $C_{n,u}$  as the product of the six functionals  $\varphi_{a,r}^{X_r}, \varphi_{b,r}^{Y_r}$  applied to  $\mathcal{P}$ . Its degree is  $6u + 36n + 15$ . Scaling the three  $X$  variables by  $a$  shows divisibility by  $a^{3u+9n+3}$ , and scaling the  $Y$  variables gives the same power of  $b$ .

We next prove divisibility by  $(b - a)^{18n+9}$ . Away from  $a = 0$ ,

$$\partial_a \varphi_{a,r}(f) = \varphi_{a,r}(\partial_a f) + a^{-1} \varphi_{a,r}(\theta f).$$

After  $v$  derivatives with respect to  $a$ , each term therefore applies the six moment functionals to

$$\prod_{r=0}^2 \theta_{X_r}^{v_r} \partial_a^j \mathcal{P}, \quad j + v_0 + v_1 + v_2 \leq v,$$

up to a scalar regular at any fixed nonzero  $a$ . Expand each power of  $\theta_{X_r}$  in the monic polynomial basis

$$T_{r,k}(\theta) = \begin{cases} (\theta + 1)^k, & k \leq r + 1, \\ (\theta + 1)^{r+1} \theta^{k-r-1}, & k \geq r + 1. \end{cases}$$

Only  $k_r \leq v_r$  occur. On setting  $b = a$ , if some  $k_r < r + 1$ , the composed  $X_r$  functional equals the  $Y_{r-k_r}$  functional. The polynomial  $\partial_a^j \mathcal{P}$  is antisymmetric in these variables: the factor outside its Vandermonde is symmetric, even after parameter differentiation. The two identical functionals consequently give zero.

If all  $k_r \geq r + 1$ , cancellation of the three inverse Euler operators leaves evaluation at  $X_0 = X_1 = X_2 = a$  after an operator of total differential order  $q = \sum_r k_r - 6$ . Before parameter differentiation and on  $b = a$ , the polynomial has total vanishing order at least  $18n + 3$  in  $(X_0 - a, X_1 - a, X_2 - a)$ : the three powers contribute  $18n$  and their Vandermonde contributes 3. The  $j$  parameter derivatives lower that order by at most  $j$ . Thus the evaluation is zero when

$$q + j \leq v - 6 < 18n + 3.$$

All derivatives of order  $v < 18n + 9$  vanish on the diagonal, proving the asserted factor. Comparing degrees gives

$$C_{n,u}(a, b) = \kappa_{n,u}(ab)^{3u+9n+3} (b - a)^{18n+9}. \quad (\text{K.5})$$

It remains to show  $\kappa_{n,u} \neq 0$ . Put

$$J_{n,u} = \det \left( \int_0^1 t^{u+h} (t - 1)^{3n} \frac{(-\log t)^r}{r!} dt \right)_{0 \leq r, h \leq 2}.$$

The determinant expansion and symmetrization express this as  $1/3!$  times an integral over  $(0, 1)^3$  whose integrand is

$$\prod_{j=1}^3 t_j^u (t_j - 1)^{3n} \det(t_j^h)_{h,j} \det\left(\frac{(-\log t_j)^r}{r!}\right)_{r,j}.$$

On  $0 < t_1 < t_2 < t_3 < 1$ , the ordinary Vandermonde is positive and the logarithmic Vandermonde has sign  $(-1)^3$ . The weight also has constant sign. Their product is symmetric and nonzero off the diagonals, so the integral is nonzero. In (K.5), set  $b = 1$ , divide by  $a^{3u+9n+3}$ , and let  $a \rightarrow 0$  after scaling the  $X$  variables. The two three-variable blocks separate, giving

$$\kappa_{n,u} = (-1)^{9n} J_{n,u} J_{n,u+3n+3} \neq 0.$$

Equations (K.3)–(K.5) now give

$$\Delta = \frac{L_n H(-1)^6 \kappa_{n,n}}{(n-1)!^{18}} (a_1 a_2)^{12n+6} (a_2 - a_1)^{18n+9} \neq 0,$$

as required.  $\square$

*Proof of Theorem K.1.* We first bound the common denominators in the preceding construction. Write  $d_j = Q_j/R^j$ , where the four integer sequences  $Q_j$  are

$$\frac{(6j)!}{(3j)!(j!)^3}, \quad \frac{(4j)!}{(j!)^4}, \quad \frac{(3j)!(2j)!}{(j!)^5}, \quad \frac{(2j)!^3}{(j!)^6}.$$

Put  $\mathcal{L}(x) = \text{lcm}(1, \dots, [x])$ . Legendre's factorial formula expresses the valuation at each prime  $p$  as a sum, over  $p^v$ , of the step functions

$$[6u] - [3u], \quad [4u], \quad [3u] + [2u], \quad 3[2u], \quad u = \{j/p^v\}.$$

The thresholds for contributions 1, 2, 3 are respectively

$$(1/6, 1/2, 5/6), \quad (1/4, 1/2, 3/4), \quad (1/3, 1/2, 2/3), \quad (1/2, 1/2, 1/2).$$

For  $j \leq N$ , a contribution of at least  $r$  implies  $p^v \leq N/t_r$ , with the corresponding threshold  $t_r$ . Consequently every  $Q_j$ ,  $1 \leq j \leq N$ , divides the indicated integer in this list:

$$\begin{aligned} &\mathcal{L}(6N)\mathcal{L}(2N)\mathcal{L}(6N/5), \quad \mathcal{L}(4N)\mathcal{L}(2N)\mathcal{L}(4N/3), \\ &\mathcal{L}(3N)\mathcal{L}(2N)\mathcal{L}(3N/2), \quad \mathcal{L}(2N)^3. \end{aligned}$$

Let  $D_N$  and  $E_N$  be the least common denominators of, respectively,  $\{d_j^{-1} : 1 \leq j \leq N\}$  and  $\{d_j : 1 \leq j \leq N\}$ . Since  $E_N \mid R^N$ , the prime number theorem gives

$$\log D_N \leq \Lambda_\ell N + o(N), \quad \log E_N \leq N \log R_\ell. \quad (\text{K.6})$$

Take the polynomials from Lemma K.2 and  $N = 6n + 7$ . The binomial factors and  $a_i$  are integers;  $D_N$  therefore clears the coefficients of every  $P_h$ . Applying  $\psi_{i,q}$  to the divided differences introduces only coefficients  $c_k$  with  $k + 1 \leq N$ , so  $D_N E_N$  clears all the  $Q_{h,i,q}$  as well. Evaluation at integer  $b$  introduces no further denominator. By (K.6),

$$\log(D_N E_N) \leq 6n(\Lambda_\ell + \log R_\ell) + o(n). \quad (\text{K.7})$$

We estimate the remainders directly. The coefficient of  $t^k$  in  $G_h(t)$  has absolute value at most  $2^{6n} M^{6n+h-k}$ . Its diagonal binomial factor is at most  $\binom{7n+6}{n-1}^3$ . Stirling's formula and the polynomial growth of  $c_k, c_k^{-1}$  give, for a fixed integer  $K$ , a bound

$$|\psi_{i,q}(t^{n+j} P_h(t))| \leq e^{U n + o(n)} (n+j+1)^K M^{7n+h+j+1}.$$

Indeed, the exponent  $-k$  from  $G_h$  cancels the  $k$  in  $a_i^{k+n+j+1}$ ; the remaining coefficient dependence is polynomial. As  $b > M$ , summing (K.2) gives, uniformly over the seven choices of  $h$ ,

$$|P_h(b) \mathcal{F}_q(a_i/b) - Q_{h,i,q}(b)| \leq \exp\{n[U + 7 \log M - \log b] + o(n)\}. \quad (\text{K.8})$$

The polynomial in  $j$  is harmless in this fixed-ratio geometric sum.

Suppose a nonzero rational relation exists among  $1, \mathcal{F}_q(a_i/b)$ . Clearing denominators gives integer coefficients  $\lambda_0, \lambda_{i,q}$ . By the nonzero determinant, for every  $n$  at least one  $h$  gives a nonzero rational number

$$L_h = \lambda_0 P_h(b) + \sum_{i,q} \lambda_{i,q} Q_{h,i,q}(b).$$

The assumed relation writes  $L_h$  as minus the corresponding sum of remainders. Since  $D_N E_N L_h$  is a nonzero integer, (K.7) and (K.8) imply

$$1 \leq |D_N E_N L_h| \leq \exp\{n[T_\ell + 7 \log M - \log b] + o(n)\} \longrightarrow 0,$$

contradicting (K.1). The replacement  $\mathcal{F}_0 = F_s - 1$  is an invertible rational change of the value vector and proves the stated result.  $\square$

**Corollary K.3** (An explicit opposite-sign exclusion). *At level  $\ell$ , let  $b$  be a positive integer. If*

$$b \geq 10^{50}, \quad 10^{40}, \quad 10^{35}, \quad 10^{33} \quad \text{for } \ell = 1, 2, 3, 4 \text{ respectively,}$$

*then the two denominators  $C = R_\ell b$  and  $C = -R_\ell b$  cannot both support standard identities with nonzero algebraic multipliers whose ratio is rational. In particular, they cannot both support identities with rational nonzero multipliers.*

*Proof.* Take  $(a_1, a_2) = (1, -1)$ , so  $M = 1$ . The four displayed integers exceed  $\exp(T_\ell)$ . For an exact check, use  $e < 11/4$  and

$$\exp(T_\ell) = \frac{2^6 7^{21} R_\ell^6}{6^{18}} e^{6\Lambda_\ell} < \frac{2^6 7^{21} R_\ell^6}{6^{18}} (11/4)^{\lceil 6\Lambda_\ell \rceil};$$

comparison with each indicated power of ten is an integer inequality. Subtracting a rational multiple of one asserted identity from the other would give a nonzero rational relation among the two copies of  $F_s, \theta F_s$ , contrary to Theorem K.1.  $\square$

This result imposes no degree or height bound on the individual multipliers; it restricts their ratio to  $\mathbb{Q}$ . It excludes a pair of identities and does not exclude an isolated non-CM identity. The seven-number independence proved here is over  $\mathbb{Q}$ , not over  $\overline{\mathbb{Q}}$ . The determinant argument therefore does not supply unconditional CM forcing.

**K.2. Complete converses within two rational-certificate families.** A rational Wilf–Zeilberger certificate supplies a second way to impose arithmetic restrictions without assuming a period conjecture. The restriction is to an explicitly specified deformation of the summand. We do not assume, or prove, that an arbitrary scalar identity admits such a deformation.

We first allow three affine parameter shifts and a varying linear weight. For  $0 < s < 1$ ,  $z \in \mathbb{C} \setminus \{0, 1\}$ , and  $\lambda \in \mathbb{C}^\times$ , put

$$U(n, k) = z^n \frac{(1/2 + k)_n (s)_n (1 - s)_n}{(1 + k)_n (1 + 2k)_n (1)_n}, \quad (\text{K.9})$$

$$H(k) = \lambda^k \frac{\Gamma(k+1)^2 \Gamma(s) \Gamma(1-s)}{\Gamma(k+s) \Gamma(k+1-s)}, \quad T(n, k) = \frac{U(n, k)}{H(k)}. \quad (\text{K.10})$$

Here a determination of  $\lambda^k$  is fixed; only its unit-shift ratio is used in the rational calculation. At  $k = 0$  one has  $H(0) = 1$  and  $U(n, 0) = c_s(n) z^n$ . The level-three specialization of this deformation occurs in Guillera's parameterized WZ identity.<sup>1</sup> For  $P(n, k) = A + Bn + Dk$ , a rational certificate means an arbitrary  $R \in \mathbb{C}(n, k)$  such that

$$\begin{aligned} P(n, k+1)T(n, k+1) - P(n, k)T(n, k) &= R(n+1, k)T(n+1, k) \\ &\quad - R(n, k)T(n, k). \end{aligned} \quad (\text{K.11})$$

The equation is interpreted after division by  $T(n, k)$  as a rational identity. No bound on the numerator or denominator degrees of  $R$  is imposed.

**Theorem K.4** (An affine-shift certificate converse). *Assume  $(A, B) \neq (0, 0)$ . A rational certificate (K.11) exists precisely in the following two cases, up to a common nonzero scalar multiplying  $A, B, D, R$ :*

| $s$        | $z$ | $\lambda$ | $(A, B, D)$     |
|------------|-----|-----------|-----------------|
| 1/3 or 2/3 | 1/2 | 1         | (1/6, 1, 1)     |
| 1/3 or 2/3 | -4  | 1         | (4/15, 1, 2/5). |

*The corresponding certificates, respectively, are unique and equal to*

$$R = \frac{2n(3k + 3n + 2)}{3(2k + n + 1)}, \quad R = \frac{n(12k + 3n + 5)}{15(2k + n + 1)}.$$

*Among the four standard families, ordinary convergence leaves only  $z = 1/2$ ,  $s = 1/3$ ,  $C = 216$ , with primitive coefficient pair  $(1, 6)$ . Its elliptic fiber has CM of discriminant  $-24$ .*

<sup>1</sup>J. Guillera, *On WZ-pairs which prove Ramanujan series*, arXiv:0904.0406v3, equation (37), <https://arxiv.org/abs/0904.0406v3>. That formula is a forward identity; the rational-certificate converse below is proved directly.

*Proof.* Set  $L = 4/\lambda$ . Direct unit shifts give

$$q := \frac{T(n+1, k)}{T(n, k)} = \frac{z(n+k+1/2)(n+s)(n+1-s)}{(n+1)(n+k+1)(n+2k+1)},$$

$$K := \frac{T(n, k+1)}{T(n, k)} = \frac{L(n+k+1/2)(k+s)(k+1-s)}{(n+k+1)(n+2k+1)(n+2k+2)}.$$

Make the invertible rational substitution  $R = nX/(n+2k+1)$ . Clearing denominators yields

$$\begin{aligned} & z(n+k+1/2)(n+s)(n+1-s)X(n+1, k) \\ & - n(n+k+1)(n+2k+2)X(n, k) \\ & = L(n+k+1/2)(k+s)(k+1-s)(A+Bn+D(k+1)) \\ & - (A+Bn+Dk)(n+k+1)(n+2k+1)(n+2k+2). \end{aligned} \quad (\text{K.12})$$

We show that  $X$  is a polynomial in  $n$ . Regard  $k$  as transcendental and work over an algebraic closure of  $\mathbb{C}(k)$ . If  $X$  has a pole, choose the first and last poles in one finite chain under integer translation. Cancellation at the endpoints requires a root of the first coefficient of  $X$  in (K.12), shifted by a positive integer, to equal a root of the second coefficient. Those root sets are

$$\{-k-1/2, -s, s-1\}, \quad \{0, -k-1, -2k-2\}.$$

The constant roots in the first set lie strictly between  $-1$  and  $0$ . The only nonconstant pair having equal coefficients of  $k$  has difference  $-1/2$ . All remaining comparisons have unequal coefficients of the transcendental  $k$ . Thus none gives the required positive integral difference, and  $X$  has no pole.

If  $d = \deg_n X$ , the left side of (K.12) has degree  $d+3$ , with leading coefficient  $(z-1)\text{lc}_n(X)$ ; its right side has degree at most four. Hence  $X = u(k)n + v(k)$ . The coefficients of  $n^4, n^3$  force

$$\begin{aligned} u &= -\frac{B}{z-1}, \\ v &= -\frac{2A(z-1) + 8Bkz - 4Bk + 3Bz - 2B + 2Dk(z-1)}{2(z-1)^2}. \end{aligned} \quad (\text{K.13})$$

Let  $E(n, k)$  be  $(z-1)^2$  times the left side minus the right side of (K.12), after (K.13) is substituted, and write  $E_{ij} = [n^i k^j]E$ . The classification is now a finite coefficient calculation, as follows.

First suppose  $B \neq 0$  and divide by  $B$  so that  $B = 1$ . Two coefficients are

$$E_{04} = -D(L-4)(z-1)^2, \quad E_{03} = -\frac{1}{2}(2A+5D)(L-4)(z-1)^2.$$

If  $L \neq 4$ , they force  $A = D = 0$ . At this specialization  $E_{12} - 3E_{22}/2 = z$ , contradicting  $z \neq 0$ . Thus  $L = 4$  and  $\lambda = 1$ . Next,

$$E_{22} = 2(2z-1)((z-1)D+2).$$

If  $z = 1/2$ , the equation  $E_{21} = 0$  gives  $D = 1$ . With  $w = s^2 - s$ , the coefficients

$$E_{20} = (-2A + 12w + 3)/16, \quad E_{02} = (4A + 3w)/4$$

give  $w = -2/9$  and  $A = 1/6$ , hence  $s = 1/3$  or  $2/3$ .

If  $z \neq 1/2$ , one instead has  $D = -2/(z-1)$ . Then

$$\begin{aligned} E_{21} &= \frac{4A(z-1)(2z-1) - z^2 + 7z - 4}{2}, \\ E_{12} &= 2A(z-1)(2z-1) + 5z - 4. \end{aligned}$$

Their difference is  $(z+4)(z-1)/2$ . Consequently  $z = -4$ , and  $E_{12} = 0$  gives  $A = 4/15$ . Finally,

$$E_{02} = 4(z-1)\{A(z-1) + (s^2 - s)(z-2)\}$$

gives  $s^2 - s = -2/9$  once more.

If  $B = 0$ , then  $A \neq 0$ . The same  $E_{04}, E_{03}$  force  $L = 4$ . In this case

$$\begin{aligned} E_{22} &= 2D(z-1)(2z-1), \\ E_{21} &= \frac{1}{2}\{4A(z-1)(2z-1) + D(z-1)(5z-2)\}. \end{aligned}$$

Their vanishing, with  $z \neq 1$ , implies  $D = 0, z = 1/2$ . But then  $E_{20} = -A/8 \neq 0$ . Thus no additional case occurs.

Substitution in the full polynomial identity proves sufficiency and gives the two displayed certificates. For uniqueness, subtract two certificates. The same pole argument makes their difference correspond to a polynomial  $X$ , whose nonzero leading term could not vanish in the homogeneous version of (K.12) because  $z \neq 1$ .

At  $k = 0$ , the  $z = -4$  survivor has  $B \neq 0$  and its summands have magnitude proportional to  $4^n n^{-1/2}$ ; they do not tend to zero. It cannot define an ordinarily convergent standard identity. The other survivor has  $C = 108/(1/2) = 216$ . Formula (4.3) gives

$$P_3(J, 216) = J^2 - 4834944J + 14670139392 = H_{-24}(J).$$

Both corresponding elliptic invariants are therefore singular moduli of discriminant  $-24$ .  $\square$

For completeness the convergent identity follows directly from the certificate, without analytic continuation in its auxiliary parameter. This also specifies the analytic step that a rational certificate alone does not supply.

**Corollary K.5.** *The ordinary sum at the convergent survivor is*

$$(1 + 6\vartheta)F_{1/3}(1/2) = \frac{3\sqrt{3}}{\pi}.$$

*Proof.* Use  $(A, B, D) = (1, 6, 6)$  and  $s = 1/3, z = 1/2, \lambda = 1$ . For every nonnegative integer  $k$ , the series

$$S(k) = \sum_{n \geq 0} (1 + 6n + 6k) \frac{U(n, k)}{H(k)}$$

converges absolutely. The certificate has zero boundary at  $n = 0$  and at infinity: its rational factor has polynomial growth, whereas the summand ratio tends to  $1/2$ . Telescoping therefore gives  $S(k+1) = S(k)$  on the nonnegative integers.

For integers  $k \geq 1$  and  $0 < s < 1$ ,

$$\frac{(k+1/2)_n}{(k+1)_n} \leq 1, \quad \frac{(s)_n}{n!} \leq 1, \quad \frac{(1-s)_n}{(1+2k)_n} \leq 1.$$

Thus  $0 < U(n, k) \leq 2^{-n}$ ; for fixed  $n \geq 1$  one also has  $U(n, k) \rightarrow 0$  as  $k \rightarrow \infty$ . The terms  $(1 + 6n + 6k)U(n, k)/k$  are dominated by  $(7+6n)2^{-n}$ , a summable bound independent of the integer  $k \geq 1$ . Dominated convergence and the gamma-ratio asymptotic give, respectively,

$$\frac{1}{k} \sum_{n \geq 0} (1 + 6n + 6k)U(n, k) \rightarrow 6,$$

$$\frac{H(k)}{k} \rightarrow \Gamma(s)\Gamma(1-s) = \frac{\pi}{\sin \pi s}.$$

Hence  $S(0) = \lim_{k \rightarrow \infty} S(k) = 6 \sin(\pi/3)/\pi$ . Only integer values of  $k$  have been used; no claim that a one-periodic analytic function must be constant enters the proof.  $\square$

The second converse allows an unrestricted rational dependence of the auxiliary coefficients and of the normalization ratio. It shows that such a change does not enlarge the earlier terminating mechanism.

**Theorem K.6** (Rational auxiliary gauges of the terminating family). *Let  $0 < s < 1$ , let  $(a, b, t)$  be a permutation of  $(1/2, s, 1-s)$ , and let  $z \in \mathbb{C} \setminus \{0, 1\}$ . Set*

$$U(n, k) = z^n \frac{(a)_n (b)_n (-k)_n}{(1)_n^2 (1+t+k)_n}, \quad T(n, k) = U(n, k)/H(k),$$

where  $h(k) = H(k+1)/H(k) \in \mathbb{C}(k)^\times$ . Let  $A, B \in \mathbb{C}(k)$  be not both identically zero. An arbitrary rational  $R(n, k)$  satisfying

$$\begin{aligned} [A(k+1) + nB(k+1)]T(n, k+1) - [A(k) + nB(k)]T(n, k) \\ = R(n+1, k)T(n+1, k) - R(n, k)T(n, k) \end{aligned} \quad (\text{K.14})$$

exists if and only if

$$\begin{aligned} s = a = b = t = 1/2, \quad z = -1, \quad A(k) = B(k)/4, \\ h(k) = \frac{k+3/2}{k+1} \frac{B(k+1)}{B(k)}, \quad R(n, k) = -\frac{B(k)n^2}{2(k+1-n)}, \end{aligned}$$

with  $B(k) \neq 0$  as a rational function. The certificate is unique.

*Proof.* Put  $x = k+1$  and  $L(k) = (k+1)(k+1+t)/h(k)$ . The shift ratios are

$$\begin{aligned} \frac{T(n+1, k)}{T(n, k)} &= \frac{z(n+a)(n+b)(n-k)}{(n+1)^2(n+k+1+t)}, \\ \frac{T(n, k+1)}{T(n, k)} &= \frac{L(k)}{(k+1-n)(n+k+1+t)}. \end{aligned}$$

Substitute  $R = n^2 X / (k + 1 - n)$  and clear denominators to obtain

$$\begin{aligned}
& -z(n+a)(n+b)(k+1-n)X(n+1, k) \\
& -n^2(n+k+1+t)X(n, k) \\
& = L(k)\{A(k+1) + nB(k+1)\} \\
& - \{A(k) + nB(k)\}(k+1-n)(n+k+1+t).
\end{aligned} \tag{K.15}$$

The same pole-endpoint argument applies, now to the root sets  $\{-a, -b, k+1\}$  and  $\{0, -k-1-t\}$ . No root of the first set reaches a root of the second by a positive integer translation:  $0 < a, b < 1$ , and the nonconstant comparisons have incompatible coefficients of  $k$ . Thus  $X$  is polynomial in  $n$ . The leading coefficient on the left is  $(z-1)\text{lc}_n(X)$  and the right side has degree at most three, so  $X$  is constant in  $n$ . If  $B = 0$  identically, comparison of degrees forces  $X = 0$ , and the  $n^2$  coefficient then forces  $A = 0$ , a contradiction. Consequently

$$X = \frac{B(k)}{z-1}, \quad \frac{A(k)}{B(k)} = w(x) := \frac{z(a+b-t) - (z+1)x}{z-1}.$$

The coefficients of  $n$  and  $1$  in (K.15) give

$$L(k) \frac{B(k+1)}{B(k)} = M(x), \quad L(k) \frac{A(k+1)}{B(k)} = N(x),$$

where

$$\begin{aligned}
M(x) &= \frac{z(ab - (a+b)x)}{z-1} - tw(x) + x(x+t), \\
N(x) &= w(x)x(x+t) - \frac{zabx}{z-1}.
\end{aligned}$$

Their necessary compatibility condition is the polynomial identity

$$\mathcal{Q}(x) := (z-1)^2 \{N(x) - w(x+1)M(x)\} = 0.$$

If  $t = 1/2$ , take  $a = s, b = 1 - s$ . The coefficients of  $x^2, x, 1$  in  $\mathcal{Q}$  are, respectively,

$$\frac{z^2 - 1}{2}, \quad -\frac{z(2s-1)^2}{2}, \quad -\frac{z(2s-1)^2(z+2)}{8}.$$

They imply  $s = 1/2, z = -1$ . Otherwise, symmetry in  $a, b$  permits  $a = 1/2, b = 1 - t$ . The coefficients are

$$\begin{aligned}
& \frac{(z+1)(4tz + 2t - z - 2)}{2}, \\
& \frac{z(2t-1)(12tz + 8t - 3z + 2)}{4}, \quad \frac{z(2t-1)^2(4tz - z + 2)}{4}.
\end{aligned}$$

If  $t \neq 1/2$ , the last gives  $4tz - z = -2$  and the middle then gives  $8t - 4 = 0$ , a contradiction. The sole survivor is therefore  $s = a = b = t = 1/2, z = -1$ . At this survivor  $w = 1/4$  and  $M = x^2$ , which give the stated formulas for  $A, h, R$ . Their direct substitution proves sufficiency. The homogeneous pole and leading coefficient argument proves uniqueness as before.  $\square$

The last theorem identifies all its certificates with rational auxiliary gauges of Bauer's mechanism. A one-periodic factor in  $H$  is invisible to its rational unit-shift ratio, and is not fixed by the algebraic classification. An analytic specialization still requires regularity and the relevant convergence argument. Neither of these two theorems supplies a converse from a bare scalar value to the existence of a rational WZ certificate. Their CM conclusions are complete within the displayed certificate families, while unconditional CM forcing for an arbitrary standard identity remains unresolved.

#### APPENDIX L. FINITE EXCLUSIONS AND A MODULAR-JET FAMILY

**L.1. Certified finite exclusion in the level-four family.** Put  $U_C = \pi F_{1/2}(64/C)$  and  $V_C = \pi \vartheta F_{1/2}(64/C)$ . The following elementary lattice observation turns bounded numerical searches into exact finite exclusions.

**Lemma L.1** (A certificate excluding small integer relations). *Let  $x_1, \dots, x_n \in \mathbb{R}$ , and suppose that integers  $a_1, \dots, a_n, Q$ , with  $Q > 0$ , satisfy  $|a_i - Qx_i| \leq 1$ . Let  $\Lambda \subset \mathbb{Z}^{n+1}$  be the lattice with row basis  $[I_n \mid a]$ . If an integer basis of  $\Lambda$  has every Gram-Schmidt squared length strictly larger than  $n(n+1)H^2$ , then*

$$\sum_{i=1}^n h_i x_i = 0, \quad h_i \in \mathbb{Z}, \quad \max_i |h_i| \leq H \implies h_1 = \dots = h_n = 0.$$

*Proof.* A nonzero relation would yield the nonzero lattice vector  $(h_1, \dots, h_n, \sum_i h_i a_i)$ , whose last coordinate has absolute value at most  $nH$  and whose squared norm is therefore at most  $n(n+1)H^2$ . For any integer basis, the squared norm of a nonzero lattice vector is at least the least Gram–Schmidt squared length: take the last nonzero integer basis coefficient and project onto the corresponding orthogonalized direction. This is a contradiction.  $\square$

**Proposition L.2** (Finite exclusion certified by exact arithmetic). *For every integer  $C$  with  $65 \leq |C| \leq 1024$  and  $C \notin \{-512, 256\}$ , no nonzero polynomial  $P \in \mathbb{Z}[X, Y]$  of total degree at most two and coefficient height at most  $10^{11}$  vanishes at  $(U_C, V_C)$ . Consequently there is no identity*

$$(A + B\vartheta)F_{1/2}(64/C) = \alpha/\pi$$

*in that denominator range with  $(A, B) \neq (0, 0)$ ,  $|A|, |B| \leq 1000$ , and  $\alpha$  annihilated by a nonzero integer polynomial of degree at most two and coefficient height at most 50000.*

*Proof with a reproducible finite certificate.* For each of the 1918 indicated parameters, enclose the six monomials

$$U_C^2, U_C V_C, V_C^2, U_C, V_C, 1$$

using outward-rounded integer intervals of scale  $10^{120}$ . These enclosures follow from

$$\frac{\binom{2n+2}{n+1}^3 |C|^{-n-1}}{\binom{2n}{n}^3 |C|^{-n}} = \frac{8(2n+1)^3}{(n+1)^3 |C|} \leq \frac{64}{|C|}$$

and geometric bounds for the tails of both the series and its Euler derivative. An exact alternating-series enclosure from Machin’s formula encloses  $\pi$ . With  $Q = 10^{80}$ , nearest-midpoint rounding gives integers  $a_i$ ; both endpoint inequalities  $|a_i - Qx_i| \leq 1$  are checked exactly. Every stored integral basis is obtained by a matrix of determinant  $\pm 1$ , with the matrix identity checked. Independent rational Gram–Schmidt computations show in all 1918 cases that the least squared length exceeds  $42(10^{11})^2$ .

The accompanying scripts and the parameter-by-parameter integer bases are stored in the research archive under the stem `20260918C_numeric_level4_lattice`; verifying a stored basis does not require trusting lattice reduction. The lemma proves the first assertion. Substituting  $AX + BY$  into an annihilating polynomial for  $\alpha$  gives a nonzero polynomial in  $X, Y$ , of total degree at most two and coefficient height at most  $2 \cdot 1000^2 \cdot 50000 = 10^{11}$ , proving the second.  $\square$

This is a bounded arithmetic exclusion, not a proof of CM forcing with unrestricted coefficients or algebraic multipliers. The three analogous tests at  $C = -512, 256, 4096$  correctly fail to exclude their known small polynomial relations.

**L.2. Additional exact finite searches.** The interval method also gives exclusions for multiplier sets with unbounded numerator. At level four, take

$$65 \leq |C| \leq 10000, \quad |A| \leq 20, \quad 0 \leq B \leq 100, \quad \gcd(A, B) = 1,$$

using  $(A, B) = (1, 0)$  as the representative when  $B = 0$ . There are 19872 denominators and 2500 coefficient pairs, hence 49680000 cases. If  $\alpha$  is rational of reduced denominator at most 100, the only surviving identities are the known rows  $(1, 6, 256)$  and  $(5, 42, 4096)$ . If instead  $\alpha^2$  is rational of reduced denominator at most 100, the only additional row is  $(1, 6, -512)$ . Each surviving row is already established exactly.

For each of levels one to three, take the 127 denominators

$$\{R_\ell + 1, \dots, R_\ell + 63\} \cup \{-R_\ell, -R_\ell - 1, \dots, -R_\ell - 63\},$$

and the 502 primitive representatives with  $|A| \leq 10$ ,  $1 \leq B \leq 40$ , together with  $(1, 0)$ . No identity in this set has either  $\alpha^2$  or  $\alpha^4$  rational with reduced denominator at most 1000. These are 382524 power tests. The exact CM class-polynomial certificate verifies that every parameter in this last set is non-CM.

Here are the certification principles. Rational recurrences and geometric tail bounds give outward enclosures for the series and its Euler derivative. Near the boundary the quadratic Clausen parameter has absolute value at most  $1/2$ , so its Gauss series gives rapidly convergent enclosures, including the negative endpoint. Machin’s alternating arctangent formula supplies an independent rational enclosure for  $\pi$ . For an enclosure of  $\alpha^k$  and a denominator bound  $M$ , an exact rational nearest-fraction calculation decides whether the enclosure contains a reduced fraction with denominator at most  $M$ . The enclosures are narrower than  $1/M^2$ , the minimum separation between two such distinct fractions. Exclusion therefore does not require an upper bound on their numerators.

All endpoints and comparisons are rational. A separate integer implementation recomputes the enclosures and verifies the rational detector. The scripts and certificates, including their input and source hashes, are stored under `20260918C_numeric_level4` and `20260918C_numeric_levels123` in the attempt archive.

Higher-degree numerical integer-relation searches likewise found only known CM controls. Their null results are evidence only and are not used in any theorem or certificate above.

**L.3. An explicit non-CM test family and an exact infinite exclusion.** The affine two-jet formulation defines a family in which a counterexample would have to meet an additional arithmetic condition. The rational constants of bounded denominator below can be excluded unconditionally. Put  $J(y) = j(iy)$  for  $y > 1$ , with primes on  $J$  denoting real  $y$ -derivatives.

**Proposition L.3.** *The function  $H(y) = J''(y)/J'(y)$  is a strictly decreasing bijection from  $(1, \infty)$  to  $(2\pi, \infty)$ . For  $c > 2\pi$ , let  $H(y_c) = c$ ,  $C_c = J(y_c)$ , and  $t_c = \sqrt{1 - 1728/C_c}$ . Then  $C_c > 1728$  is strictly decreasing in  $c$ , and*

$$[(864 - C_c) + (C_c - 1728)\vartheta]F_{1/6}(1728/C_c) = -\frac{c(C_c - 1728)}{2\pi t_c}. \quad (\text{L.1})$$

The series converges absolutely. If  $c$  is algebraic, its elliptic fiber is non-CM. Nevertheless,  $C_c \notin \mathbb{Z}$  for every integer  $c > 2\pi$ .

*Proof.* Writing  $q = e^{-2\pi y}$ , positivity of the coefficients in  $qj = E_4^3 \prod_{n \geq 1} (1 - q^n)^{-24}$  gives

$$J = q^{-1} + 744 + \sum_{n \geq 1} a_n q^n, \quad a_n > 0, \quad U = \sum_{n \geq 1} n a_n q^{n+1}, \quad V = \sum_{n \geq 1} n^2 a_n q^{n+1}.$$

Consequently  $J' = 2\pi q^{-1}(1 - U)$  and  $H = 2\pi(1 + V)/(1 - U)$ . Since  $j'(i) = 0$ , one has  $U(1) = 1$ . Both  $U$  and  $V$  strictly decrease with  $y$ , proving that  $H$  strictly decreases from infinity to  $2\pi$ . Also  $J$  strictly increases from 1728 to infinity. These facts prove existence, uniqueness, and monotonicity.

For the standard level-one model  $g_2 = 3$ ,  $g_3 = t$ ,

$$F_{1/6} = \frac{3\omega^2}{2\pi^2}, \quad \frac{\vartheta F_{1/6}}{F_{1/6}} = -\frac{1}{6} - \frac{h}{3t\omega}.$$

Thus the quadratic period coefficients for  $(A + B\vartheta)F_{1/6}$  are  $a = 3A/2 - B/4$  and  $b = -B/(2t)$ . The period-jet dictionary gives

$$bj'' + \left(\frac{a}{3jt} - bR(j)\right)j'^2 + i\alpha j' = 0, \quad R(j) = \frac{2}{3j} + \frac{1}{2(j - 1728)}.$$

Its  $j'^2$  coefficient vanishes precisely when  $A/B = -(j - 864)/(j - 1728)$ . Since  $j''(iy_c)/j'(iy_c) = -ic$ , choosing  $A = 864 - C_c$ ,  $B = C_c - 1728$  yields  $\alpha = bc = -c(C_c - 1728)/(2t_c)$ , proving (L.1).

To exclude CM when  $c$  is algebraic, use the exact formula

$$\frac{j''}{j'} = \left(R(j) - \frac{s_2}{6j}\right)j' + \frac{i}{y}, \quad s_2 = \frac{E_4}{E_6} \left(E_2 - \frac{3}{\pi y}\right).$$

At a CM point on this branch,  $j, s_2, y$  are algebraic, while  $j'$  is transcendental. Here  $E_6 \neq 0$ , so the algebraicity assertion for  $s_2$  in [22, Theorem 8.1] applies. Comparing with  $j''/j' = -ic$  would force  $-c = 1/y$ , which is impossible.

It remains to exclude integer  $C_c$  for integer  $c$ . This last claim uses a finite exact certificate. The archived verifier `20260918C_explicit_jet_search.py` and its accompanying JSON file contain rational brackets  $q_- < q_+$  for all  $c = 7, \dots, 200$ . The brackets have denominator  $10^{65}$ . Their validity is checked by 384-bit directed dyadic interval arithmetic using

$$H = \frac{\pi}{3} \left(4 \frac{E_6}{E_4} + 3 \frac{E_4^2}{E_6} - E_2\right), \quad J = \frac{1728 E_4^3}{E_4^3 - E_6^2}.$$

The Eisenstein sums are truncated after 50 terms. For  $k = 1, 3, 5$  and  $0 < q < 1/500$ , their omitted sums satisfy the elementary bound

$$0 \leq \sum_{n > 50} \sigma_k(n) q^n \leq 2 \cdot 51^{k+1} q^{51},$$

since  $\sigma_k(n) \leq n^{k+1}$  and the successive majorant ratios are at most  $(1/500)(52/51)^{k+1} < 1/2$ . A rational enclosure of  $\pi$  follows from Machin's formula using 110 alternating terms in each arctangent. Every bracket has certified  $E_6 > 0$ , so it lies on the required branch. For each of the 194 brackets, the verifier establishes  $H(q_-) < c < H(q_+)$  and places  $[J(q_+), J(q_-)]$  strictly between consecutive integers. Approximate arithmetic proposes the brackets only; their replay uses exact integers. The smallest certified distance to an integer over this range exceeds  $0.0032177275685462482644977288$ , at  $c = 156$ . In particular,

$$1728.6093005038821554072482832073 < C_{200} < 1728.6093005038821554072482832074 < 1729.$$

Finally, every integer  $c > 2\pi$  is at least 7, and monotonicity gives  $1728 < C_c \leq C_{200} < 1729$  for  $c \geq 200$ . The finite certificate and this last inequality cover all integer  $c$ .  $\square$

The proposition excludes a genuine candidate mechanism, rather than a normalization defect. If an algebraic noninteger  $c > 2\pi$  had integer  $C_c$ , division of  $A = 864 - C_c$  and  $B = C_c - 1728$  by  $\gcd(C_c - 864, C_c - 1728)$  in (L.1) would yield a primitive standard identity at a non-CM fiber. Algebraicity of such a  $c$  remains unexcluded. Equivalently, the unconditional result above proves  $H(y(C)) \notin \mathbb{Z}$  for integer  $C > 1728$ ; it does not prove  $H(y(C)) \notin \overline{\mathbb{Q}}$ .

**Proposition L.4.** *In the notation of Proposition L.3, if  $C_c$  is an integer and  $c = a/b$  is rational in lowest terms, with  $b > 0$ , then  $b > 5$ . More generally, for all rational  $c = a/b$  in this family,*

$$b \gg C_c^{125/444} \quad (C_c \rightarrow \infty).$$

*Consequently, for fixed  $L > 0$  and  $\gamma < 125/444$ , only finitely many integer fibers in this family can have  $b \leq LC_c^\gamma$ . The implied constant is not made effective here.*

*Proof.* The certificate 20260918C\_jet\_rational.json contains all 1938 reduced rational numbers  $c$  with  $1 \leq b \leq 5$  and  $2\pi < c \leq 200$ . The endpoint list is determined using the exact Machin bounds from Proposition L.3; its smallest element is  $19/3$ . The accompanying verifier checks every rational root bracket by the dyadic method of that proposition and by a separately implemented decimal-grid interval method. The latter uses 45 direct Eisenstein terms and the uniform tail  $2 \cdot 46^{k+1}/500^{46}$  for  $k = 1, 3, 5$ . Both verifiers prove that each  $C_c$  lies strictly between consecutive integers. This adds 1744 nonintegral rational constants to the earlier 194 integer constants. The smallest certified distance to an integer is greater than

$$0.000048699599821220308647971801,$$

at  $c = 623/4$ . In particular,

$$1728.9999513004001787796913520281 < C_{623/4} < 1728.9999513004001787796913520282 < 1729.$$

Monotonicity now excludes integer  $C_c$  for every real  $c \geq 623/4$ ; together with the finite certificate, this proves  $b > 5$ .

For the growth statement, the cusp expansion

$$J = q^{-1} + 744 + 196884q + O(q^2)$$

and the functions  $U, V$  in Proposition L.3 give  $U = 196884q^2 + O(q^3)$  and  $V = 196884q^2 + O(q^3)$ . Therefore

$$H = 2\pi \frac{1+V}{1-U} = 2\pi + 787536\pi q^2 + O(q^3).$$

Since  $q = C_c^{-1} + O(C_c^{-2})$ , a rational value  $H = a/b$  satisfies

$$0 < \frac{a}{2b} - \pi = \frac{393768\pi}{C_c^2} + O(C_c^{-3}).$$

The denominator of  $a/(2b)$  in lowest terms is at most  $2b$ . Applying the Zeilberger–Zudilin bound (E.3), with  $\mu_* = 7.104 = 888/125$ , yields

$$c_*(2b)^{-888/125} \ll C_c^{-2},$$

and hence  $b \gg C_c^{125/444}$ . If  $b \leq LC_c^\gamma$  with  $\gamma < 125/444$ , this bounds  $C_c$ ; integrality and the uniqueness of  $c$  at each fiber prove finiteness.  $\square$

The primitive slopes in this family have a simple description. Put  $d = \gcd(C_c - 1728, 864)$  and  $k = 864/d$ . Then

$$(A, B) = (-B - k, B), \quad C_c = 1728 + \frac{864}{k}B, \quad \gcd(B, k) = 1,$$

where  $k$  ranges over the 24 positive divisors of 864. For rational  $c$ , the proposed multiplier satisfies  $\alpha^2 = c^2 C_c (C_c - 1728)/(4d^2) \in \mathbb{Q}$ . Thus this restriction addresses a specific part of the quadratic multiplier case. It does not exclude rational  $c$  of arbitrary denominator, or algebraic irrational  $c$ .

#### DISCLOSURE OF AI ASSISTANCE

OpenAI Codex assisted with mathematical exploration and critique, literature cross-checks, proposed proofs and counterexamples, symbolic and numerical computations, manuscript drafting and revision, and adversarial review in separate AI sessions. These reviews are AI-assisted checks, not independent human peer review or formal proof verification. The text distinguishes cited results, proved deductions, conditional statements, and finite computational certificates. The author attribution is retained at the author's request; no claim of subsequent human validation is implied. The original 118-page manuscript and its research records are preserved separately.

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